Words of Concern

Point to be As Allan Bloom has said "Education is the movement from darkness to light". Through this handbook, I have tried to illuminate what might otherwise appear as black boxes to some. In doing so, I have used references from several other authors to synthesize or simplify or elaborate information. This is not possible without omitting details that I deem trivial while dilating the data that I consider relevant to topic. Every effort has been made to avoid errors. In spite of this, some errors might have crept in. Any errors or discrepancies noted maybe brought to my notice which I shall rectify in my next revision.

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Thanks,
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References

1. The-Morgan-Kaufmann-Series-in-Data-Management-Systems-Jiawei-Han-Micheline-Kamber-Jian-Pei-Data-Mining.-Concepts-and-Techniques-3rd-Edition-Morgan-Kaufmann-2011.
2. https://www.geeksforgeeks.org/data-warehousing/

Syllabus:
Course Code: DSC18
Course Title: Data Warehouse and Data Mining
Credits - 04
Sem - VI
Hours- 52 Theory
Formative Assessment Marks: 40
Summative Assessment Marks: 60
Duration of SEA/Exam: 02 hrs.

Course Outcomes
After the successful completion of the course, the student will be able to:

1. To understand the principles of data warehouses and Data Mining.
2. To be familiar with the Data warehouse architecture and its implementation.
3. To know the architecture of a Data mining system.
4. To understand the various Data preprocessing Methods.

Unit Content **Duration**

Unit 1
Data Warehousing and Business Analysis: Data Warehousing Components - Building a Data warehouse-Data Warehouse Architecture - DBMS Schemas for Decision Support - Data Extraction, Cleanup, and Transformation Tools - Metadata reporting Query tools and Applications - Online Analytical Processing (OLAP)- OLAP and Multidimensional Data Analysis. 10 Hr

UNIT-2:
Data Mining: Data Mining Functionalities Data Preprocessing Data Cleaning Data Integration and Transformation - Data Reduction - Data Discretization and Concept Hierarchy Generation- Architecture Of A Typical Data Mining Systems- Classification Of Data Mining Systems. Association Rule Mining: Efficient and Scalable Frequent Item Set Mining Methods - Mining Various Kinds of Association Rules-Association Mining to Correlation Analysis - Constraint-Based Association Mining. 12 Hr

UNIT-3:
Classification and Prediction: Issues Regarding Classification and Prediction - Classification by Decision Tree Introduction - Bayesian Classification - Rule Based Classification - Classification by Backpropagation Support Vector Machines - Associative Classification - Lazy Learners - Other Classification Methods - Prediction - Accuracy and Error Measures - Evaluating the Accuracy of a Classifier or Predictor Ensemble Methods - Model Section. 10 Hr

UNIT-4:

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Cluster Analysis: Types of Data in Cluster Analysis - A Categorization of Major Clustering Methods Partitioning Methods - Hierarchical methods - Density-Based Methods - Grid-Based Methods- Model-Based Clustering Methods Clustering High-Dimensional Data Constraint-Based Cluster Analysis-Outlier Analysis. 10 Hr

UNIT-5:

Mining Object, Spatial, Multimedia, Text, and Web Data: Multidimensional Analysis and Descriptive Mining of Complex Data Objects - Spatial Data Mining - Multimedia Data Mining - Text Mining-Mining the World Wide Web. 10 Hr

52 Hr

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UNIT 1

DATA WAREHOUSING AND BUSINESS ANALYSIS

A data warehouse, or enterprise data warehouse (EDW), is a system that aggregates data from different sources into a single, central, consistent data store to support data analysis, data mining, artificial intelligence (AI), and machine learning.

Data warehousing provides architectures and tools for business executives to systematically organize, understand, and use their data to make strategic decisions.

Data warehouse systems are valuable tools in today's competitive, fast-evolving world. In the last several years, many firms have spent millions of dollars in building enterprise-wide data warehouses. Many people feel that with competition mounting in every industry, data warehousing is the latest must-have marketing weapon, a way to retain customers by learning more about their needs.

Data warehouses have been defined in many ways, making it difficult to formulate a rigorous definition. Loosely speaking, **a data warehouse refers to a database that is maintained separately from an organization's operational databases. Data warehouse systems allow for the integration of a variety of application systems. They support information processing by providing a solid platform of consolidated historical data for analysis.**

Data warehousing tools

- **Amazon Redshift**
- **BigQuery**
- **Teradata**
- **Snowflake**
- **Amazon S3**

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How are organizations using the information from data warehouses?

Many organizations use this information to support business decision-making activities, including

(1) Increasing customer focus, which includes the analysis of customer buying patterns (such as buying preference, buying time, budget cycles, and appetites for spending)

(2) Repositioning products and managing product portfolios by comparing the performance of sales by quarter, by year, and by geographic regions to fine-tune production strategies

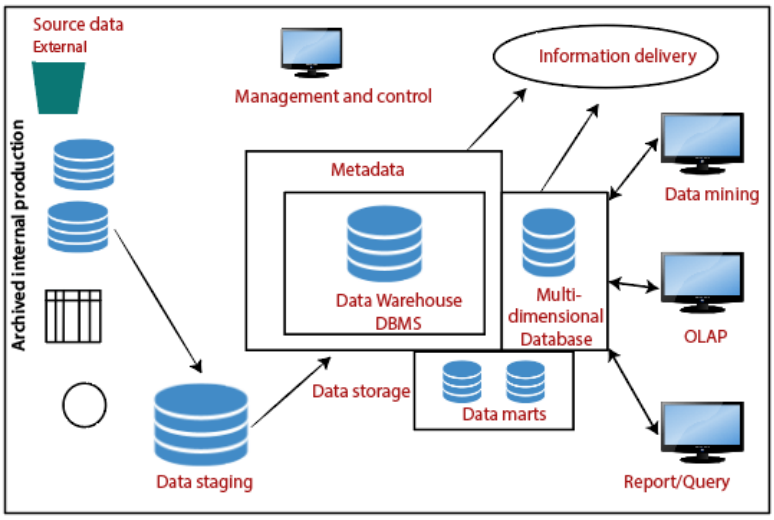
(3) Analyzing operations and looking for sources of profit.

(4) Managing customer relationships, making environmental corrections, and managing the cost of corporate assets.

A Data Warehouse is a centralized repository that stores integrated data from multiple sources. It is used for querying, reporting, and data analysis to support decision-making.

DATA WAREHOUSING COMPONENTS

A Data Warehouse (DW) is a relational database that is designed for query and analysis rather than transaction processing. It includes historical data derived from transaction data from single and multiple sources.
A Data Warehouse provides integrated, enterprise-wide, historical data and focuses on providing support for decision-makers for data modeling and analysis.



Components or Building Blocks of Data Warehouse

Source Data Component

- **Production Data:** This type of data comes from the different operating systems of the enterprise. Based on the data requirements in the data warehouse, we choose segments of the data from the various operational modes.
- **Internal Data:** In each organization, the client keeps their "private" spreadsheets, reports, customer profiles, and sometimes even department databases. This is the internal data, part of which could be useful in a data warehouse.
- **Archived Data:** Operational systems are mainly intended to run the current business. In every operational system, we periodically take the old data and store it in achieved files.
- **External Data:** Most executives depend on information from external sources for a large percentage of the information they use. They use statistics associated with their industry produced by the external department.

Data Staging Component

- After we have extracted data from various operational systems and external sources, we have to prepare the files for storage in the data warehouse. The extracted data coming from several different sources need to be changed, converted, and made ready in a format that is relevant to be saved for querying and analysis.

ETL (Extract, Transform, Load)

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- These are tools and processes that extract data from source systems, transform it into a consistent format, and load it into the data warehouse.

Data Storage Components

- Data storage for data warehousing is a split repository. The data repositories for the operational systems generally include only the current data. Also, these data repositories include the data structured in highly normalized for fast and efficient processing.

Metadata Component

- Metadata in a data warehouse is equal to the data dictionary or the data catalog in a database management system. In the data dictionary, we keep the data about the logical data structures, the data about the records and addresses, the information about the indexes, and so on.

Data Mart

- A subset of the data warehouse that is focused on a specific business function or department. Data marts are often used to provide quicker access to relevant data for specific user groups.

OLAP Engine
Allows fast querying and analysis of multidimensional data.

Enables slicing, dicing, drill-down, and roll-up operations.

Data Access Tools

- (Ex: Microsoft Power BI, Oracle OLAP, and Tableau) These are tools that allow users to query, analyze, and report on the data in the data warehouse. They can include reporting tools, OLAP (Online Analytical Processing) tools, data mining tools (Ex: Teradata, oracle data mining), and visualization tools (Ex: Tableau, power B I, Google Charts).

Data Management and Administration

- This includes activities such as data quality monitoring, security management, performance optimization, and user management.

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BUILDING A DATA WAREHOUSE

The Design of a Data Warehouse: A Business Analysis Framework

- First, having a data warehouse may provide a competitive advantage by presenting relevant information from which to measure performance and make critical adjustments to help win over competitors.
- Second, a data warehouse can enhance business productivity because it can quickly and efficiently gather information that accurately describes the organization.
- Third, a data warehouse facilitates customer relationship management because it provides a consistent view of customers and items across all lines of business, all departments, and all markets.
- Finally, a data warehouse may bring about cost reduction by tracking trends, patterns, and exceptions over long periods consistently and reliably.

- The construction of a large and complex information system can be viewed as the construction of a large and complex building, for which the owner, architect, and builder have different views.
- These views are combined to form a complex framework that represents the top-down, business-driven, or owner's perspective, as well as the bottom-up, builder-driven, or implementor's view of the information system.

Four different views regarding the design of a data warehouse must be considered: the top-down view, the data source view, the data warehouse view, and the business query view.

- The **top-down view** allows the selection of the relevant information necessary for the data warehouse. This information matches current and future business needs.
- The **data source view** exposes the information being captured, stored, and managed by operational systems. This information may be documented at various levels of detail and accuracy, from individual data source tables to integrated data source tables. Data sources are often modeled by traditional data modeling techniques, such as
    - The entity-relationship model or
    - CASE (computer-aided software engineering) tools.
- The **data warehouse view** includes fact tables and dimension tables. It represents the information that is stored inside the data warehouse, including precalculated totals and counts, as well as information regarding the source, date, and time of origin, added to provide historical context.
- Finally, **the business query view** is the data perspective in the data warehouse from the end-user's viewpoint.

- Building and using a data warehouse is a complex task because it requires business skills, technology skills, and program management skills.
- Regarding business skills, building a data warehouse involves understanding how such systems store and manage their data, how to build extractors that transfer data from the operational system to the data warehouse, and how to build warehouse refresh software that keeps the data warehouse reasonably up-to-date with the operational system's data.
- Using a data warehouse involves understanding the significance of the data it contains, as well as understanding and translating the business requirements into queries that can be satisfied by the data warehouse.

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The Process of Data Warehouse Design

A data warehouse can be built using a top-down approach, a bottom-up approach, or a combination of both.

- The **top-down approach** starts with overall design and planning. It is useful in cases where the technology is mature and well-known, and where the business problems that must be solved are clear and well-understood.
- The **bottom-up approach** starts with experiments and prototypes. This is useful in the early stage of business modeling and technology development. It allows an

organization to move forward at considerably less expense and to evaluate the technological benefits before making significant commitments.

- In the **combined approach,** an organization can exploit the planned and strategic nature of the top-down approach while retaining the rapid implementation and opportunistic application of the bottom-up approach.

In general, the warehouse design process consists of the following steps:

1. Choose a business process to model,
   - for example, orders, invoices, shipments, inventory, account administration, sales, or the general ledger.
   - If the business process is organizational and involves multiple complex object collections, a data warehouse model should be followed.
   - However, if the process is departmental and focuses on the analysis of one kind of business process, a data mart model should be chosen.
2. Choose the grain of the business process.
   - The grain is the fundamental, atomic level of data to be represented in the fact table for this process, for example, individual transactions, individual daily snapshots, and so on.
3. Choose the dimensions that will apply to each fact table record.
   - Typical dimensions are time, item, customer, supplier, warehouse, transaction type, and status.
4. Choose the measures that will populate each fact table record.
   - Typical measures are numeric additive quantities like dollars sold and units sold.

DATA WAREHOUSE ARCHITECTURE

Multi-Tiered Data Warehouse Architecture

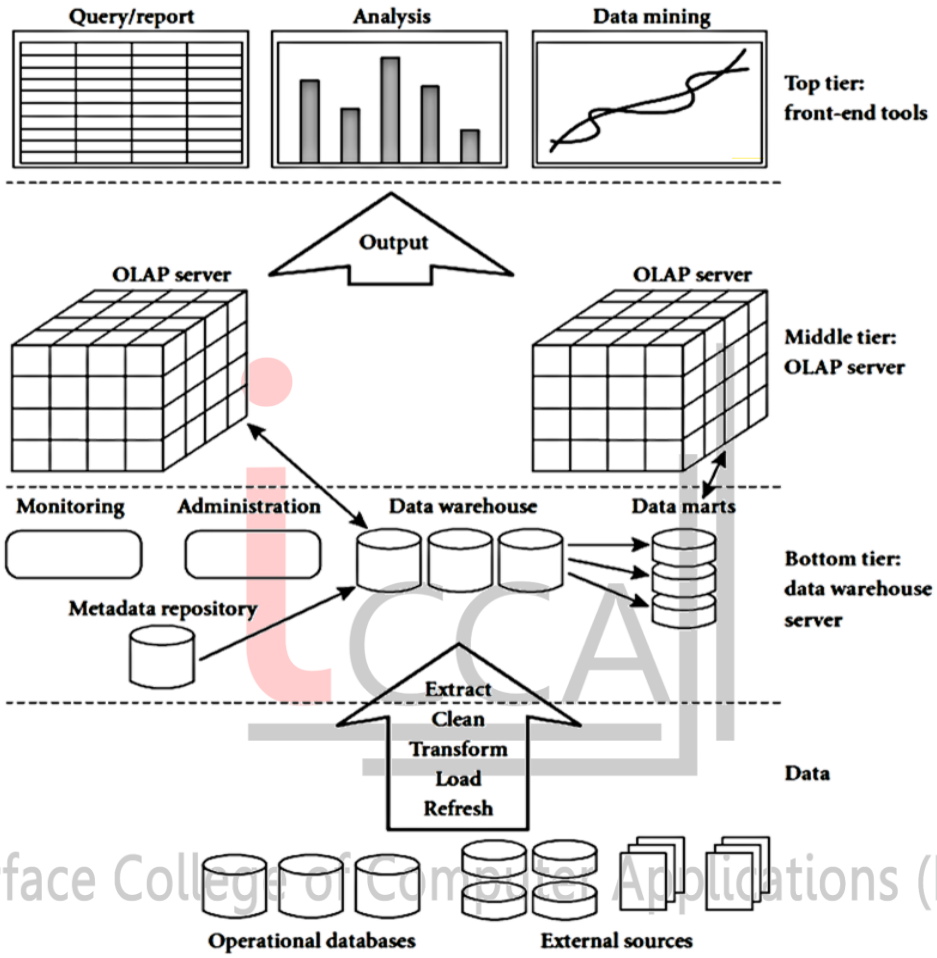
1. **The bottom tier** is a warehouse database server that is almost always a relational database system.
   - Back-end tools and utilities are used to feed data into the bottom tier from operational databases or other external sources (such as customer profile information provided by external consultants).
   - These tools and utilities perform data extraction, cleaning, and transformation (e.g., to merge similar data from different sources into a unified format), as well as load and refresh functions to update the data warehouse.
   - The data are extracted using application program interfaces known as gateways. The underlying DBMS supports a gateway and allows client programs to generate SQL code to be executed on a server.
   - Examples of gateways include ODBC (Open Database Connection) and OLEDB (Open Linking and Embedding for Databases) by Microsoft and JDBC (Java Database Connection).
   - This tier also contains a metadata repository, which stores information about the data warehouse and its contents.

2. The middle tier is an OLAP server that is typically implemented using either

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(1) a relational OLAP (ROLAP) model, that is, an extended relational DBMS that maps operations on multidimensional data to standard relational operations; or
(2) a multidimensional OLAP (MOLAP) model, that is, a special-purpose server that directly implements multidimensional data and operations.

3. The top tier is a front-end client layer, which contains query and reporting tools, analysis tools, and/or data mining tools (e.g., trend analysis, prediction, and so on).



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Figure: A Multi-tiered Data Warehouse Architecture

DBMS SCHEMAS FOR DECISION SUPPORT

Schema is a logical description of the entire database. It includes the name and description of records of all record types including all associated data items and aggregates. Much like a database, a data warehouse also requires to maintain a schema. A database uses a relational model, while a data warehouse uses Star, Snowflake, and Fact Constellation schemas

Star Schema

- A star schema is a type of relational database schema that is composed of a single, central fact table that is surrounded by dimension tables.
- Each dimension in a star schema is represented with only one-dimension table.
- This dimension table contains the set of attributes.
- The following diagram shows the sales data of a company concerning the four dimensions, namely time, item, branch, and location.
- There is a fact table at the center. It contains the keys to each of the four dimensions.
- The fact table also contains the attributes, namely dollars sold and units sold.

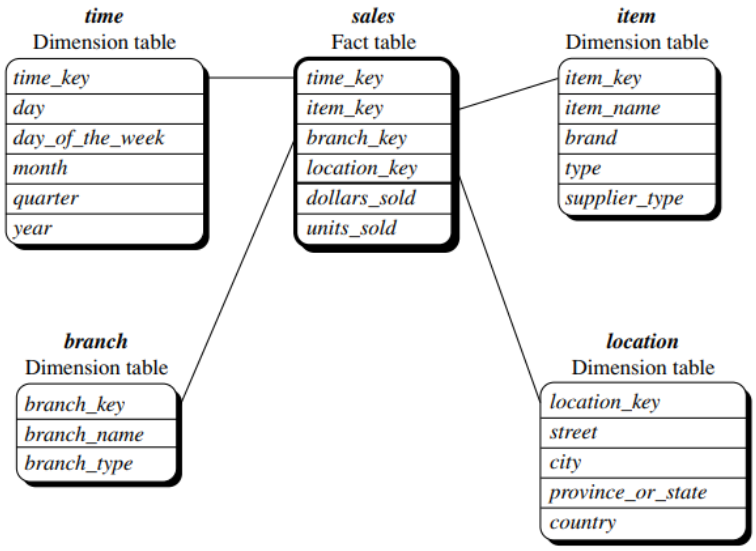


Figure: Star schema of a sales data warehouse

Snowflake Schema

- A snowflake schema is a multi-dimensional data model that is an extension of a star schema, where dimension tables are broken down into subdimensions. Snowflake schemas are commonly used for business intelligence and reporting in OLAP data warehouses, data marts, and relational databases.
- Some dimension tables in the Snowflake schema are normalized.
- The normalization splits up the data into additional tables.
- Unlike Star schema, the dimensions table in a snowflake schema is normalized. For example, the item dimension table in a star schema is normalized and split into two dimension tables, namely item and supplier table.
- Now the item dimension table contains the attributes item_key, item_name, type, brand, and supplier-key.
- The supplier key is linked to the supplier dimension table. The supplier dimension table contains the attributes supplier_key and supplier_type.

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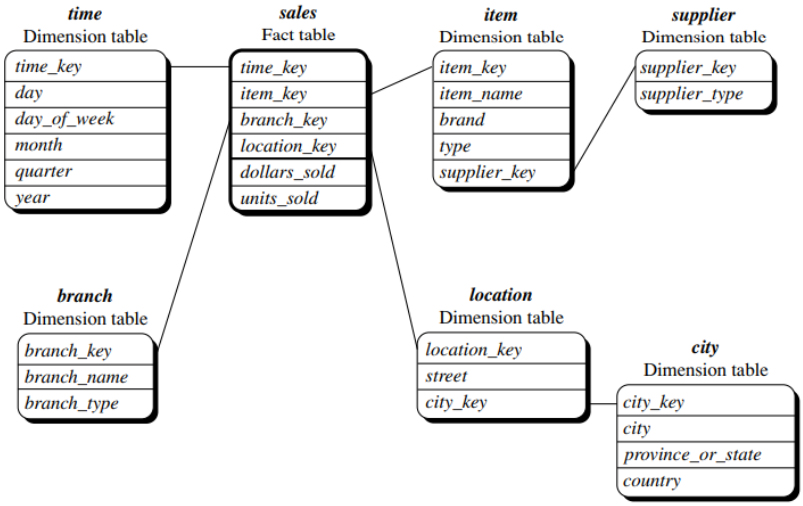


Figure: Snowflake schema of a sales data warehouse

Fact Constellation Schema

- Fact Constellation is a schema for representing multidimensional model. It is a collection of multiple fact tables having some common dimension tables. It can be viewed as a collection of several star schemas
- A fact constellation has multiple fact tables. It is also known as galaxy schema.
- The following diagram shows two fact tables, namely sales and shipping.
- The sales fact table is the same as that in the star schema.
- The shipping fact table has five dimensions, namely item_key, time_key, shipper_key, from_location, to_location.
- The shipping fact table also contains two measures, namely dollars sold and units sold.
- It is also possible to share dimension tables between fact tables. For example, time, item, and location dimension tables are shared between the sales and shipping fact tables.

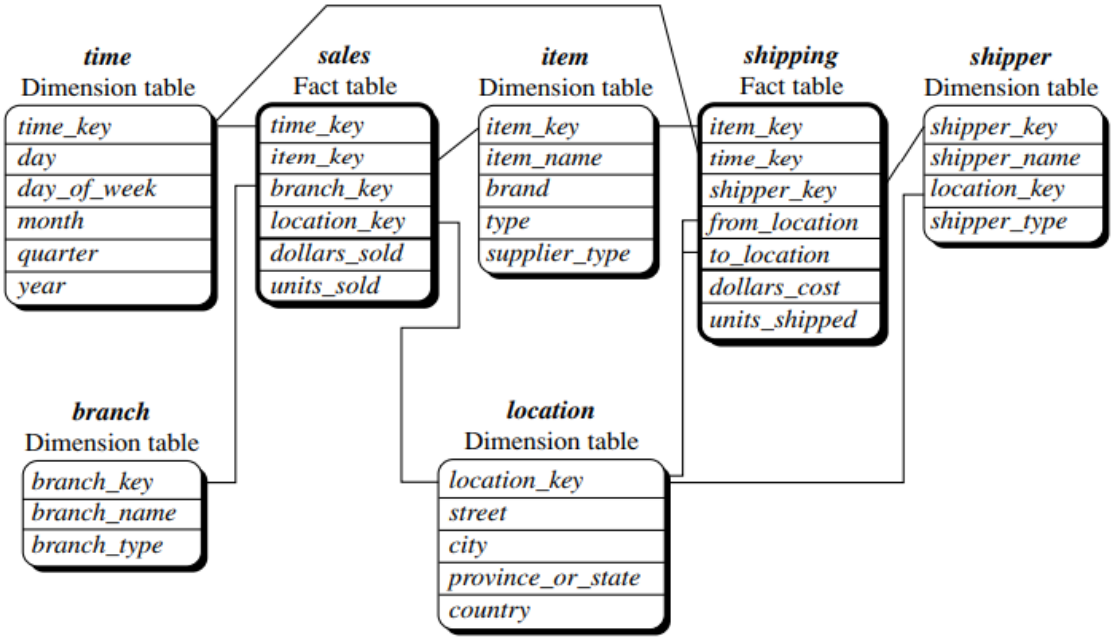


Figure: Fact Constellation schema of a sales shipping data warehouse

DATA EXTRACTION, CLEANUP, AND TRANSFORMATION TOOLS

Data Extraction Tools

Data extraction is the act or process of retrieving data out of (usually unstructured or poorly structured) data sources for further data processing or data storage.

- **Octoparse:** A user-friendly web scraping tool that allows users to extract data from websites without coding knowledge.
- **ParseHub:** Another web scraping tool that offers both point-and-click interface and advanced features for extracting data from dynamic websites.
- **WebHarvy:** A visual web scraping software that enables users to scrape data from web pages and save it in various formats like CSV, Excel, XML, and more.
- **Import.io:** A powerful data extraction platform that offers both automatic and custom data extraction solutions for businesses.
- **Scrapy:** An open-source and collaborative web crawling framework for Python. It allows users to extract data from websites and save it in structured formats like JSON, CSV, or XML.

Data Clean Up Tools

Data cleaning is the process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset.

- **OpenRefine:** Clean and organize messy data sets with features like clustering and faceting.
- **Trifacta Wrangler:** Visually explore, clean, and transform complex data sets with automated suggestions.
- **Talend Data Preparation:** Self-service tool for cleansing, enriching, and blending data from various sources.
- **DataCleaner:** Open-source solution to discover and correct errors in data with profiling and validation rules.
- **Data Ladder:** Identify and eliminate duplicate records with features like fuzzy matching and intelligent clustering.

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Transformation tools in data warehouse

In the context of a data warehouse, transformation tools are essential for shaping and preparing data for analysis and reporting. These tools are typically part of the broader Extract, Transform, Load (ETL) process. Here are some transformation tools commonly used in the context of data warehousing:

- **Apache NiFi:** Open-source tool for automating data flows between systems, great for ingesting and transforming data within data warehouses.
- **Informatica PowerCenter:** Enterprise-level ETL tool for integrating and transforming data from various sources, ideal for large-scale data warehousing projects.

- **AWS Glue:** Fully managed ETL service by AWS, simplifies data preparation and loading into data warehouses, especially those on AWS.
- **Matillion:** Cloud-native ETL solution for cloud data warehouses like Amazon Redshift and Google BigQuery, with a user-friendly interface for building transformation workflows.
- **Snowflake Data Cloud:** Cloud-based data warehouse platform with built-in data transformation capabilities, allowing users to perform transformations directly within Snowflake.

Metadata reporting Query tools and Applications

Metadata is simply defined as data about data. The data that is used to represent other data is known as metadata. For example, the index of a book serves as metadata for the contents of the book. In other words, we can say that metadata is the summarized data that leads us to detailed data.

Metadata reporting, the process of gathering and presenting information about your data itself, plays a crucial role in data governance, data quality, and understanding your data landscape. Here's a breakdown of tools and applications you can leverage:

Query Tools:

- **SQL:** Standard SQL queries can be used to extract metadata from databases, especially for structured data. Tools like **MySQL Workbench, pgAdmin, or Microsoft SQL Server Management Studio** allow you to run queries and view results.
- **Metadata APIs:** Many data platforms and tools offer APIs specifically for accessing metadata. Platforms like **Databricks, Snowflake, and Azure Synapse** provide programmatic access to schema information, lineage, and other metadata details.
- **Data Catalog Tools:** Dedicated data catalog tools like **Alteryx Data Catalog, Collibra, and Azure Purview** offer user-friendly interfaces and comprehensive search functionalities to query and explore metadata across diverse data sources.

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Reporting Tools:

- **Spreadsheet Software:** While basic, tools like Microsoft Excel or Google Sheets can be used to structure and present metadata reports with manual data extraction and formatting.
- **Data Visualization Tools:** Data visualization tools like Tableau, Power BI, and Looker can create visually appealing and interactive reports based on metadata, showcasing data relationships, lineage, and distribution.
- **Reporting Automation Tools**: Dedicated metadata reporting tools like Ambeo, Metadata.io, and Collibra Data Lineage automate report generation and scheduling, ensuring consistency and reducing manual effort.

Applications:

- **Data Governance:** Metadata reporting facilitates data governance practices by providing insights into data ownership, access controls, and usage patterns.
- **Data Quality Management**: Reports on data completeness, consistency, and validity help identify and address data quality issues.
- **Data Migration and Integration**: Analyzing lineage and relationships through metadata reports empowers informed decisions during data migration and integration projects.
- **Compliance and Security:** Metadata reports are crucial for demonstrating compliance with regulations and ensuring data security by tracking access and usage.

ONLINE ANALYTICAL PROCESSING (OLAP)

OLAP stands for Online Analytical Processing Server. It is a software technology that allows users to analyze information from multiple database systems at the same time. It is based on a multidimensional data model and allows the user to query multi-dimensional data.

Types of OLAP Servers

- **Relational OLAP (ROLAP)**
- **Multidimensional OLAP (MOLAP)**
- **Hybrid OLAP (HOLAP)**

ROLAP (Relational OLAP):

- These are the intermediate servers that stand in between a relational back-end server and client front-end tools.
- They use a relational or extended-relational DBMS to store and manage warehouse data, and queries are processed using SQL.
- ROLAP servers offer greater flexibility in terms of data storage and support for ad-hoc queries but may suffer from slower query performance compared to MOLAP.

MOLAP (Multidimensional OLAP):

- These servers support multidimensional data views through array-based multidimensional storage engines. They map multidimensional views directly to data cube array structures.
- Data is pre-aggregated and stored in multidimensional cubes, allowing for rapid analysis and exploration of data along multiple dimensions.
- MOLAP servers typically provide high performance and scalability but may require significant storage space for pre-aggregated data.

HOLAP (Hybrid OLAP):

- HOLAP servers combine features of both MOLAP and ROLAP.
- They use a combination of multidimensional cubes and relational tables to store and process data.
- HOLAP servers aim to leverage the performance benefits of MOLAP for aggregated data while providing the flexibility of ROLAP for detailed data analysis.

OLAP Operations

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Five basic analytical operations can be performed on an OLAP cube:

Drill down: It navigates from less detailed data to more detailed data.
It allows users to explore data hierarchies and gain insights at different levels of granularity. For example, drilling down from yearly sales to quarterly sales, then to monthly sales, and finally to daily sales.

Roll-up: Roll-up is the opposite of drill-down, involving summarizing detailed data to higher levels of aggregation. It allows users to view data at more aggregated levels to understand overall trends and patterns. For example, rolling up daily sales to monthly sales, then to quarterly sales, and finally to yearly sales.

Dice: It selects a sub-cube from the OLAP cube by selecting two or more dimensions. In the cube given in the overview section, a sub-cube is selected.
It allows users to view data from different perspectives simultaneously. For example, dicing sales data by product category and time period to analyze sales trends.

Slice: It selects a single dimension from the OLAP cube which results in a new sub-cube creation.
It allows users to filter the data to focus on specific criteria or conditions. For example, slicing sales data by a particular product category or region.

Pivot: It is also known as rotation operation as it rotates the current view to get a new view of the representation. In the sub-cube obtained after the slice operation, performing the pivot operation gives a new view of it.

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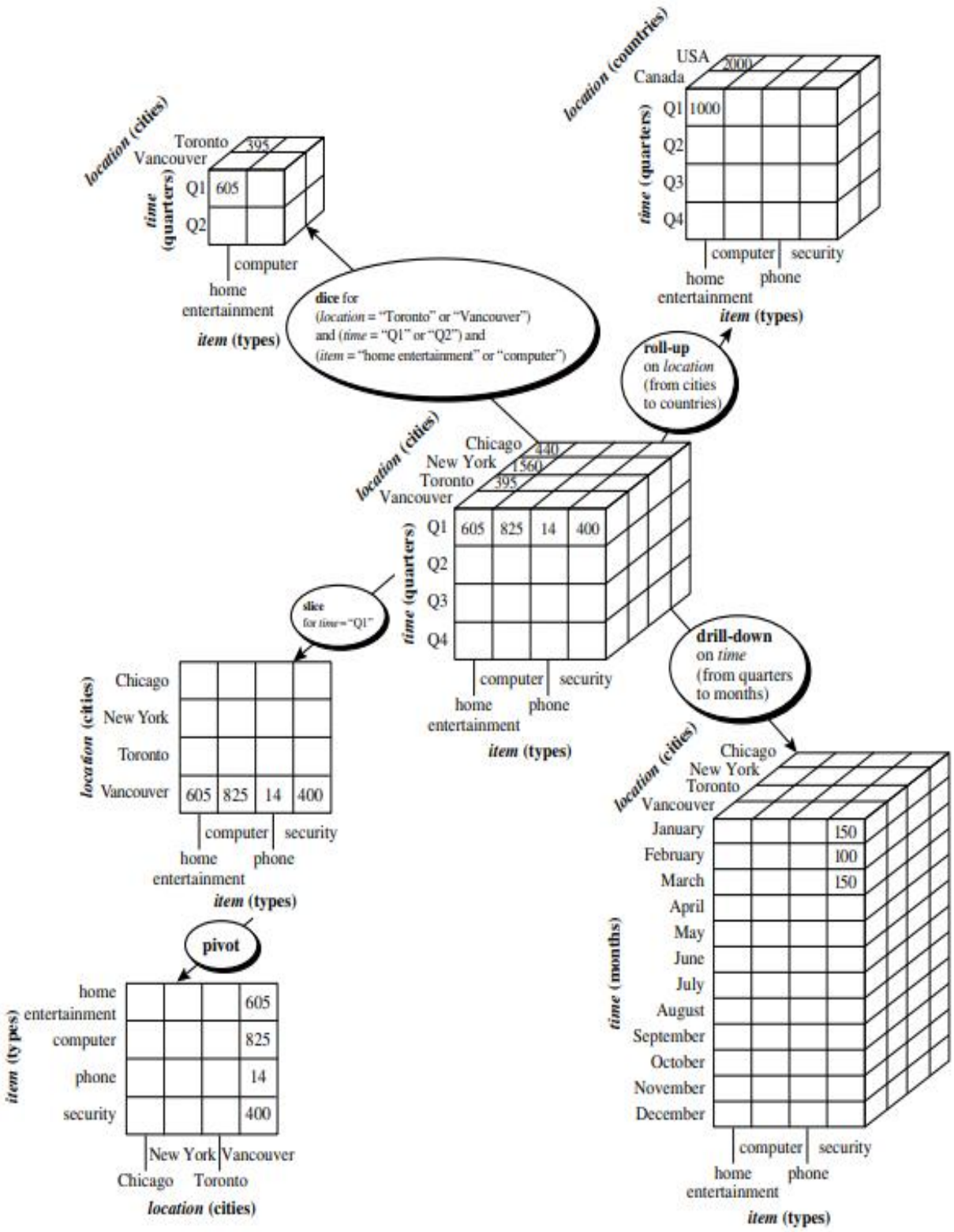


Figure: Online Analytical Processing Operations

OLAP AND MULTIDIMENSIONAL DATA ANALYSIS

OLAP (Online Analytical Processing) and multidimensional data analysis are key concepts in the field of data warehousing and business intelligence. Here's a concise overview of each:

OLAP (Online Analytical Processing)
OLAP refers to a category of data processing that enables users to interactively analyze multidimensional data from multiple perspectives. It provides the capability for complex calculations, trend analysis, and sophisticated data modeling.

Key Features of OLAP:

1. **Multidimensional Analysis:** Allows users to view data across multiple dimensions (e.g., time, geography, product categories).
2. **Complex Calculations:** Supports advanced calculations like aggregations, percentages, and comparisons.
3. **Interactive Data Exploration:** Enables dynamic querying and drilling down into data for more detailed insights.
4. **Fast Query Performance:** Optimized for quick retrieval of data from large datasets.

Types of OLAP:

1. **MOLAP (Multidimensional OLAP):** Uses multidimensional data cubes to store data. Example: Microsoft Analysis Services.
2. **ROLAP (Relational OLAP):** Queries data directly from relational databases. Example: IBM Cognos.
3. **HOLAP (Hybrid OLAP):** Combines features of MOLAP and ROLAP to leverage both multidimensional cubes and relational databases.

Multidimensional Data Analysis
Multidimensional data analysis involves examining data across multiple dimensions to gain a comprehensive view of performance and trends. It typically involves:

1. **Data Cubes:** A data structure that allows data to be viewed from different angles or dimensions. Each cell in a cube represents a measure, while each dimension represents different attributes of the data (e.g., sales figures across time, geography, and product types).
2. **Dimensions:** Aspects or perspectives from which data can be analyzed (e.g., time, location, product).
3. **Measures:** Quantitative metrics used in analysis (e.g., sales revenue, profit margins).
4. **Slicing and Dicing:** Techniques for focusing on specific portions of the data cube (e.g., viewing sales data for a specific region or time period).
5. **Drill Down and Roll Up:** Techniques for navigating between levels of data granularity. Drilling down means moving to more detailed data, while rolling up involves summarizing data to a higher level.

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Benefits of Multidimensional Data Analysis:

- **Enhanced Data Insights:** Provides a more nuanced view of data, helping to identify trends, anomalies, and patterns.
- **Improved Decision-Making:** Facilitates data-driven decisions by offering various perspectives and detailed analyses.
- **Increased Efficiency:** Streamlines complex queries and analysis processes, enabling faster insights.

Tools for OLAP and Multidimensional Analysis:

- **Microsoft SQL Server Analysis Services (SSAS):** For MOLAP and OLAP solutions.
- **IBM Cognos:** Provides ROLAP and multidimensional analysis capabilities.
- **SAP BusinessObjects:** Offers comprehensive OLAP functionality.
- **Oracle OLAP:** Part of Oracle's database solutions, providing multidimensional analysis features.

Difference Between ROLAP, MOLAP, HOLAP

Basis	ROLAP	MOLAP	HOLAP
Storage location for summary aggregation	Relational Database is used as storage location for summary aggregation.	Multidimensional Database is used as storage location for summary aggregation.	Multidimensional Database is used as storage location for summary aggregation.
Processing time	Processing time of ROLAP is very slow.	Processing time of MOLAP is fast.	Processing time of HOLAP is fast.
Storage space requirement	Large storage space requirement in ROLAP as compare to MOLAP and HOLAP.	Medium storage space requirement in MOLAP as compare to ROLAP and HOLAP.	Small storage space requirement in HOLAP as compare to MOLAP and ROLAP.
Storage location for detail data	Relational database is used as storage location for detail data.	Multidimensional database is used as storage location for detail data.	Relational database is used as storage location for detail data.
Latency	Low latency in ROLAP as compared to MOLAP and HOLAP.	High latency in MOLAP as compared to ROLAP and HOLAP.	Medium latency in HOLAP as compared to MOLAP and ROLAP.

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Basis	ROLAP	MOLAP	HOLAP
Query response time	Slow query response time in ROLAP as compared to MOLAP and HOLAP.	Fast query response time in MOLAP as compared to ROLAP and HOLAP.	Medium query response time in HOLAP as compare to MOLAP and ROLAP.

DIFFERENCE BETWEEN OLAP AND OLTP

Criteria	OLAP	OLTP
Purpose	OLAP helps you analyze large volumes of data to support decision-making.	OLTP helps you manage and process real-time transactions.
Data source	OLAP uses historical and aggregated data from multiple sources.	OLTP uses real-time and transactional data from a single source.
Data structure	OLAP uses multidimensional (cubes) or relational databases.	OLTP uses relational databases.
Data model	OLAP uses star schema, snowflake schema, or other analytical models.	OLTP uses normalized or denormalized models.
Volume of data	OLAP has large storage requirements. Think terabytes (TB) and petabytes (PB).	OLTP has comparatively smaller storage requirements. Think gigabytes (GB).
Response time	OLAP has longer response times, typically in seconds or minutes.	OLTP has shorter response times, typically in milliseconds
Example applications	OLAP is good for analyzing trends, predicting customer behavior, and identifying profitability.	OLTP is good for processing payments, customer data management, and order processing.

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UNIT-2:

DATA MINING:

The process of extracting information to identify patterns, trends, and useful data that would allow the business to make decisions decision from huge sets of data is called Data Mining.

- In other words, we can say that **Data Mining is the process of investigating hidden patterns of information to various perspectives for categorization into useful data,** which is collected and assembled in particular areas such as data warehouses, efficient analysis, data mining algorithms, helping decision making and other data requirement to eventually cost-cutting and generating revenue.
- Data Mining is a process used by organizations to extract specific data from huge databases to solve business problems. It primarily turns raw data into useful information.
- Data mining is one of the most useful techniques that help entrepreneurs, researchers, and individuals to extract valuable information from huge sets of data. Data mining is also called Knowledge Discovery in Database (KDD). The knowledge discovery process includes Data cleaning, Data integration, Data selection, Data transformation, Data mining, Pattern evaluation, and Knowledge presentation.

DATA MINING FUNCTIONALITIES

Data mining functionalities are used to specify the kind of patterns to be found in data mining tasks.
The functionalities include characterization and discrimination; the mining of frequent patterns, associations, and correlations; classification and regression; cluster analysis; and outlier detection.

In general, data mining tasks can be classified into two categories:
Descriptive and Predictive.

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- Descriptive mining tasks characterize the general properties of the data in the database.
- Predictive mining tasks perform inference on the current data to make predictions.

Class/Concept Description: Characterization and Discrimination
Data entries can be associated with classes or concepts.
For example, in the AllElectronics store, classes of items for sale include computers and printers, and the concepts of customers include bigSpenders and budgetSpenders. It can be useful to describe individual classes and concepts in summarized, concise, and yet precise terms. Such descriptions of a class or a concept are called class/concept descriptions.

These descriptions can be derived using
(1) data characterization, by summarizing the data of the class under study (often called the target class) in general terms, or

(2) data discrimination, by comparison of the target class with one or a set of comparative classes (often called the contrasting classes)

Data characterization

Data characterization is a summarization of the general characteristics or features of a target class of data. The data corresponding to the user-specified class are typically collected by a query.

- For example, to study the characteristics of software products with sales that increased by 10% in the previous year, the data related to such products can be collected by executing an SQL query on the sales database.

Data discrimination

Data discrimination is a comparison of the general features of target class data objects with the general features of objects from one or a set of contrasting classes. The target and contrasting classes can be specified by the user, and the corresponding data objects retrieved through database queries

Association analysis

- **Association analysis is a data mining method for identifying correlations between objects within a database. Based on frequencies, it can be used to determine whether different combinations occur with a certain probability.**

Association analysis. Suppose, as a marketing manager of AllElectronics, you would like to determine which items are frequently purchased together within the same transactions.
An example of such a rule is mined from the AllElectronics transactional database.

A confidence, or certainty, of 50% means that if a customer buys a computer, there is a 50% chance that she will buy software as well. A 1% support means that 1% of all of the transactions under analysis showed that computers and software were purchased together. This association rule involves a single attribute or predicate (i.e., buys) that repeats. Association rules that contain a single predicate are referred to as single-dimensional association rules. Dropping the predicate notation, the above rule can be written simply as “computer ⇒ software [1%, 50%]”.

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Classification

Classification is a technique in data mining that involves categorizing or classifying data objects into predefined classes, categories, or groups based on their features or attributes.

Prediction

In the prediction method, we need to predict the missing data for a new observation, depending on the previous data. Or we can say that the predictive models use comprehended outcomes to create a model that can be used to predict values for new data.

Regression analysis is a statistical methodology that is most often used for numeric prediction, although other methods exist as well. Prediction also encompasses the identification of distribution trends based on the available data.

Cluster Analysis

Unlike classification and prediction, which analyze class-labeled data objects, clustering analyzes data objects without consulting a known class label.

- clustering is a method used when we don't have labels (class labels) for our data initially. It helps group similar things based on their similarities and differences. The idea is to form clusters where things within the same cluster are very similar to each other but different from things in other clusters.
- Think of it like sorting a pile of mixed-up puzzle pieces into different groups based on their shapes and patterns. Each group represents a cluster, and we can use these clusters to create labels or categories for the puzzle pieces. This process helps us organize things and find patterns, making it easier to understand and work with the data

Cluster analysis. Cluster analysis can be performed on AllElectronics customer data to identify homogeneous subpopulations of customers. These clusters may represent individual target groups for marketing.

Outlier Analysis

- A database may contain data objects that do not comply with the general behavior or model of the data. These data objects are outliers.
- Most data mining methods discard outliers as noise or exceptions. However, in some applications such as fraud detection, the rare events can be more interesting than the more regularly occurring ones.
- The analysis of outlier data is referred to as outlier mining.
- Outliers are like the odd ones out in a group of things. There are different ways to spot them. One way is by using statistical tests or measuring distances between points. If something is far away from the rest, it might be an outlier. Another way is by looking at how much something deviates or differs from the usual characteristics of the group. So, outliers are like the unusual members that stand out because they're different from the majority.

Evolution Analysis

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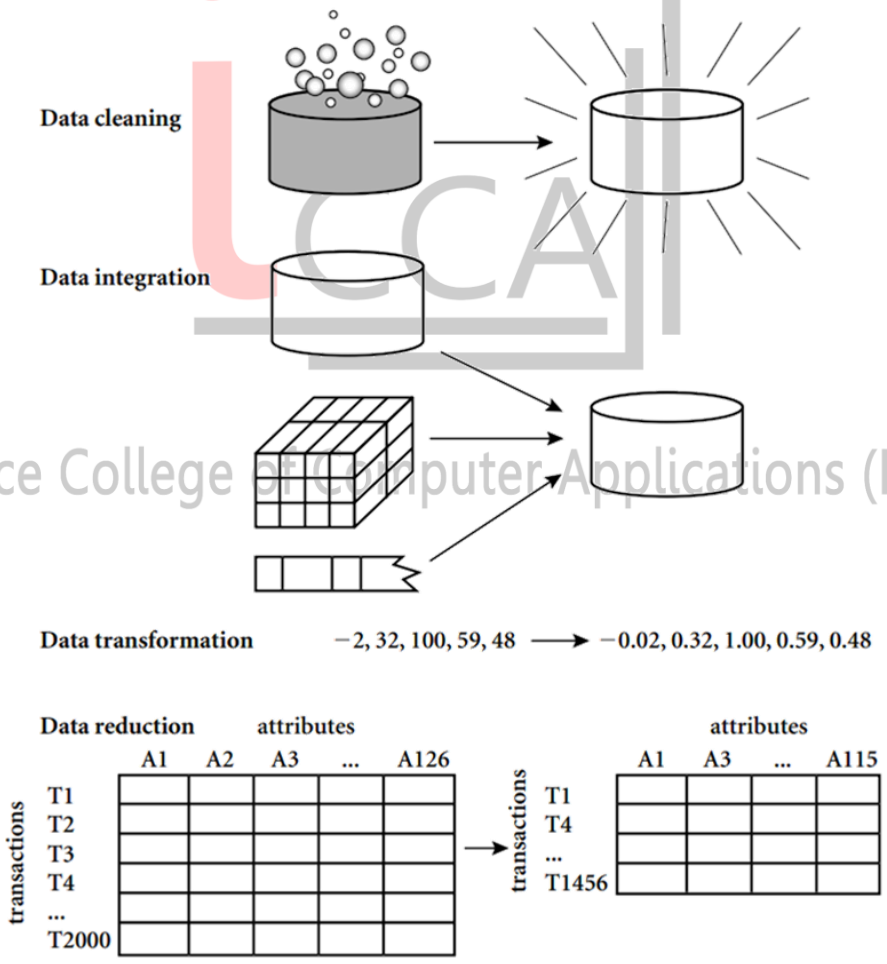
- Data evolution analysis describes and models regularities or trends for objects whose behavior changes over time. Although this may include characterization, discrimination, association and correlation analysis, classification, prediction, or clustering of time-related data, distinct features of such an analysis include time-series data analysis, sequence or periodicity pattern matching, and similarity-based data analysis.
- Ex: Suppose that you have the major stock market (time-series) data of the last several years available from the New York Stock Exchange and you would like to invest in shares of high-tech industrial companies. A data mining study of stock exchange data may identify stock evolution regularities for overall stocks and the stocks of particular companies. Such regularities may help predict future trends in stock market prices, contributing to your decision-making regarding stock investments.

DATA PREPROCESSING

Data preprocessing is an important step in the data mining process. **It refers to cleaning, transforming, and integrating data to prepare it for analysis. The goal of data preprocessing is to improve the data quality and make it more suitable for the specific data mining task.**

Incomplete, noisy, and inconsistent data are commonplace properties of large real-world databases and data warehouses. Incomplete data can occur for several reasons.

- Attributes of interest may not always be available, such as customer information for sales transaction data.
- Some data may not be included simply because it was not considered important at the time of entry. Relevant data may not be recorded due to a misunderstanding, or because of equipment malfunctions.
- Data that were inconsistent with other recorded data may have been deleted.
- The data collection instruments used may be faulty. There may have been human or computer errors occurring at data entry.
- Errors in data transmission can also occur.
- Incorrect data may also result from inconsistencies in naming conventions or data codes used, or inconsistent formats for input fields, such as date. Duplicate tuples also require data cleaning



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Fig: Forms of data preprocessing

Data cleaning

- Data cleaning help us remove inaccurate, incomplete and incorrect data from the dataset.
- Clean the data by filling in missing values, smoothing noisy data, identifying or removing outliers, and resolving inconsistencies.
- Dirty data can cause confusion for the mining procedure, resulting in unreliable output.
- **Some techniques used in data cleaning are**
    - **Handling missing values**
        - This type of scenario occurs when some data is missing.
    - **Noisy Data**
        - Noisy data are the data that cannot be interpreted by machine and It can be generated due to faulty data collection, data entry errors.

Data integration.

- This involves combining data from multiple sources to create a unified dataset. Data integration can be challenging as it requires handling data with different formats, structures, and semantics. Techniques such as record linkage and data fusion can be used for data integration.
- Some attributes representing a given concept may have different names in different databases, causing inconsistencies and redundancies.
- For example, the attribute for customer identification may be referred to as customer ID in one data store and cust ID in another.
- Having a large amount of redundant data may slow down or confuse the knowledge discovery process.
- Additional data cleaning can be performed to detect and remove redundancies that may have resulted from data integration.

Data transformation

- Change in format or structure of data to transform the data suitable for mining process.
- Common techniques used in data transformation include normalization, standardization, and discretization.
- Normalization is used to scale the data to a common range.
- Standardization is used to transform the data to have zero mean and unit variance.
- Discretization is used to convert continuous data into discrete categories.

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Data reduction

It helps in increasing storage efficiency and reducing data storage to make the analysis easier by producing almost the same results. The analysis becomes harder while working with huge amounts of data, so the reduction is used to get rid of that.

There are several strategies for data reduction.

- These include data aggregation (e.g., building a data cube)
- Attribute subset selection (e.g., removing irrelevant attributes through correlation analysis)
- Numerosity reduction (e.g., “replacing” the data by alternative, smaller representations such as clusters or parametric models).
- Data discretization This involves dividing continuous data into discrete categories or intervals.

In the real world, data can be messy and not perfect. Using data preprocessing, we clean up the data, fix errors, and make it better. This helps ensure that the information we use for decision-making is accurate and efficient. Taking care of data issues early on payoffs big when making important decisions.

DATA CLEANING

Real-world data tend to be incomplete, noisy, and inconsistent. Data cleaning (or data cleansing) routines attempt to fill in missing values, smooth out noise while identifying outliers, and correct inconsistencies in the data.

1. **Missing value**
   Imagine that you need to analyze AllElectronics sales and customer data. You note that many tuples have no recorded value for several attributes, such as customer income. How can you go about filling in the missing values for this attribute?

Let's look at the following methods:

- **Ignore the tuple:** This is usually done when the class label is missing (assuming the mining task involves classification). This method is not very effective unless the tuple contains several attributes with missing values. It is especially poor when the percentage of missing values per attribute varies considerably.
- **Fill in the missing value manually:** In general, this approach is time-consuming and may not be feasible given a large data set with many missing values.
- **Use a global constant to fill in the missing value:** Replace all missing attribute values with the same constant, such as a label like "Unknown" or $-\infty$. If missing values are replaced by, say, "Unknown," then the mining program may mistakenly think that they form an interesting concept since they all have a value in common—that of "Unknown." Hence, although this method is simple, it is not foolproof.
- **Use the attribute mean to fill in the missing value:** For example, suppose that the average income of AllElectronics customers is $56,000. Use this value to replace the missing value for income.
- **Use the attribute mean for all samples belonging to the same class as the given tuple:** For example, if classifying customers according to credit risk, replace the missing value with the average income value for customers in the same credit risk category as that of the given tuple.
- **Use the most probable value to fill in the missing value:** This may be determined with regression, inference-based tools using a Bayesian formalism, or decision tree induction. For example, using the other customer attributes in your data set, you may construct a decision tree to predict the missing values for income

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Noisy Data

Noisy data is a meaningless data that can't be interpreted by machines.It can be generated due to faulty data collection, data entry errors etc. It can be handled in following ways :

1. **Binning Method:**
   This method works on sorted data in order to smooth it. The whole data is divided into segments of equal size and then various methods are performed to complete the task.

Each segmented is handled separately. One can replace all data in a segment by its mean or boundary values can be used to complete the task.

- **Smoothing by bin mean:** each value in a bin is replaced by the mean value of the bin.
  For example, the mean of the values 4, 8, and 15 in Bin 1 is 9. Therefore, each original value in this bin is replaced by the value 9.
- **Smoothing by bin median:** each bin value is replaced by the bin median.
- **Smoothing by bin boundry:** the minimum and maximum values in a given bin are identified as the bin boundaries. Each bin value is then replaced by the closest boundary value. In general, the larger the width, the greater the effect of the smoothing.

Sorted data for *price* (in dollars): 4, 8, 15, 21, 21, 24, 25, 28, 34

Partition into (equal-frequency) bins:
Bin 1: 4, 8, 15
Bin 2: 21, 21, 24
Bin 3: 25, 28, 34

Smoothing by bin means:
Bin 1: 9, 9, 9
Bin 2: 22, 22, 22
Bin 3: 29, 29, 29

Smoothing by bin boundaries:
Bin 1: 4, 4, 15
Bin 2: 21, 21, 24
Bin 3: 25, 25, 34

2. Regression:

Data smoothing can also be done by regression, a technique that conforms data values to a function. Linear regression involves finding the "best" line to fit two attributes (or variables) so that one attribute can be used to predict the other. Multiple linear regression is an extension of linear regression, where more than two attributes are involved and the data are fit to a multidimensional surface.

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3.Outlier analysis:

Outliers may be detected by clustering, for example, where similar values are organized into groups, or "clusters." Intuitively, values that fall outside of the set of clusters may be considered outliers

Data Cleaning as a Process

Missing values, noise, and inconsistencies contribute to inaccurate data.

The first step in data cleaning as a process is discrepancy detection

- Discrepancies can be caused by several factors, including poorly designed data entry forms that have many optional fields, human error in data entry, deliberate errors (e.g., respondents not wanting to divulge information about themselves), and data decay (e.g., outdated addresses).

- Discrepancies may also arise from inconsistent data representations and the inconsistent use of codes.
- Errors in instrumentation devices that record data, and system errors, are other sources of discrepancies.
- Errors can also occur when the data are (inadequately) used for purposes other than originally intended.
- There may also be inconsistencies due to data integration (e.g., where a given attribute can have different names in different databases).

Tools for data cleaning

1. Openrefine
2. Trifecta wrangler
3. IBM data refinery
4. Microsoft Excel

DATA REDUCTION

Data reduction techniques can be applied to obtain a reduced representation of the data set that is much smaller in volume, yet closely maintains the integrity of the original data. That is, mining on the reduced data set should be more efficient yet produce the same (or almost the same) analytical results.

Techniques of Data Reduction

1. **Dimensionality Reduction**
2. **Numerosity Reduction**
3. **Data cube Aggregation**
4. **Data compression**
5. **Discretization operation**

1. Dimensionality Reduction

Whenever we encounter weakly important data, we use the attribute required for our analysis. Dimensionality reduction eliminates the attributes from the data set under consideration, thereby reducing the volume of original data. It reduces data size as it eliminates outdated or redundant features. Here are three methods of dimensionality reduction.

i. **Wavelet Transform:** In the wavelet transform, suppose a data vector A is transformed into a numerically different data vector A' such that both A and A' vectors are of the same length. Then how it is useful in reducing data because the data obtained from the wavelet transform can be truncated. The compressed data is obtained by retaining the smallest fragment of the strongest wavelet coefficients. Wavelet transform can be applied to data cubes, sparse data, or skewed data.

ii. **Principal Component Analysis:** Suppose we have a data set to be analyzed that has tuples with n attributes. The principal component analysis identifies k independent tuples with n attributes that can represent the data set. In this way, the original data can be cast on a much smaller space, and dimensionality

reduction can be achieved. Principal component analysis can be applied to sparse and skewed data.

iii. **Attribute Subset Selection:** The large data set has many attributes, some of which are irrelevant to data mining or some are redundant. The core attribute subset selection reduces the data volume and dimensionality. The attribute subset selection reduces the volume of data by eliminating redundant and irrelevant attributes. The attribute subset selection ensures that we get a good subset of original attributes even after eliminating the unwanted attributes. The resulting probability of data distribution is as close as possible to the original data distribution using all the attributes.

2. Numerosity Reduction

The numerosity reduction reduces the original data volume and represents it in a much smaller form. This technique includes two types parametric and non-parametric numerosity reduction.

i. **Parametric:** Parametric numerosity reduction incorporates storing only data parameters instead of the original data. One method of parametric numerosity reduction is the regression and log-linear method.

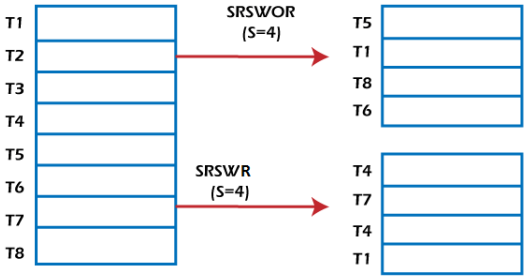
- **Regression and Log-Linear:** Linear regression models a relationship between the two attributes by modeling a linear equation to the data set.

i. Log-linear model discovers the relation between two or more discrete attributes in the database. Suppose we have a set of tuples presented in n-dimensional space. Then the log-linear model is used to study the probability of each tuple in a multidimensional space.
Regression and log-linear methods can be used for sparse data and skewed data.

ii. **Non-Parametric:** A non-parametric numerosity reduction technique does not assume any model. The non-Parametric technique results in a more uniform reduction, irrespective of data size, but it may not achieve a high volume of data reduction like the parametric. There are at least four types of Non-Parametric data reduction techniques, Histogram, Clustering, Sampling, Data Cube Aggregation, and Data Compression.

- **Histogram:** A histogram is a graph that represents frequency distribution which describes how often a value appears in the data. Histogram uses the binning method to represent an attribute's data distribution. It uses a disjoint subset which we call bin or buckets. A histogram can represent a dense, sparse, uniform, or skewed data. Instead of only one attribute, the histogram can be implemented for multiple attributes. It can effectively represent up to five attributes.
- **Clustering:** Clustering techniques groups similar objects from the data so that the objects in a cluster are similar to each other, but they are dissimilar to objects in another cluster. How much similar are the objects inside a cluster can be calculated using a distance function. More is the similarity between the objects in a cluster closer they appear in the cluster. The quality of the cluster depends on the diameter of the cluster, i.e., the max distance between any two objects in the cluster. The cluster representation replaces the original data. This technique is more effective if the present data can be classified into a distinct clustered.
- **Sampling:** One of the methods used for data reduction is sampling, as it can reduce the large data set into a much smaller data sample. Below we will discuss the different methods in which we can sample a large data set D containing N tuples:

a. **Simple random sample without replacement (SRSWOR) of size s:** In this s, some tuples are drawn from N tuples such that in the data set D (s<N). The probability of drawing any tuple from the data set D is 1/N. This means all tuples have an equal probability of getting sampled.
b. **Simple random sample with replacement (SRSWR) of size s:** It is similar to the SRSWOR, but the tuple is drawn from data set D, is recorded, and then replaced into the data set D so that it can be drawn again.

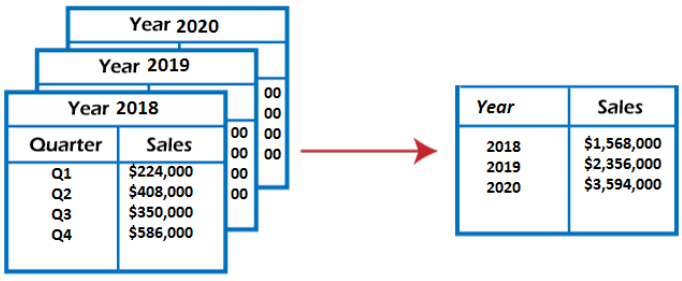


c. **Cluster sample:** The tuples in data set D are clustered into M mutually disjoint subsets. The data reduction can be applied by implementing SRSWOR on these clusters. A simple random sample of size s could be generated from these clusters where s<M.
d. **Stratified sample:** The large data set D is partitioned into mutually disjoint sets called 'strata'. A simple random sample is taken from each stratum to get stratified data. This method is effective for skewed data.

3. Data Cube Aggregation

This technique is used to aggregate data in a simpler form. Data Cube Aggregation is a multidimensional aggregation that uses aggregation at various levels of a data cube to represent the original data set, thus achieving data reduction.

- For example, suppose you have the data of All Electronics sales per quarter for the year 2018 to the year 2022. If you want to get the annual sale per year, you just have to aggregate the sales per quarter for each year. In this way, aggregation provides you with the required data, which is much smaller in size, and thereby we achieve data reduction even without losing any data.



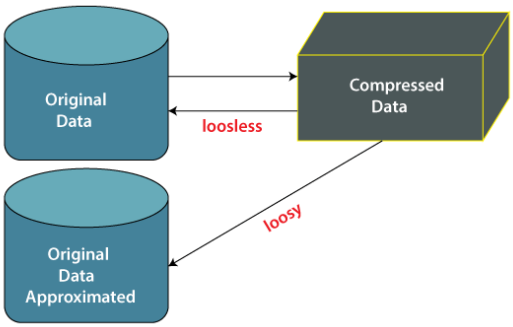
Aggregated Data

The data cube aggregation is a multidimensional aggregation that eases multidimensional analysis. The data cube present precomputed and summarized data which eases the data mining into fast access.

4. Data Compression

Data compression employs modification, encoding, or converting the structure of data in a way that consumes less space. Data compression involves building a compact representation of

information by removing redundancy and representing data in binary form. Data that can be restored successfully from its compressed form is called Lossless compression. In contrast, the opposite where it is not possible to restore the original form from the compressed form is Lossy compression. Dimensionality and numerosity reduction method are also used for data compression.



This technique reduces the size of the files using different encoding mechanisms, such as Huffman Encoding and run-length Encoding. We can divide it into two types based on their compression techniques.

i. **Lossless Compression:** Encoding techniques (Run Length Encoding) allow a simple and minimal data size reduction. Lossless data compression uses algorithms to restore the precise original data from the compressed data.
ii. **Lossy Compression: In** lossy-data compression, the decompressed data may differ from the original data but are useful enough to retrieve information from them. For example, the JPEG image format is a lossy compression, but we can find the meaning equivalent to the original image. Methods such as the Discrete Wavelet transform technique PCA (principal component analysis) are examples of this compression.

5. Discretization Operation

The data discretization technique is used to divide the attributes of the continuous nature into data with intervals. We replace many constant values of the attributes with labels of small intervals. This means that mining results are shown in a concise and easily understandable way.

i. **Top-down discretization:** If you first consider one or a couple of points (so-called breakpoints or split points) to divide the whole set of attributes and repeat this method up to the end, then the process is known as top-down discretization, also known as splitting.
ii. **Bottom-up discretization:** If you first consider all the constant values as split-points, some are discarded through a combination of the neighborhood values in the interval. That process is called bottom-up discretization.

DATA DISCRETIZATION AND CONCEPT HIERARCHY GENERATION

Data discretization techniques can be used to reduce the number of values for a given continuous attribute by dividing the range of the attribute into intervals.
Data discretization refers to a method of converting a huge number of data values into smaller ones so that the evaluation and management of data become easy. In other words, data discretization is a method of converting attributes values of continuous data into a finite set of intervals with minimum data loss.

1. **Discretization by Binning**

Binning refers to a data smoothing technique that helps to group a huge number of continuous values into smaller values. For data discretization and the development of idea hierarchy, this technique can also be used.

2. **Discretization by Histogram Analysis**

Histogram refers to a plot used to represent the underlying frequency distribution of a continuous data set. Histogram assists the data inspection for data distribution. For example, Outliers, skewness representation, normal distribution representation, etc.

3. **Discretization by Cluster**

clustering can be applied to discretize numeric attributes. It partitions the values into different clusters or groups by following a top-down or bottom-up strategy

4. **Discretization by Decision Tree**

Data discretization refers to a decision tree analysis using a top-down slicing technique. It is done through a supervised procedure. In a numeric attribute discretization, first, you need to select the attribute with the least entropy and then run it with the help of a recursive process. The recursive process divides it into discretized disjoint intervals, from top to bottom, using the same splitting criterion.

5. **Discretization by Correlation Analyses**

Discretizing data by linear regression technique, you can get the best neighboring interval, and then the large intervals are combined to develop a larger overlap to form the final 20 overlapping intervals. It is a supervised procedure.

Concept hierarchy generation

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The term hierarchy represents an organizational structure or mapping in which items are ranked according to their levels of importance. In other words, we can say that a hierarchy concept refers to a sequence of mappings with a set of more general concepts to complex concepts.

- It means mapping is done from low-level concepts to high-level concepts.
- For example, in computer science, there are different types of hierarchical systems. A document is placed in a folder in windows at a specific place in the tree structure is the best example of a computer hierarchical tree model. There are two types of hierarchy: top-down mapping and the second one is bottom-up mapping.

Top-down mapping

Top-down mapping generally starts with the top with some general information and ends with the bottom to the specialized information.

Bottom-up mapping

Bottom-up mapping generally starts with the bottom with some specialized information and ends with the top to the generalized information.

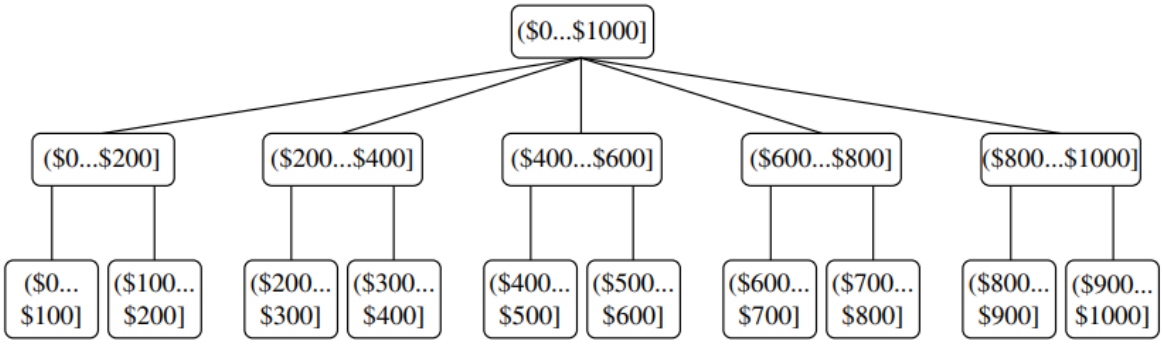


Figure:Concept hierarchy for the attribute price

Types of Concept Hierarchies

1. **Schema Hierarchy:** Schema Hierarchy is a type of concept hierarchy that is used to organize the schema of a database in a logical and meaningful way, grouping similar objects together. A schema hierarchy can be used to organize different types of data, such as tables, attributes, and relationships, in a logical and meaningful way.

2. **Set-Grouping Hierarchy:** Set-Grouping Hierarchy is a type of concept hierarchy that is based on set theory, where each set in the hierarchy is defined in terms of its membership in other sets. Set-grouping hierarchy can be used for data cleaning, data pre-processing and data integration. This type of hierarchy can be used to identify and remove outliers, noise, or inconsistencies from the data and to integrate data from multiple sources.

3. **Operation-Derived Hierarchy:** An Operation-Derived Hierarchy is a type of concept hierarchy that is used to organize data by applying a series of operations or transformations to the data. The operations are applied in a top-down fashion, with each level of the hierarchy representing a more general or abstract view of the data than the level below it. This type of hierarchy is typically used in data mining tasks such as clustering and dimensionality reduction. The operations applied can be mathematical or statistical operations such as aggregation, normalization.

4. **Rule-based Hierarchy:** Rule-based Hierarchy is a type of concept hierarchy that is used to organize data by applying a set of rules or conditions to the data. This type of hierarchy is useful in data mining tasks such as classification, decision-making, and data exploration. It allows to the assignment of a class label or decision to each data point based on its characteristics and identifies patterns and relationships between different attributes of the data.

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ARCHITECTURE OF A TYPICAL DATA MINING SYSTEMS

The significant components of data mining systems are a data source, data mining engine, data warehouse server, pattern evaluation module, graphical user interface, and knowledge base.

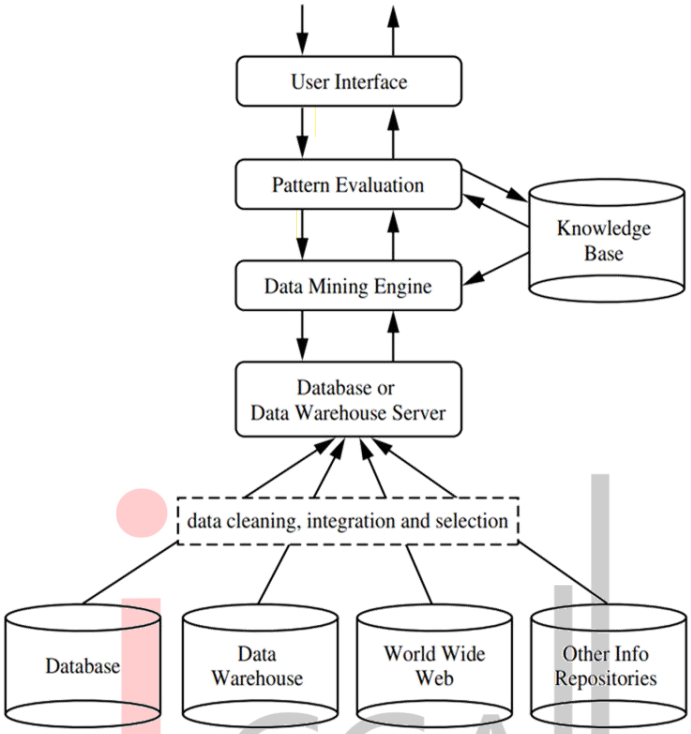


Fig: Architecture of a typical data mining system

Data Source: The actual source of data is the Database, data warehouse, World Wide Web (WWW), text files, and other documents. You need a huge amount of historical data for data mining to be successful. Organizations typically store data in databases or data warehouses. Data warehouses may comprise one or more databases, text files spreadsheets, or other repositories of data. Sometimes, even plain text files or spreadsheets may contain information. Another primary source of data is the World Wide Web or the internet.

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Database or Data Warehouse Server: The database or data warehouse server consists of the original data that is ready to be processed. Hence, the server is cause for retrieving the relevant data that is based on data mining as per user request.

Data Mining Engine: The data mining engine is a major component of any data mining system. It contains several modules for operating data mining tasks, including association, characterization, classification, clustering, prediction, time-series analysis, etc.
In other words, we can say data mining engine is the root of our data mining architecture. It comprises instruments and software used to obtain insights and knowledge from data collected from various data sources and stored within the data warehouse.

Pattern evaluation module: This component typically employs interestingness measures and interacts with the data mining modules to focus the search toward interesting patterns. It may use interestingness thresholds to filter out discovered patterns. Alternatively, the pattern evaluation module may be integrated with the mining module, depending on the implementation of the data mining method used. For efficient data mining, it is highly recommended to push the evaluation of pattern interestingness as deep as possible into the mining process so as to confine the search to only the interesting patterns.

Knowledge base: This is the domain knowledge that is used to guide the search or evaluate the interestingness of resulting patterns. Such knowledge can include concept hierarchies, used to organize attributes or attribute values into different levels of abstraction. Knowledge such as user beliefs, which can be used to assess a pattern's interestingness based on its unexpectedness, may also be included.

User interface: This module communicates between users and the data mining system, allowing the user to interact with the system by specifying a data mining query or task, providing information to help focus the search, and performing exploratory data mining based on the intermediate data mining results. In addition, this component allows the user to browse database and data warehouse schemas or data structures, evaluate mined patterns, and visualize the patterns in different forms.

CLASSIFICATION OF DATA MINING SYSTEMS

- Data mining is an interdisciplinary field, the confluence of a set of disciplines, including database systems, statistics, machine learning, visualization, and information science. Moreover, depending on the data mining approach used, techniques from other disciplines may be applied, such as neural networks, fuzzy and/or rough set theory, knowledge representation, inductive logic programming, or high-performance computing.
- Depending on the kinds of data to be mined or on the given data mining application, the data mining system may also integrate techniques relating to data analysis, information retrieval, pattern recognition, image analysis, signal processing, computer graphics, Web technology, economics, business, bioinformatics, or psychology. Because of the diversity of disciplines contributing to data mining, data mining research is expected to generate a large variety of data mining systems.
- Therefore, it is necessary to provide a clear classification of data mining systems, which may help potential users distinguish between such systems and identify those that best match their needs

Classification according to the kinds of databases mined: A data mining system can be classified according to the kinds of databases mined. Database systems can be classified according to different criteria (such as data models, or the types of data or applications involved), each of which may require its data mining technique. Data mining systems can therefore be classified accordingly.

- For instance, if classifying according to data models, we may have a relational, transactional, object-relational, or data warehouse mining system. If classifying

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according to the special types of data handled, we may have a relate , time-series, text, stream data, multimedia data mining system, or a World Wide Web mining system.

Classification according to the kinds of knowledge mined: Data mining systems can be categorized according to the kinds of knowledge they mine, that is, based on data mining functionalities, such as characterization, discrimination, association and correlation analysis, classification, prediction, clustering, outlier analysis, and evolution analysis. A comprehensive data mining system usually provides multiple and/or integrated data mining functionalities.

Classification according to the kinds of techniques utilized: Data mining systems can be categorized according to the underlying data mining techniques employed. These techniques can be described according to the degree of user interaction involved (e.g., autonomous systems, interactive exploratory systems, query-driven systems) or the methods of data analysis employed (e.g., database-oriented or data warehouse– oriented techniques, machine learning, statistics, visualization, pattern recognition, neural networks, and so on). A sophisticated data mining system will often adopt multiple data mining techniques or work out an effective, integrated technique that combines the merits of a few individual approaches.

Classification according to the applications adapted: Data mining systems can also be categorized according to the applications they adapt. For example, data mining systems may be tailored specifically for finance, telecommunications, DNA, stock markets, e-mail, and so on. Different applications often require the integration of application-specific methods. Therefore, a generic, all-purpose data mining system may not fit domain-specific mining tasks.

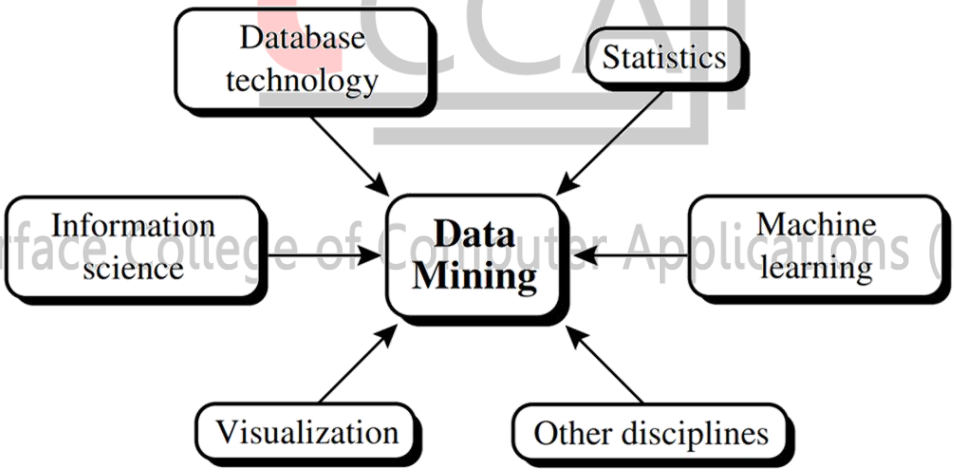


Fig: Data mining as a confluence of multiple disciplines.

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ASSOCIATION RULE MINING: -

Association rule mining is a data mining technique used to discover interesting relationships or associations between variables in large datasets. These relationships are often expressed as "if-then" statements, known as association rules.

What is Association Rule Mining?

Association Rule Mining finds patterns, correlations, or relationships among a set of items in transactional databases, relational databases, or other data repositories.

For example, in a supermarket, ARM can help identify rules like:
"If a customer buys bread, they are likely to buy butter."
"Customers who purchase a laptop often buy a mouse as well."

These rules help businesses optimize marketing strategies, inventory management, and customer engagement.

2. Terminologies in Association Rule Mining

1. Itemset

An **itemset** is a collection of one or more items. For example, in a supermarket dataset, {Milk, Bread, Butter} is an itemset representing a group of products frequently bought together.

- **Singleton Itemset**: An itemset with only one item, e.g., {Milk}.
- **K-itemset**: An itemset containing *k* items, e.g., {Milk, Bread, Butter} is a **3-itemset**.

2. Support

- Support measures the frequency of an itemset in the dataset. It indicates how often an itemset appears in the transaction database.

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$$\text{Support (A)} = \frac{\text{Number of transactions containing A}}{\text{Total number of transactions}}$$

For example, if {Milk, Bread} appears in 50 out of 1000 transactions: 0.05

A high support value means that the itemset is frequently occurring in the dataset.

3. Confidence

Confidence measures the reliability of a rule. It is the proportion of transactions that contain itemset **A** which also contain itemset **B** when considering a rule **A** → **B**.

$$\text{Confidence}(A \rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

For example, if {Milk, Bread} appears in 50 transactions and {Milk} appears in 100 transactions:
A high confidence value indicates a strong correlation between **A** and **B**.

5. **Lift**

Lift measures how much more likely **B** is to be bought when **A** is bought, compared to when **B** is bought independently.

$$\text{Lift}(A \rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A) \times \text{Support}(B)}$$

- **Lift = 1**: No correlation between **A** and **B**.
- **Lift > 1**: Positive correlation (A increases the likelihood of B).
- **Lift < 1**: Negative correlation (A reduces the likelihood of B).

For example, if:

- Support(Milk) = 0.2
- Support(Bread) = 0.3
- Support(Milk, Bread) = 0.1

$$\text{Lift}(Milk \rightarrow Bread) = \frac{0.1}{0.2 \times 0.3} = \frac{0.1}{0.06} = 1.67$$

Since Lift > 1, Milk and Bread are positively correlated.

Apriori Algorithm -

The Apriori algorithm consists of three main phases:

Step 1: Set Minimum Support and Minimum Confidence

- **Minimum Support (min_sup):** Defines how frequently an itemset should appear to be considered significant.
- **Minimum Confidence (min_conf):** Determines the strength of an association rule.

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For example:

- `min_sup = 50%` (An itemset must appear in at least 50% of transactions).
- `min_conf = 60%` (The rule should have at least 60% confidence).

Step 2: Find Frequent Itemsets Using Iteration

The algorithm follows an iterative approach called **level-wise search**:

Iteration 1: Find Frequent 1-itemsets (L1)

- Count the occurrence of each individual item in the transactions.

- Remove items that do not meet the **min_sup**.

Iteration 2: Generate Candidate 2-itemsets (C2) and Find Frequent 2-itemsets (L2)

- Create **C2 (candidate 2-itemsets)** by combining frequent 1-itemsets.
- Count the occurrences of each candidate itemset in the transactions.
- Remove itemsets that do not meet **min_sup**, forming **L2**.

Iteration 3: Generate Candidate 3-itemsets (C3) and Find Frequent 3-itemsets (L3)

- Use **L2** to generate **C3 (candidate 3-itemsets)**.
- Count the occurrences and filter itemsets below **min_sup** to get **L3**.
- **Repeat** this process until no more frequent itemsets are found.

Step 3: Generate Association Rules

Once **frequent itemsets** are found, association rules are generated using **confidence**:

For each frequent itemset **L**, generate rules of the form:

$$A \rightarrow B$$

where:

$$\text{Confidence}(A \rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

- If **confidence ≥ min_conf**, the rule is considered strong.
- **Lift** can be calculated to measure the rule's significance.

Explain the generation of candidate itemset and frequent pattern with minimum support count = 2 by using Apriori algorithm with below example.

Example

TID	Item Set
10	A, D, E
20	B, C, E
30	A, B, C, E
40	B, E

Step 1: Given Transaction Data

We have a transaction dataset with a **minimum support count = 2**.

	Item Set
10	A, D, E
20	B, C, E
30	A, B, C, E
40	B, E

Step 2: Generate Candidate 1-itemsets (C1)

We first count the occurrences of individual items across all transactions.

Item	Count
A	2
B	3
C	2
D	1
E	4

Since **minimum support count = 2**, we remove item **D** (as it appears only once).

Frequent 1-itemset (L1):

L1={(A:2),(B:3),(C:2),(E:4)}

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Step 3: Generate Candidate 2-itemsets (C2)

We generate all possible **pairs** (combinations of two items) from **L1** and count their occurrences.

TID	Items
10	A, D, E
20	B, C, E
30	A, B, C, E
40	B, E

Now, let's count each **2-itemset**:

Candidate 2-Itemset	Transactions Containing It	Count
(A, B)	{30}	1
(A, C)	{30}	1
(A, E)	{10, 30}	2
(B, C)	{20, 30}	2
(B, E)	{20, 30, 40}	3
(C, E)	{20, 30}	2

Itemset Count

(A, B) 1

(A, C) 1

(A, E) 2

(B, C) 2

(B, E) 3

(C, E) 2

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Filtering 2-Itemsets with Minimum Support Count

The given **minimum support count** is **2**, so we discard the itemsets that appear less than 2 times.

Itemset Count Meets Min Support?

(A, B) 1 (Discarded)

(A, C) 1 (Discarded)

Itemset	Count	Meets Min Support?
(A, E)	2	(Accepted)
(B, C)	2	(Accepted)
(B, E)	3	(Accepted)
(C, E)	2	(Accepted)

So, the **Frequent 2-Itemset (L2)** is:

L2={(A,E:2),(B,C:2),(B,E:3),(C,E:2)}

- We generated **candidate 2-itemsets (C2)** by forming **all possible pairs** from the frequent 1-itemsets (L1).
- We **scanned the database** to count their occurrences.
- We **filtered out** itemsets with **support count < 2**, leaving us with **frequent 2-itemsets (L2)**.
- The frequent 2-itemsets will be used in **Step 4** to generate **candidate 3-itemsets (C3)**.

Step 4: Generating Candidate 3-Itemsets (C3)

- Now that we have **frequent 2-itemsets (L2)**, we move to the next step of the **Apriori Algorithm**: generating **candidate 3-itemsets (C3)** and filtering them based on the minimum support count.
- We generate **3-item combinations** from **L2**.
- From **Step 3**, we obtained the **frequent 2-itemsets (L2):**
- L2={(A,E:2),(B,C:2),(B,E:3),(C,E:2)}

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Now, we form **3-item combinations** from these **L2** sets:

- Combine (B, C) and (B, E) → (B, C, E)
- Combine (B, C) and (C, E) → (B, C, E) (Already considered)

Thus, the only **unique candidate 3-itemset (C3)** is:

C3={(B,C,E)}

Counting the Occurrences of Candidate 3-Itemsets (C3)

We check how many times (B, C, E) appears in the transactions.

TID	Item Set	Contains (B, C, E)?
10	A, D, E	(No B, C)
20	B, C, E	(Yes)
30	A, B, C, E	(Yes)
40	B, E	(No C)

Now, we count the occurrences:

Itemset	Count
(B, C, E)	2

Step 5: Generating Candidate 4-Itemsets (C4)

Final Frequent Itemsets

- **L1** (Frequent 1-itemsets): {(A),(B),(C),(E)}
- **L2** (Frequent 2-itemsets): {(A,E),(B,C),(B,E),(C,E)}
- **L3** (Frequent 3-itemsets): {(B,C,E)}

To generate candidate 4 itemset (c4), we need at least two frequent 3 itemset that share a common subset of two items. However in this case L3 contains only one frequent 3

FP Growth in Data Mining

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The FP Growth algorithm is a popular method for frequent pattern mining in data mining. It works by constructing a frequent pattern tree (FP-tree) from the input dataset. The FP-tree is a compressed representation of the dataset that captures the frequency and association information of the items in the data.

The algorithm first scans the dataset and maps each transaction to a path in the tree. Items are ordered in each transaction based on their frequency, with the most frequent items appearing first. Once the FP tree is constructed, frequent itemsets can be generated by recursively mining

the tree. This is done by starting at the bottom of the tree and working upwards, finding all combinations of itemsets that satisfy the minimum support threshold.

Working on FP Growth Algorithm

The working of the FP Growth algorithm in data mining can be summarized in the following steps:

The working of the FP Growth algorithm in data mining can be summarized in the following steps:

- **Scanthedatabase:**

  In this step, the algorithm scans the input dataset to determine the frequency of each item. This determines the order in which items are added to the FP tree, with the most frequent items added first.

- **Sortitems:**

  In this step, the items in the dataset are sorted in descending order of frequency. The infrequent items that do not meet the minimum support threshold are removed from the dataset. This helps to reduce the dataset's size and improve the algorithm's efficiency.

- **Construct the FP-tree:**

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  In this step, the FP-tree is constructed. The FP-tree is a compact data structure that stores the frequent itemsets and their support counts.

- **Generate frequent itemsets:**

  Once the FP-tree has been constructed, frequent itemsets can be generated by recursively mining the tree. Starting at the bottom of the tree, the algorithm finds all combinations of frequent item sets that satisfy the minimum support threshold.

- **Generate association rules:**

  Once all frequent item sets have been generated, the algorithm post-processes the

generated frequent item sets to generate association rules, which can be used to identify interesting relationships between the items in the dataset.

FP Tree

The **FP-tree (Frequent Pattern tree)** is a data structure used in the FP Growth algorithm for frequent pattern mining. It represents the frequent itemsets in the input dataset compactly and efficiently. The FP tree consists of the following components:

- **RootNode:**

  The root node of the FP-tree represents an empty set. It has no associated item but a pointer to the first node of each item in the tree.

- **ItemNode:**

  Each item node in the FP-tree represents a unique item in the dataset. It stores the item name and the frequency count of the item in the dataset.

- **HeaderTable:**

  The header table lists all the unique items in the dataset, along with their frequency count. It is used to track each item's location in the FP tree.

- **ChildNode:**

  Each child node of an item node represents an item that co-occurs with the item the parent node represents in at least one transaction in the dataset.

- **NodeLink:**

  The node-link is a pointer that connects each item in the header table to the first node of that item in the FP-tree. It is used to traverse the conditional pattern base of each item during the mining process.

The FP tree is constructed by scanning the input dataset and inserting each transaction into the tree one at a time. For each transaction, the items are sorted in descending order of frequency count and then added to the tree in that order. If an item exists in the tree, its frequency count

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is incremented, and a new path is created from the existing node. If an item does not exist in the tree, a new node is created for that item, and a new path is added to the tree. We will understand in detail how FP-tree is constructed in the next section.

Algorithm by Han

Let's understand with an example how the FP Growth algorithm in data mining can be used to mine frequent itemsets. Suppose we have a dataset of transactions as shown below:

Transaction ID	Items
T1	{M, N, O, E, K, Y}
T2	{D, O, E, N, Y, K}
T3	{K, A, M, E}
T4	{M, C, U, Y, K}
T5	{C, O, K, E, I}

Let's scan the above database and compute the frequency of each item as shown in the below table.

Item	Frequency
A	1
C	2
D	1
E	4
I	1
K	5
M	3
N	2

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O	3
U	1
Y	3

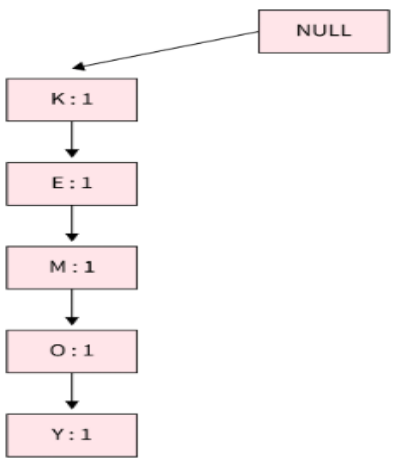
Let's consider minimum support as 3. After removing all the items below minimum support in the above table, we would remain with these items - {K: 5, E: 4, M : 3, O : 3, Y : 3}. Let's re-order the transaction database based on the items above minimum support. In this step, in each transaction, we will remove infrequent items and re-order them in the descending order of their frequency, as shown in the table below.

Transaction ID	Items	Ordered Itemset
T1	{M, N, O, E, K, Y}	{K, E, M, O, Y}
T2	{D, O, E, N, Y, K}	{K, E, O, Y}
T3	{K, A, M, E}	{K, E, M}
T4	{M, C, U, Y, K}	{K, M, Y}
T5	{C, O, K, E, I}	{K, E, O}

Now we will use the ordered itemset in each transaction to build the FP tree. Each transaction will be inserted individually to build the FP tree, as shown below -

- **First Transaction {K, E, M, O, Y}:**

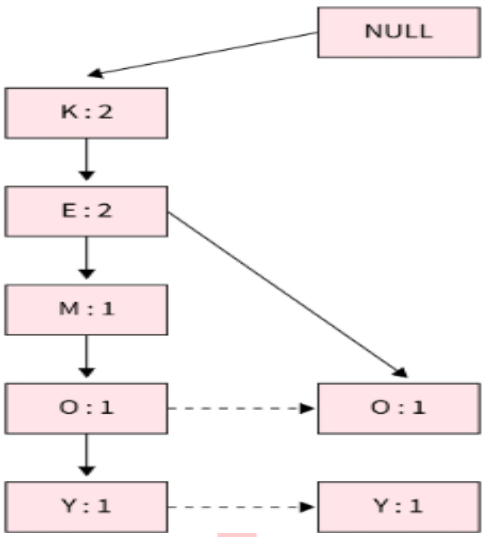
  In this transaction, all items are simply linked, and their support count is initialized as 1.



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- **Second Transaction {K, E, O, Y}:**

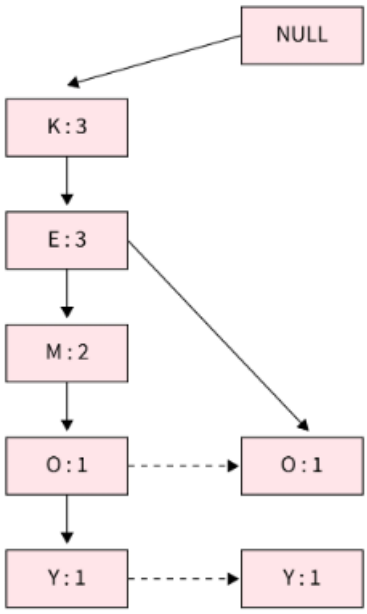
  In this transaction, we will increase the support count of K and E in the tree to 2. As **no direct link** is available from E to O, we will insert a new path for O and Y and initialize their support count as 1.



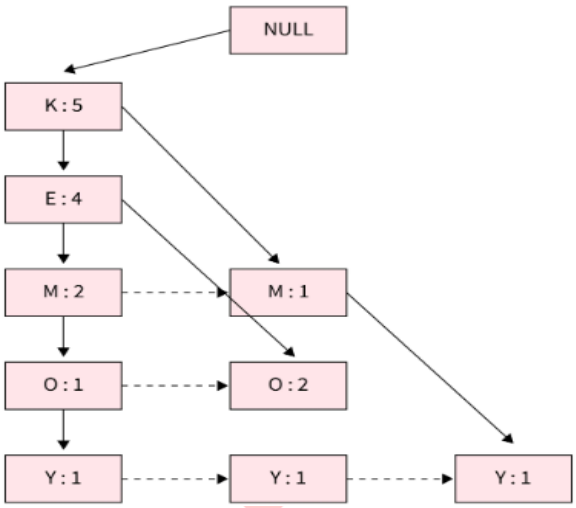
- **Third Transaction {K, E, M}:**

  After inserting this transaction, the tree will look as shown below. We will increase the support count for K and E to 3 and for M to 2.

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- **Fourth Transaction {K, M, Y}** and **Fifth Transaction {K, E, O}:** After inserting the last two transactions, the FP-tree will look like as shown below:



Now we will create a **Conditional Pattern Base** for all the items. The conditional pattern base is the path in the tree ending at the given frequent item. For example, for item O, the paths {K, E, M} and {K, E} will result in item O. The conditional pattern base for all items will look like as shown below table:

Item	Conditional Pattern Base
Y	{K, E, M, O : 1}, {K, E, O : 1}, {K, M : 1}
O	{K, E, M : 1}, {K, E : 2}
M	{K, E : 2}, {K : 1}
E	{K : 4}
K	

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Now for each item, we will build a conditional frequent pattern tree. It is computed by identifying the set of elements common in all the paths in the conditional pattern base of a given frequent item and computing its support count by summing the support counts of all the

paths in the conditional pattern base. The conditional frequent pattern tree will look like this as shown below table:

Item	Conditional Pattern Base	Conditional FP Tree
Y	{K, E, M, O : 1}, {K, E, O : 1}, {K, M : 1}	{K : 3}
O	{K, E, M : 1}, {K, E : 2}	{K, E : 3}
M	{K, E : 2}, {K: 1}	{K : 3}
E	{K: 4}	{K: 4}
K		

From the above conditional FP tree, we will generate the frequent itemsets as shown in the below table:

Item	Frequent Patterns
Y	{K, Y - 3}
O	{K, O - 3}, {E, O - 3}, {K, E, O - 3}
M	{K, M - 3}
E	{K, E - 4}

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Mining Frequent Itemsets using the Vertical Data Format (Eclat)

Eclat (Equivalence Class Transformation) is a depth-first search algorithm for discovering frequent itemsets in a transaction database. Unlike the Apriori algorithm which uses a horizontal data format, Eclat uses a vertical data format, which makes it significantly more efficient for many datasets.

The Vertical Data Format

In traditional horizontal data format, data is represented as:

- Transaction ID → List of items

In vertical data format used by Eclat:

- Item → List of transaction IDs (TID-list)

This vertical representation allows for much faster computation of support counts through simple set intersections.

How Eclat Works

1. Initial Transformation: Convert the transaction database from horizontal to vertical format
2. Support Counting: Compute the support of an itemset by intersecting the TID-lists
3. Candidate Generation: Generate candidate itemsets using a depth-first approach
4. Pruning: Eliminate candidates that don't meet the minimum support threshold

Algorithm Steps in Detail

1. Convert to Vertical Format

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For each item in the database, create a list of transaction IDs where the item appears.

Example:

Horizontal format:

T1: {A, B, C}

T2: {A, C}

T3: {A, D}

T4: {B, C, D}

Vertical format:

A: {T1, T2, T3}

B: {T1, T4}

C: {T1, T2, T4}

D: {T3, T4}

2. Find Frequent 1-itemsets

Count the length of each TID-list and keep only items that meet the minimum support threshold.

3. Generate Frequent k-itemsets ($k \geq 2$)

For each pair of frequent (k-1)-itemsets that share a common prefix of length (k-2):

- The new k-itemset is formed by joining the two (k-1)-itemsets
- The TID-list of the new itemset is the intersection of the TID-lists of the two (k-1)-itemsets
- The support of the new itemset is the length of its TID-list

4. Recursively Continue

Repeat step 3 in a depth-first manner until no more frequent itemsets can be generated.

Example Walkthrough

Let's use the example database from above with minimum support = 2:

Step 1: Convert to vertical format

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A: {T1, T2, T3} (support = 3)

B: {T1, T4} (support = 2)

C: {T1, T2, T4} (support = 3)

D: {T3, T4} (support = 2)

All 1-itemsets are frequent (support ≥ 2).

Step 2: Generate frequent 2-itemsets

AB: {T1} = {T1, T2, T3} ∩ {T1, T4} (support = 1) - Not frequent

AC: {T1, T2} = {T1, T2, T3} ∩ {T1, T2, T4} (support = 2) - Frequent

AD: {T3} = {T1, T2, T3} ∩ {T3, T4} (support = 1) - Not frequent

BC: {T1, T4} = {T1, T4} ∩ {T1, T2, T4} (support = 2) - Frequent

BD: {T4} = {T1, T4} ∩ {T3, T4} (support = 1) - Not frequent

CD: {T4} = {T1, T2, T4} ∩ {T3, T4} (support = 1) - Not frequent

Step 3: Generate frequent 3-itemsets Since we have only two frequent 2-itemsets (AC and BC) and they don't share a common prefix, we can't generate any 3-itemsets.

Result: Frequent itemsets

A (support = 3)

B (support = 2)

C (support = 3)

D (support = 2)

AC (support = 2)

BC (support = 2)

Advantages of Eclat

1. Efficiency: Significantly faster than Apriori for many datasets due to:
    - Avoiding multiple database scans
    - Computing support through simple set intersections
    - Depth-first search approach reducing memory requirements
2. Memory Usage: The vertical data format is often more compact, especially for sparse datasets
3. Simplicity: The algorithm is conceptually simpler than many other frequent itemset mining algorithms

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Variations and Optimizations

1. dEclat: Uses diffsets (difference of TID-lists) instead of intersections to further improve performance
2. Parallel Eclat: Distributes the computation across multiple processors
3. Memory optimization: Various techniques to reduce the memory footprint when dealing with very large datasets

MINING VARIOUS KINDS OF ASSOCIATION RULES

Mining Multilevel Association Rules

- For many applications, it is difficult to find strong associations among data items at low or primitive levels of abstraction due to the sparsity of data at those levels.
- Strong associations discovered at high levels of abstraction may represent commonsense knowledge.
- Moreover, what may represent common sense to one user may seem novel to another.
- Therefore, data mining systems should provide capabilities for mining association rules at multiple levels of abstraction, with sufficient flexibility for easy traversal among different abstraction spaces.
- Using uniform support for all levels.
- Using reduced minimum support at lower levels.
- Using item group-based minimum support.

Mining Multidimensional Association Rules from Relational Databases and Data Warehouses

Mining multidimensional association rules from relational databases and data warehouses involves discovering associations among multiple dimensions or attributes in a dataset that is structured in a relational or multidimensional database format.

Mining Multidimensional Association Rules Using Static Discretization of Quantitative Attributes

Mining multidimensional association rules using static discretization of quantitative attributes involves transforming continuous numerical data into discrete intervals and then discovering association rules among these discretized attributes across multiple dimensions.

- The intervals are created statically according to the user.

Mining Quantitative Association Rules

Mining quantitative association rules involves discovering interesting relationships between quantitative variables in a dataset.

- The intervals ae created dynamically means generated by the algorithms by using binning methods.

ASSOCIATION MINING TO CORRELATION ANALYSIS

- As we have seen so far, the support and confidence measures are insufficient at filtering out uninteresting association rules.
- To tackle this weakness, a correlation measure can be used to augment the support–confidence framework for association rules.
- This leads to correlation rules of the form

A ⇒ B [support, confidence, correlation].

- That is, a correlation rule is measured not only by its support and confidence but also by the correlation between itemsets A and B.

Lift is a simple correlation measure that is given as follows.

- The occurrence of itemset A is independent of the occurrence of itemset B if P(A ∪B) = P(A)P(B); otherwise, itemsets A and B are dependent and correlated as events.
- This definition can easily be extended to more than two item sets.
- The lift between the occurrence of A and B can be measured by computing

$$lift(A, B) = \frac{P(A \cup B)}{P(A)P(B)}.$$

- If the resulting value of equation is less than 1, then the occurrence of A is negatively correlated with the occurrence of B, meaning that the occurrence of one likely leads to the absence of the other one.
- If the resulting value is greater than 1, then A and B are positively correlated, meaning that the occurrence of one implies the occurrence of the other.
- If the resulting value is equal to 1, then A and B are independent and have no correlation.

CONSTRAINT-BASED ASSOCIATION MINING.

A data mining process may uncover thousands of rules from a given set of data, most of which end up being unrelated or uninteresting to the users. Often, users have a good sense of which "direction" of mining may lead to interesting patterns and the "form" of the patterns or rules they would like to find. Thus, a good heuristic is to have the users specify such intuition or expectations as constraints to confine the search space. This strategy is known as constraint-based mining.

The constraints can include the following:

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- **Knowledge type constraints:** These specify the type of knowledge to be mined, such as association, regression or correlation.

- **Data constraints:** Specifies the type of data on which you want to generate the rules. These specify the set of task-relevant data.

- **Dimension/level constraints:** These specify the desired dimensions (or attributes) of the data, or levels of the concept hierarchies, to be used in mining.

- **Interestingness constraints:** These specify thresholds on statistical measures of rule interestingness, such as support, confidence, and correlation.

- **Rule constraints:** These specify the form of rules to be mined. Such constraints may be expressed as metarules (rule templates), as the maximum or minimum number of

predicates that can occur in the rule antecedent or consequent, or as relationships among attributes, attribute values, and/or aggregates.

ADVANTAGES OF DATA MINING

Data mining is a powerful tool used across various industries to extract meaningful patterns and knowledge from large datasets. Here are some of the key advantages:

1. **Informed Decision-Making**: Data mining helps organizations make better decisions by uncovering hidden patterns and relationships in data. This can lead to more accurate forecasting, improved strategic planning, and effective resource allocation.

2. **Improved Customer Relationships**: By analyzing customer data, companies can better understand customer preferences and behaviors. This enables personalized marketing, targeted promotions, and improved customer service, which can enhance customer satisfaction and loyalty.

3. **Enhanced Efficiency**: Data mining can identify inefficiencies and areas for improvement within processes and operations. For example, it can help streamline supply chain management, optimize production schedules, and reduce operational costs.

4. **Competitive Advantage**: Organizations can gain a competitive edge by leveraging data mining to uncover trends and insights that their competitors might not be aware of. This can lead to innovative products, services, and business strategies.

5. **Risk Management**: Data mining techniques can help in identifying potential risks and fraud. For instance, financial institutions use data mining to detect unusual transaction patterns that could indicate fraudulent activity, allowing for timely intervention.

6. **Predictive Analytics**: By analyzing historical data, data mining can provide predictive insights into future trends and behaviors. This is valuable for anticipating market changes, customer needs, and other business dynamics.

7. **Product and Service Improvement**: Data mining can reveal insights into product performance and customer feedback, leading to improvements in product design, quality, and service offerings.

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8. **Enhanced Data Quality**: Data mining can identify inconsistencies, errors, and missing values in datasets, leading to better data quality and more reliable analysis.

9. **Market Segmentation**: Businesses can use data mining to segment their markets more effectively. This allows for more precise targeting of different customer groups and tailoring of marketing strategies to meet the needs of each segment.
10. **Trend Analysis**: Data mining can help identify and analyze emerging trends in various fields, such as consumer behavior, market conditions, and technological advancements. This information can be crucial for adapting business strategies and staying ahead of the curve.

ISSUES OR CHALLENGES DATA MINING

Data mining, while powerful, also comes with a set of challenges and issues that can impact its effectiveness and ethical implications. Here are some of the key challenges:

1. **Data Quality**: Poor-quality data—such as incomplete, inaccurate, or inconsistent data—can lead to misleading or incorrect findings. Ensuring data cleanliness and accuracy is crucial for reliable results.

2. **Data Privacy**: Mining sensitive or personal data raises significant privacy concerns. There is a risk of unauthorized access and misuse of personal information, which can lead to ethical and legal issues.

3. **Data Security**: Protecting data from breaches, cyber-attacks, and other security threats is a major concern. Ensuring robust security measures is essential to safeguard sensitive information.

4. **Complexity and Scalability**: As datasets grow larger and more complex, the computational resources and time required for data mining increase. Handling large-scale data efficiently requires advanced technologies and infrastructure.

5. **Bias and Fairness**: Data mining algorithms can perpetuate or even exacerbate biases present in the data. If the data reflects historical biases or imbalances, the results can lead to unfair or discriminatory outcomes.

6. **Interpretability**: Some data mining techniques, such as deep learning models, can be complex and hard to interpret. Understanding how these models make decisions is important for trust and accountability.

7. **Ethical Considerations**: The ethical use of data mining involves ensuring that data is used responsibly and that decisions based on data mining do not harm individuals or communities. This includes considerations around consent and transparency.

8. **Legal and Regulatory Compliance**: Adhering to data protection regulations, such as GDPR in Europe or CCPA in California, is essential. Non-compliance can result in legal penalties and damage to reputation.
9. **Integration with Existing Systems**: Integrating data mining insights into existing business processes and systems can be challenging. It requires careful planning and coordination to ensure that the insights are actionable and effectively utilized.
10. **Overfitting**: Data mining models can sometimes be too closely aligned with the training data, resulting in overfitting. This means the model performs well on the training data but poorly on new, unseen data.
11. **Skill Requirements**: Effective data mining requires expertise in statistics, machine learning, and domain knowledge. Finding and retaining skilled professionals can be a challenge.
12. **Changing Data Patterns**: Data patterns can change over time due to evolving market conditions, behaviors, or other factors. Models need to be updated regularly to reflect these changes and remain accurate.

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UNIT – 3

CLASSIFICATION AND PREDICTION

CLASSIFICATION

Classification is a task in data mining that involves assigning a class label to each instance in a dataset based on its features. The goal of classification is to build a model that accurately predicts the class labels of new instances based on their features.

Or

Classfication is a machine learning technique identify the category of new observation on training data

PREDICTION

In the prediction method, we need to predict the missing data for a new observation, depending on the previous data. Or we can say that the predictive models use comprehended outcomes to create a model that can be used to predict values for new data.

ISSUES REGARDING CLASSIFICATION AND PREDICTION

- **Data Quality and Preprocessing:** Data mining often deals with large, complex datasets containing missing values, outliers, or noise. Ensuring data quality through preprocessing techniques such as outlier detection, and noise reduction is crucial for building accurate classification and prediction models.

- **Feature Selection and Dimensionality Reduction:** High-dimensional datasets can pose challenges for classification and prediction tasks. Feature selection techniques are used to identify the most relevant features.

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- **Imbalanced Datasets:** Imbalanced class distributions are common in real-world datasets, where one class is significantly more prevalent than others. Dealing with imbalanced data requires special attention to avoid biased model performance.

- **Model Selection and Evaluation:** Choosing the right classification or prediction model is critical for achieving good performance. Data miners need to consider factors such as model complexity, interpretability, scalability, and the specific characteristics of the dataset. Furthermore, robust evaluation techniques, such as cross-validation and bootstrapping, are essential for assessing model performance accurately.

- **Interpretability vs. Accuracy Trade-off:** In many data mining applications, interpretability is as important as accuracy, especially when the models are used to make decisions or provide insights to stakeholders. Balancing between model complexity and interpretability is a common challenge in classification and prediction tasks.

- **Time Complexity and Scalability:** Some classification and prediction algorithms may have high time complexity, making them impractical for large-scale datasets. Data miners need to consider algorithms' scalability and computational requirements when dealing with big data.

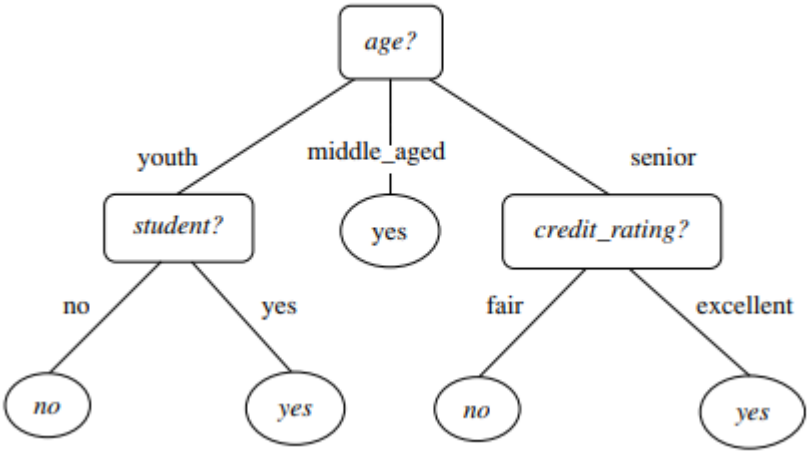
- **Model Deployment and Integration:** Deploying data mining models into production environments requires careful consideration of integration with existing systems, real-time processing requirements, and scalability. Ensuring smooth deployment and maintenance of models is essential for their practical utility.

- **Ethical and Legal Considerations:** Data mining often involves sensitive information, and the use of predictive models may raise ethical and legal concerns, such as privacy violations, discrimination, or unintended consequences. Data miners must adhere to ethical guidelines and legal regulations when collecting, processing, and using data for classification and prediction purposes.

CLASSIFICATION BY DECISION TREE INDUCTION

Decision tree induction is the learning of decision trees from class-labeled training tuples.
A decision tree is a flowchart-like tree structure, where

- each internal node (nonleaf node) denotes a test on an attribute
- each branch represents an outcome of the test
- each leaf node (or terminal node) holds a class label.
- The topmost node in a tree is the root node.



A decision tree for the concept buys a computer, indicating whether an AllElectronics customer is likely to purchase a computer. Each internal (non-leaf) node represents a test on an attribute. Each leaf node represents a class (either buys computer = yes or buys computer = no).

- The construction of decision treeclassifiers does not require any domain knowledge or parameter setting, and therefore I appropriate for exploratory knowledge discovery.
- Decision trees can handle high dimensionaldata.

- Their representation of acquired knowledge in tree formis intuitive and generallyeasy to assimilate by humans.
- The learning and classification steps of decision treeinduction are simple and fast. In general, decision tree classifiers have good accuracy.
- Decision tree induction algorithmshave been used for classification in many application areas, such as medicine,manufacturing and production, financial analysis, astronomy, and molecular biology

Algorithm: Generate decision tree. Generate a decision tree from the training tuples of data partition, D.
Input:

- Data partition, D, which is a set of training tuples and their associated class labels
- attribute list, the set of candidate attributes
- Attribute selection method, a procedure to determine the splitting criterion that "best" partitions the data tuples into individual classes. This criterion consists of a splitting attribute and, possibly, either a split-point or splitting subset.

Output: A decision tree.

Method:

```
(1)  create a node N;
(2)  if tuples in D are all of the same class, C, then
(3)      return N as a leaf node labeled with the class C;
(4)  if attribute_list is empty then
(5)      return N as a leaf node labeled with the majority class in D; // majority voting
(6)  apply Attribute_selection_method(D, attribute_list) to find the "best" splitting_criterion;
(7)  label node N with splitting_criterion;
(8)  if splitting_attribute is discrete-valued and
         multiway splits allowed then // not restricted to binary trees
(9)      attribute_list ← attribute_list − splitting_attribute; // remove splitting_attribute
(10) for each outcome j of splitting_criterion
     // partition the tuples and grow subtrees for each partition
(11)     let Dj be the set of data tuples in D satisfying outcome j; // a partition
(12)     if Dj is empty then
(13)         attach a leaf labeled with the majority class in D to node N;
(14)     else attach the node returned by Generate_decision_tree(Dj, attribute_list) to node N;
     endfor
(15) return N;
```

The basic algorithm for inducing a decision tree from training tuples.

- If the tuples in D are all of the same class, then node N becomes a leaf and is labeled with that class (steps 2 and 3).
- Note that steps 4 and 5 are terminating conditions.
- All terminating conditions are explained at the end of the algorithm.
- Otherwise, the algorithm calls Attribute selection method to determine the splitting criterion.
- The splitting criterion tells us which attribute to test at node N by determining the "best" way to separate or partition the tuples in D into individual classes (step 6).

- The splitting criterion tells us which attribute to test at node N by determining the “best” way to separate or partition the tuples in D into individual classes (step 6).
- The node N is labeled with the splitting criterion, which serves as a test at the node (step 7).
- A branch is grown from node N for each of the outcomes of the splitting criterion. The tuples in D are partitioned accordingly (steps 10 to 11).

There are three possible scenarios, as illustrated in below figure. Let A be the splitting attribute. A has v distinct values, {a1, a2,..., av }, based on the training data.

1. A is discrete-valued: In this case, the outcomes of the test at node N correspond directly to the known values of A. A branch is created for each known value, aj , of A and labeled with that value (Figure 4a).
Partition Dj is the subset of class-labeled tuples in D having value aj of A. Because all the tuples in a given partition have the same value for A, A need not be considered in any future partitioning of the tuples. Therefore, it is removed from attribute list (steps 8 and 9).

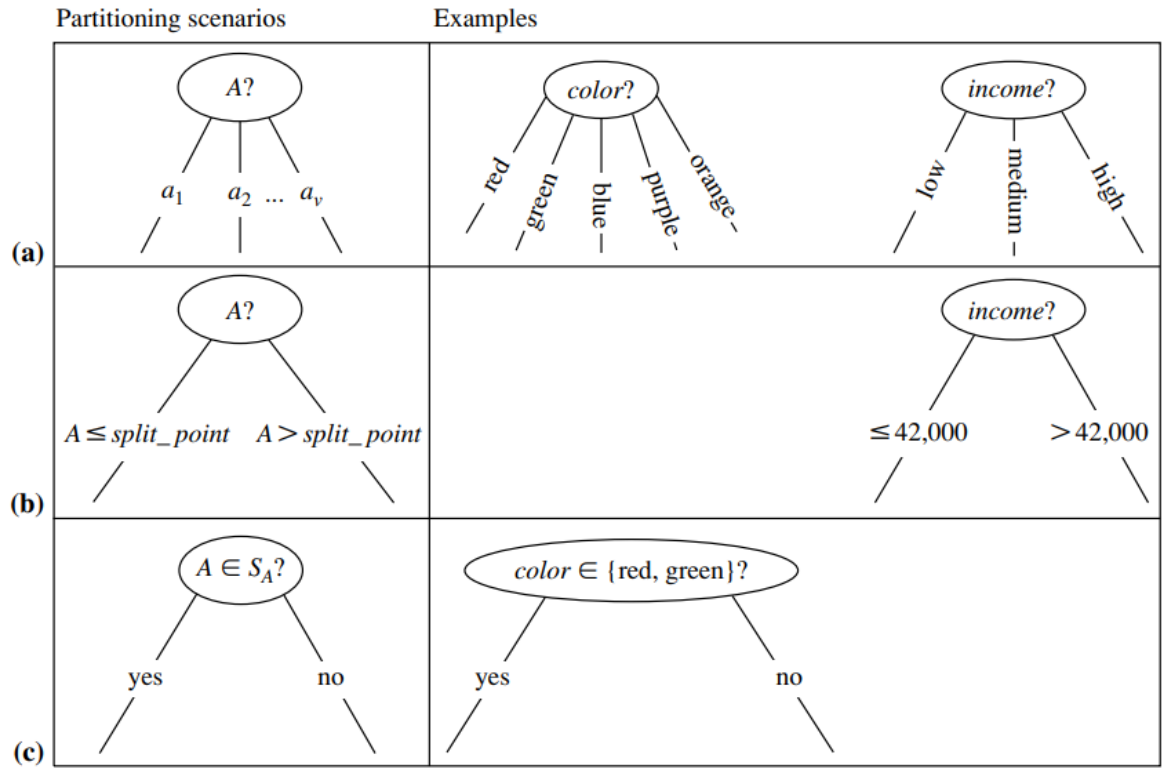
A is continuous-valued: In this case, the test at node N has two possible outcomes, corresponding to the conditions A ≤ split point and A > split point, respectively, where split point is the split-point returned by Attribute selection method as part of the splitting criterion. (In practice, the split-point, a, is often taken as the midpoint of two known adjacent values of A and therefore may not actually be a pre existing value of A from the training data.)

- Two branches are grown from N and labeled according to the previous outcomes (Figure 4b).
- The tuples are partitioned such that D1 holds the subset of class-labeled tuples in D for which A ≤ split point, while D2 holds the rest.

3. A is discrete-valued and a binary tree must be produced:

- Test at Node N: The test is structured as "A ∈ SA?". This means we check if the attribute value A of a tuple belongs to the subset SA.
- If a tuple's attribute value A is in SA (satisfies the test), it follows the left branch labeled "yes".
- If the tuple's attribute value A is not in SA (does not satisfy the test), it follows the right branch labeled "no".
- **D1**: Contains tuples where attribute A belongs to SA and are labeled according to their class.
- **D2**: Contains tuples where attribute A does not belong to SA and are also labeled according to their class.

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This figure shows three possibilities for partitioning tuples based on the splitting criterion, each with examples. Let A be the splitting attribute. (a) If A is discrete-valued, then one branch is grown for each known value of A. (b) If A is continuous-valued, then two branches are grown, corresponding to A ≤ split point and A > split point. (c) If A is discrete-valued and a binary tree must be produced, then the test is of the form A ∈ SA, where SA is the splitting subset for A.

BAYESIAN CLASSIFICATION

Bayesian classifiers are statistical classifiers. They can predict class membership probabilities such as the probability that a given tuple belongs to a particular class.

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Bayes' Theorem

Bayes' theorem is named after Thomas Bayes, who did early work in probability and decision theory during the 18th century.

- Let X be a data tuple. In Bayesian terms, X is considered "evidence." As usual, it is described by measurements made on a set of n attributes.
- Let H be some hypothesis such as that the data tuple X belongs to a specified class C.
- For classification problems, we want to determine P(H|X), the probability that the hypothesis H holds given the "evidence" or observed data tuple X.
- P(H|X) is the posterior probability of H conditioned on X.
- For example, suppose our world of data tuples is confined to customers described by the attributes age and income, respectively, and that X is a 35-year-old customer with an income of $40,000.
- Suppose that H is the hypothesis that our customer will buy a computer.
- Then P(H|X) reflects the probability that customer X will buy a computer given that we know the customer's age and income.

- In contrast, P(H) is the prior probability of H.
- For our example, this is the probability that any given customer will buy a computer, regardless of age, income, or any other information, for that matter.
- The posterior probability, P(H|X), is based on more information (e.g., customer information) than the prior probability, P(H), which is independent of X.
- Similarly, P(X|H) is the posterior probability of X conditioned on H. That is, it is the probability that a customer, X, is 35 years old and earns $40,000, given that we know the customer will buy a computer.
- P(X) is the prior probability of X.
- Using our example, it is the probability that a person from our set of customers is 35 years old and earns $40,000.

Bayes' theorem is useful in that it provides a way of calculating the posterior probability, P(H|X), from P(H), P(X|H), and P(X). Bayes' theorem is

$$P(H|\boldsymbol{X}) = \frac{P(\boldsymbol{X}|H)P(H)}{P(\boldsymbol{X})}.$$

X, H is the events
P(X/H) is the probability of X given H is true
P(H/X) is the probability of H given X is true
P(X), and P(H) are the independent probabilities of X and H

NAÏVE BAYES CLASSIFICATION

The naive Bayesian classifier, or simple Bayesian classifier, works as follows:

Let D be a training set of tuples and their associated class labels. As usual, each tuple is represented by an n-dimensional attribute vector, X = (x1, x2,..., xn), depicting n measurements made on the tuple from n attributes, respectively, A1, A2,..., An.

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Suppose that there are m classes, C1, C2,..., Cm. Given a tuple, X, the classifier will predict that X belongs to the class having the highest posterior probability, conditioned on X. That is, the na¨ıve Bayesian classifier predicts that tuple X belongs to the class Ci if and only if

$$P(C_i|\boldsymbol{X}) > P(C_j|\boldsymbol{X}) \quad \text{for } 1 \le j \le m, j \ne i.$$

Thus, we maximize P(Ci |X). The class Ci for which P(Ci |X) is maximized is called the maximum posterior hypothesis. By Bayes' theorem

$$P(C_i|\boldsymbol{X}) = \frac{P(\boldsymbol{X}|C_i)P(C_i)}{P(\boldsymbol{X})}.$$

As P(X) is constant for all classes, only P(X|Ci)P(Ci) needs to be maximized. If the class prior probabilities are not known, then it is commonly assumed that the classes are equally likely, that is, P(C1) = P(C2) = ⋯ = P(Cm), and we would therefore maximize P(X|Ci). Otherwise,

we maximize P(X|Ci)P(Ci). Note that the class prior probabilities may be estimated by P(Ci) = |Ci,D|/|D|, where |Ci,D| is the number of training tuples of class Ci in D.

For example

- let X = (35,$40,000), where A1 and A2 are the attributes age and income, respectively.
- Let the class label attribute be buys computer. The associated class label for X is yes (i.e., buys computer = yes).
- Let's suppose that age has not been discretized and therefore exists as a continuous-valued attribute.
- Suppose that from the training set, we find that customers in D who buy a computer are 38 ± 12 years of age.
- In other words, for attribute age and this class, we have μ = 38 years and σ = 12. We can plug these quantities, along with x1 = 35 for our tuple X, into Equation to estimate P(age = 35|buys computer = yes).
- To predict the class label of X, P(X|Ci)P(Ci) is evaluated for each class Ci . The classifier predicts that the class label of tuple X is the class Ci if and only if P(X|Ci)P(Ci) > P(X|Cj)P(Cj) for 1 ≤ j ≤ m, j 6= i.
- In other words, the predicted class label is the class Ci for which P(X|Ci)P(Ci) is the maximum.

RID	*age*	*income*	*student*	*credit_rating*	*Class: buys_computer*
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle_aged	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle_aged	low	yes	excellent	yes
8	youth	medium	no	fair	no
9	youth	low	yes	fair	yes
10	senior	medium	yes	fair	yes
11	youth	medium	yes	excellent	yes
12	middle_aged	medium	no	excellent	yes
13	middle_aged	high	yes	fair	yes
14	senior	medium	no	excellent	no

The training data are shown in the above Table. The data tuples are described by the attributes age, income, student, and credit rating. The class label attribute, buys computer, has two distinct values (namely, {yes, no}). Let C1 correspond to the class buys computer = yes and C2 correspond to buys computer = no. The tuple we wish to classify is

X = (age = youth, income = medium, student = yes, credit rating = fair)

We must maximize P(X|Ci)P(Ci), for i = 1, 2. P(Ci), the prior probability of each class, can be computed based on the training tuples:
P(buys computer = yes) = 9/14 = 0.643
P(buys computer = no) = 5/14 = 0.357

To compute P(X|Ci), for i = 1, 2, we compute the following conditional probabilities:

P(age = youth | buys computer = yes) = 2/9 = 0.222
P(age = youth | buys computer = no) = 3/5 = 0.600
P(income = medium | buys computer = yes) = 4/9 = 0.444
P(income = medium | buys computer = no) = 2/5 = 0.400
P(student = yes | buys computer = yes) = 6/9 = 0.667
P(student = yes | buys computer = no) = 1/5 = 0.200
P(credit rating = fair | buys computer = yes) = 6/9 = 0.667
P(credit rating = fair | buys computer = no) = 2/5 = 0.400

Using these probabilities, we obtain

P(X|buys computer = yes) = P(age = youth | buys computer = yes)
× P(income = medium | buys computer = yes)
× P(student = yes | buys computer = yes)
× P(credit rating = fair | buys computer = yes)
= 0.222 × 0.444 × 0.667 × 0.667 = 0.044.

Similarly,
P(X|buys computer = no) = 0.600 × 0.400 × 0.200 × 0.400 = 0.019.
To find the class, Ci , that maximizes P(X|Ci)P(Ci), we compute
P(X|buys computer = yes)P(buys computer = yes) = 0.044 × 0.643 = 0.028
P(X|buys computer = no)P(buys computer = no) = 0.019 × 0.357 = 0.007

Therefore, the naïve Bayesian classifier predicts buys computer = yes for tuple X

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RULE BASED CLASSIFICATION

Using IF-THEN Rules for Classification

Rules are a good way of representing information or bits of knowledge. A rule-based classifier uses a set of IF-THEN rules for classification. An IF-THEN rule is an expression of the form

IF condition THEN conclusion.

An example is rule R1

R1: IF age = youth AND student = yes THEN buys computer = yes

- The “IF” part (or left side) of a rule is known as the rule antecedent or precondition.

- The "THEN" part (or right side) is the rule consequent. In the rule antecedent, the condition consists of one or more attribute tests (e.g., age = youth and student = yes) that are logically ANDed.
- The rule's consequent contains a class prediction (in this case, we are predicting whether a customer will buy a computer)
- R1 can also be written as
- **R1: (age = youth) ∧ (student = yes) ⇒ (buys computer = yes)**

If the condition (i.e., all the attribute tests) in a rule antecedent holds true for a given tuple, we say that the rule antecedent is satisfied (or simply, that the rule is satisfied) and that the rule covers the tuple.

A rule R can be assessed by its coverage and accuracy.
Given a tuple, X, from a classlabeled data set, D, let n_{covers} be the number of tuples covered by R; $n_{correct}$ be the number of tuples correctly classified by R; and |D| be the number of tuples in D.
We can define the coverage and accuracy of R as

$$coverage(R) = \frac{n_{covers}}{|D|}$$

$$accuracy(R) = \frac{n_{correct}}{n_{covers}}.$$

- That is, a rule's coverage is the percentage of tuples that are covered by the rule (i.e., their attribute values hold true for the rule's antecedent).
- For a rule's accuracy, we look at the tuples that it covers and see what percentage of them the rule can correctly classify.

Rule Extraction from a Decision Tree

- To extract rules from a decision tree, one rule is created for each path from the root to a leaf node.
- Each splitting criterion along a given path is logically ANDed to form the rule antecedent ("IF" part).
- The leaf node holds the class prediction, forming the rule consequent ("THEN" part).

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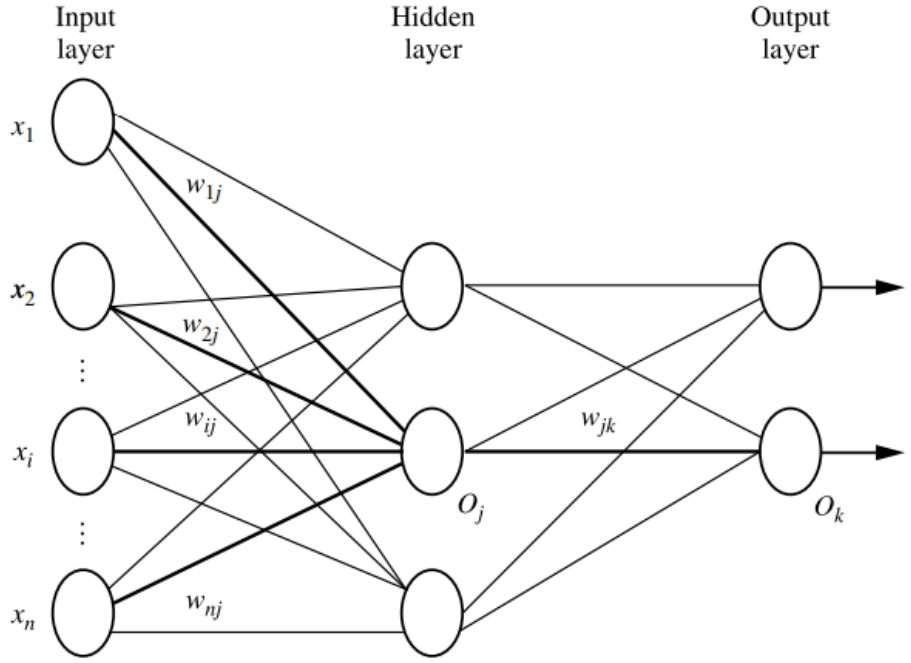
CLASSIFICATION BY BACKPROPAGATION

Backpropagation is an algorithm that backpropagates the errors from the output nodes to the input nodes. Therefore, it is simply referred to as the backward propagation of errors. It uses in the vast applications of neural networks in data mining like Character recognition, Signature verification, etc.

Neural Network:

- Neural networks are an information processing paradigm inspired by the human nervous system.

- Just like in the human nervous system, we have biological neurons in the same way in neural networks we have artificial neurons, artificial neurons are mathematical functions derived from biological neurons.
- The human brain is estimated to have about 10 billion neurons, each connected to an average of 10,000 other neurons.
- Each neuron receives a signal through a synapse, which controls the effect of the signconcerning on the neuron.



Multilayer feed-forward neural network.

Backpropagation:

- Backpropagation is a widely used algorithm for training feedforward neural networks.
- It computes the gradient of the loss function with respect to the network weights.
- It is very efficient, rather than naively directly computing the gradient concerning each weight.
- This efficiency makes it possible to use gradient methods to train multi-layer networks and update weights to minimize loss; variants such as gradient descent or stochastic gradient descent are often used.
- The backpropagation algorithm works by computing the gradient of the loss function with respect to each weight via the chain rule, computing the gradient layer by layer, and iterating backward from the last layer to avoid redundant computation of intermediate terms in the chain rule.

Features of Backpropagation:

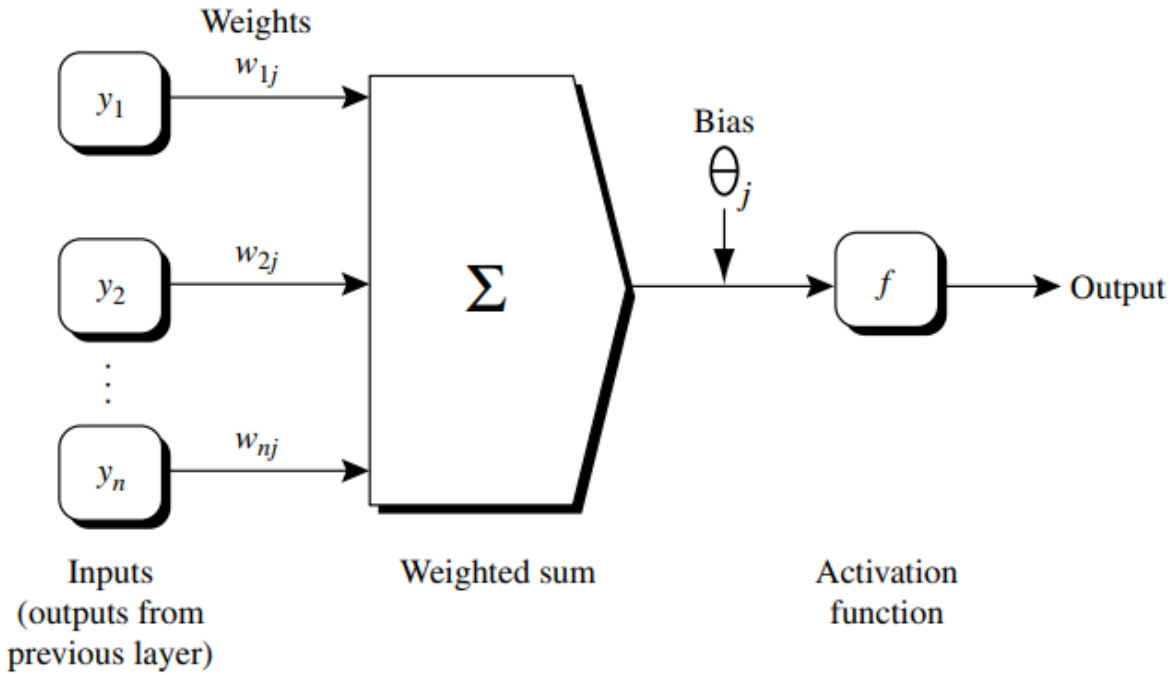
- it is the gradient descent method as used in the case of simple perceptron network with the differentiable unit.
- it is different from other networks in respect to the process by which the weights are calculated during the learning period of the network.

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- training is done in the three stages :
  - the feed-forward of input training pattern
  - the calculation and backpropagation of the error
  - updation of the weight

Backpropagation Algorithm:

Step 1: Inputs X, arrive through the preconnected path.
Step 2: The input is modeled using true weights W. Weights are usually chosen randomly.
Step 3: Calculate the output of each neuron from the input layer to the hidden layer to the output layer.
Step 4: Calculate the error in the outputs
Backpropagation Error= Actual Output – Desired Output
Step 5: From the output layer, go back to the hidden layer to adjust the weights to reduce the error.
Step 6: Repeat the process until the desired output is achieved.



SUPPORT VECTOR MACHINES

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression tasks. Its main goal is to find the best decision boundary (hyperplane) that separates data points of different classes.

Hyperplane :A line (in 2D), plane (in 3D), or surface that divides the feature space into two classes.

A support vector machine (SVM) is a supervised machine learning algorithm that classifies data by finding an optimal line or hyperplane that maximizes the distance between each class in an N-dimensional space. Support Vectors: The closest data points to the hyperplane.

- An SVM is an algorithm that works as follows.
- It uses a nonlinear mapping to transform the original training data into a higher dimension. Within this new dimension, it searches for the linear optimal separating hyperplane (i.e., a “decision boundary” separating the tuples of one class from another).

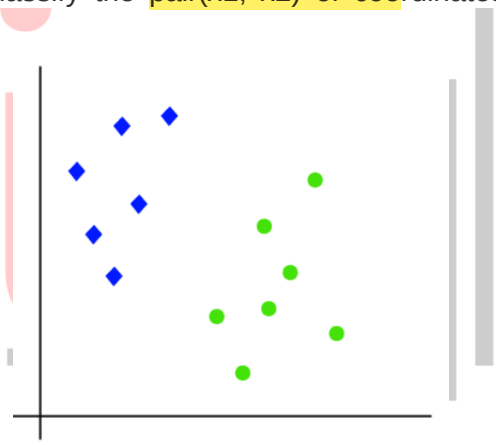
- SVMs can be used for numeric prediction as well as classification.
- They have been applied to a number of areas, including handwritten digit recognition, object recognition, and speaker identification, as well as benchmark time-series prediction tests.

Types of SVM

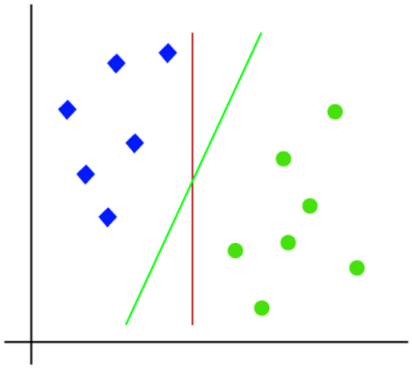
- **Linear SVM/Linearly Separable:** Linear SVM is used for linearly separable data, which means if a dataset can be classified into two classes by using a single straight line, then such data is termed as linearly separable data, and classifier is used called as Linear SVM classifier.
- **Non-linear SVM/Linearly Inseparable**: Non-Linear SVM is used for non-linearly separated data, which means if a dataset cannot be classified by using a straight line, then such data is termed as non-linear data and classifier used is called as Non-linear SVM classifier.

The Case When the Data Are Linearly Separable

The working of the SVM algorithm can be understood by using an example. Suppose we have a dataset that has two tags (green and blue), and the dataset has two features x1 and x2. We want a classifier that can classify the pair(x1, x2) of coordinates in either green or blue. Consider the below image:

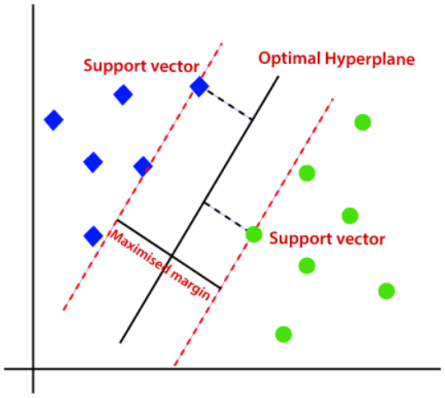


So as it is 2-d space so by just using a straight line, we can easily separate these two classes. But there can be multiple lines that can separate these classes. Consider the below image:



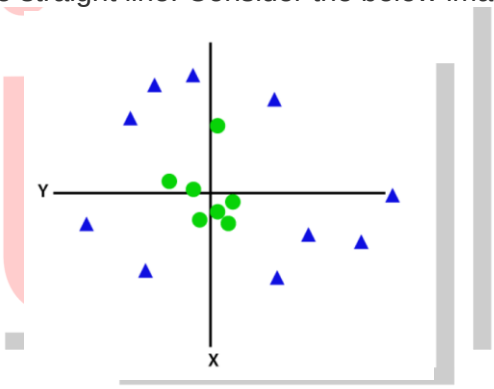
Hence, the SVM algorithm helps to find the best line or decision boundary; this best boundary or region is called as a **hyperplane**. SVM algorithm finds the closest point of the lines from both the classes. These points are called support vectors. The distance between the vectors

and the hyperplane is called as **margin**. And the goal of SVM is to maximize this margin. The **hyperplane** with maximum margin is called the **optimal hyperplane**.



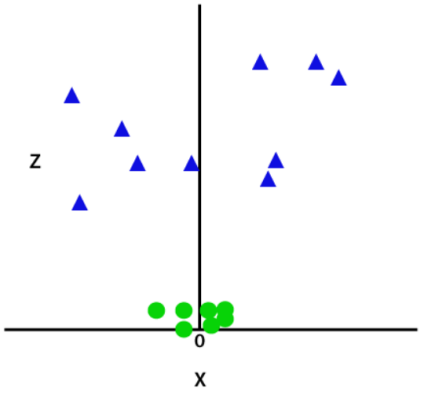
The Case When the Data Are Linearly Inseparable

If data is linearly arranged, then we can separate it by using a straight line, but for non-linear data, we cannot draw a single straight line. Consider the below image

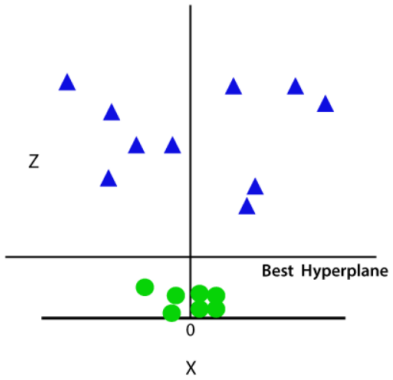


So to separate these data points, we need to add one more dimension. For linear data, we have used two dimensions x and y, so for non-linear data, we will add a third dimension z. It can be calculated as: $z=x^2+y^2$

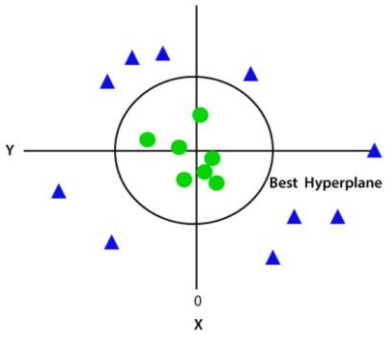
By adding the third dimension, the sample space will become as below image:



So now, SVM will divide the datasets into classes in the following way. Consider the below image:



Since we are in 3-d Space, hence it is looking like a plane parallel to the x-axis. If we convert it in 2d space with z=1, then it will become as:



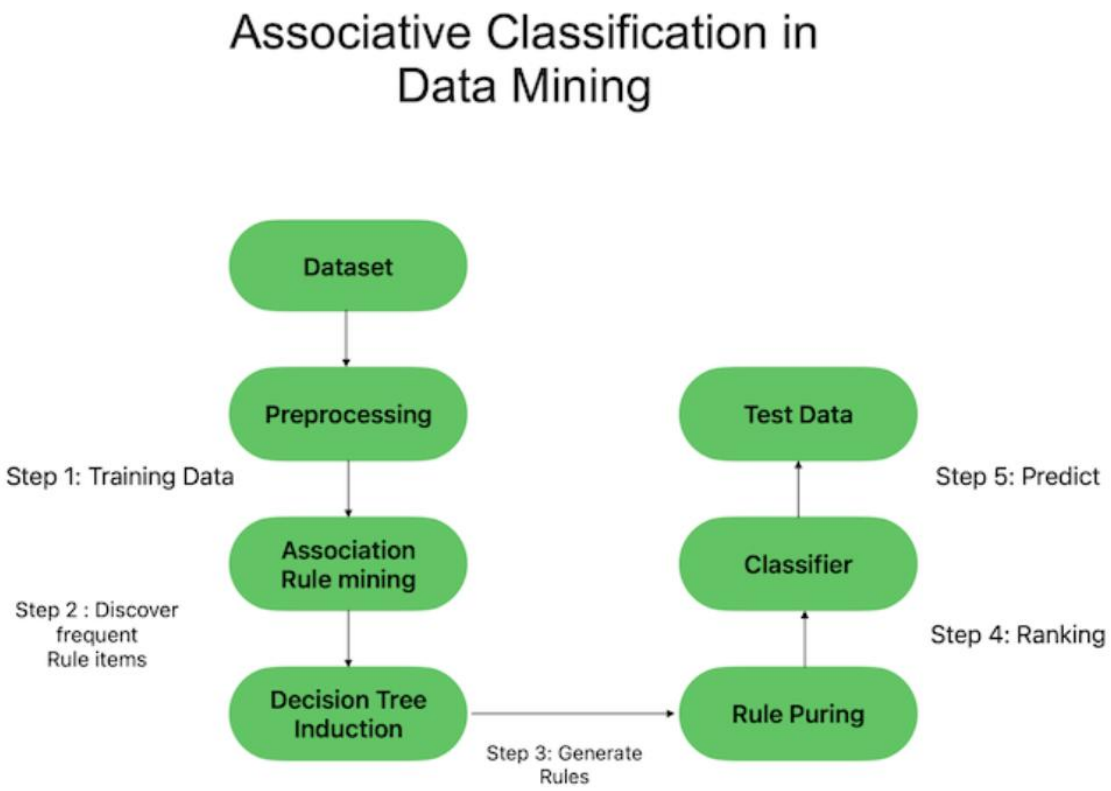
Hence we get a circumference of radius 1 in the case of non-linear data.

ASSOCIATION CLASSIFICATION (ASSOCIATIVE CLASSIFICATION)

- A crucial data mining approach called associative classification focuses on identifying connections and interactions between various variables in a dataset. Its goal is to find relationships and patterns among qualities so that future events can be predicted or new occurrences can be categorized.
- Associative classification is essential for realizing its full potential in data mining.
- Making prediction or classification jobs easier, entails identifying correlations and links between attributes in a collection.
- The fundamental goal of associative classification is to identify patterns connecting different variables by using association rule mining techniques.
- Rule creation, rule assessment, and rule selection are generally the three main steps in the process.
- When rules are developed, they are based on the dataset, however, when rules are evaluated, they are evaluated for quality and importance.
- In order to improve the accuracy and relevance of the classification process, rule selection seeks to weed out unimportant or inapplicable rules.
- A few benefits of associative categorization are its capacity to manage complicated data linkages, manage high−dimensional datasets, and give comprehensible rules.
- The computational complexity of big datasets, sensitivity to noise and irrelevant features, and a possible trade−off between accuracy and interpretability are some of its drawbacks.

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- Nevertheless, being aware of these factors enables data analysts to employ associative categorization efficiently and base choices on the discovered patterns.



Techniques and Algorithms

Apriori Algorithm and Its Role in Associative Classification

- In associative classification, the Apriori algorithm is a key method that is essential for identifying popular item sets.
- The method finds itemsets that meet a minimal support criterion via an iterative technique, creating strong correlations between qualities.
- Its main function in associative categorization is to produce a set of frequent item sets from which association rules may be derived.
- Utilizing the "apriori property," which stipulates that any non−frequent itemset must have non−frequent subsets, the method effectively prunes the search space.

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Fuzzy Association Rule Mining and Its Applications

- A development of conventional association rule mining that addresses ambiguity and imprecision in data is fuzzy association rule mining.
- In datasets where characteristics include degrees of membership rather than binary values, it enables the discovery of relationships.
- In fields like medical diagnosis or consumer behavior research, where ambiguity and vagueness are common, fuzzy association rule mining is very helpful.
- This method uses fuzzy logic to generate rules and identify correlations, allowing for more informed decision−making and the identification of patterns in large datasets.

Evaluation and Validation

Metrics of association rules

To assess the value and importance of association rules produced by associative classification, many metrics are used. The metrics lift, support, and interestingness are frequently employed. The potency of connections, the accuracy of forecasts, and the applicability of the patterns found are all quantified by these measures.

Cross−Validation and Holdout Methods for Model Evaluation

Cross−validation and holdout approaches are frequently used to confirm the efficacy of the associative classification model. By splitting the dataset into several subsets, cross−validation enables repeated training and testing on various partitions.
Holdout approaches, in contrast, divide the data into training and testing sets, utilizing the first to build the model and the latter to assess how well it performs.

Techniques for Handling Imbalanced Datasets

Associative categorization can be complicated by datasetsthat are unbalanced in terms of class distribution. Undersampling, oversampling, and ensemble procedures, among others, can be used to balance the dataset and lessen the effect of class imbalance on model performance.

LAZY LEARNERS

k-Nearest-Neighbor Classifiers

- The **K-Nearest Neighbors (K-NN)** algorithm is a simple yet powerful **supervised learning algorithm** used for **classification and regression** tasks. It works by comparing a new data point to its closest "neighbors" in the training set and making predictions based on majority voting (for classification) or averaging (for regression).

- It has since been widely used in the area of pattern recognition.
- Nearest-neighbor classifiers are based on learning by analogy, that is, by comparing a given test tuple with training tuples that are similar to it.
- The training tuples are described by n attributes.
- Each tuple represents a point in an n-dimensional space.
- In this way, all the training tuples are stored in an n-dimensional pattern space.
- When given an unknown tuple, a k-nearest-neighbor classifier searches the pattern space for the k training tuples that are closest to the unknown tuple.
- These k training tuples are the k “nearest neighbors” of the unknown tuple.

$$v' = \frac{v - min_A}{max_A - min_A},$$

Where,
min A and max A are the minimum and maximum values of attribute A

- For k-nearest-neighbor classification, the unknown tuple is assigned the most common class among its k-nearest neighbors.
- When k = 1, the unknown tuple is assigned the class of the training tuple that is closest to it in pattern space.
- Nearest neighbor classifiers can also be used for prediction, that is, to return a real-valued prediction for a given unknown tuple.
- In this case, the classifier returns the average value of the real-valued labels associated with the k nearest neighbors of the unknown tuple.

OTHER CLASSIFICATION METHODS:

GENETIC ALGORITHMS

Genetic algorithms attempt to incorporate ideas of natural evolution. In general, genetic learning starts as follows.

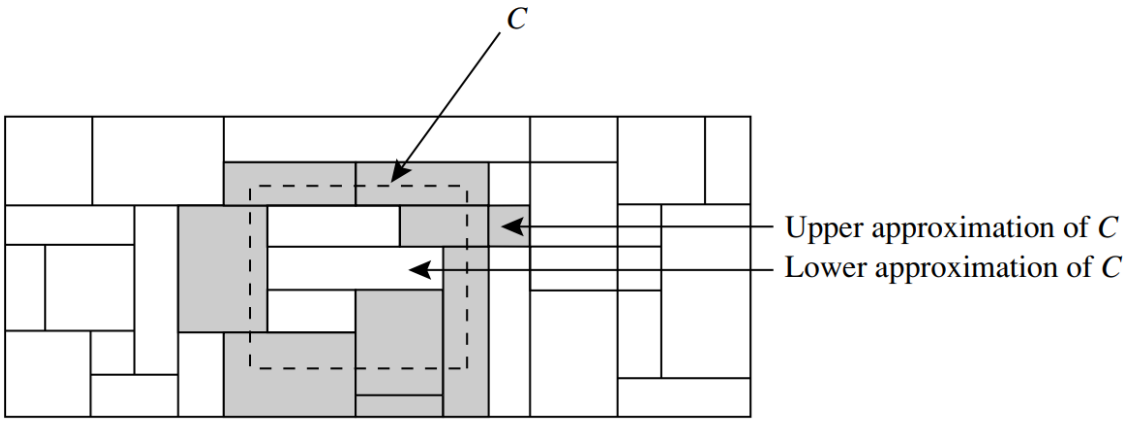
- An initial population is created consisting of randomly generated rules.
- Each rule can be represented by a string of bits.
- As a simple example, suppose that samples in a given training set are described by two Boolean attributes, A1 and A2, and that there are two classes, C1 and C2.
- The rule “IF A1 AND NOT A2 THEN C2” can be encoded as the bit string “100,” where the two leftmost bits represent attributes A1 and A2, respectively, and the rightmost bit represents the class.
- Similarly, the rule “IF NOT A1 AND NOT A2 THEN C1” can be encoded as “001.”
- If an attribute has k values, where k > 2, then k bits may be used to encode the attribute’s values.
- Classes can be encoded in a similar fashion. Based on the notion of survival of the fittest, a new population is formed to consist of the fittest rules in the current population, as well as offspring of these rules.
- Typically, the fitness of a rule is assessed by its classification accuracy on a set of training samples.
- Offspring are created by applying genetic operators such as crossover and mutation.
- In crossover, substrings from pairs of rules are swapped to form new pairs of rules.
- In mutation, randomly selected bits in a rule’s string are inverted.
- The process of generating new populations based on prior populations of rules continues until a population, P, evolves where each rule in P satisfies a prespecified fitness threshold.
- Genetic algorithms are easily parallelizable and have been used for classification and other optimization problems.
- In data mining, they may be used to evaluate the fitness of other algorithms.

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ROUGH SET APPROACH

- Rough set theory can be used for classification to discover structural relationships within imprecise or noisy data.

- It applies to discrete-valued attributes.
- Continuous-valued attributes must therefore be discretized before its use.
- Rough set theory is based on the establishment of equivalence classes within the given training data.
- All the data tuples forming an equivalence class are indiscernible, that is, the samples are identical with respect to the attributes describing the data.
- Given real-world data, it is common that some classes cannot be distinguished in terms of the available attributes.
- Rough sets can be used to approximately or “roughly” define such classes.
- A rough set definition for a given class, C, is approximated by two sets—a lower approximation of C and an upper approximation of C.
- The lower approximation of C consists of all the data tuples that, based on the knowledge of the attributes, are certain to belong to C without ambiguity.
- The upper approximation of C consists of all the tuples that, based on the knowledge of the attributes, cannot be described as not belonging to C.
- The lower and upper approximations for a class C are shown in below figure where each rectangular region represents an equivalence class. Decision rules can be generated for each class.
- Typically, a decision table is used to represent the rules.
- Rough sets can also be used for attribute subset selection (or feature reduction, where attributes that do not contribute to the classification of the given training data can be identified and removed) and relevance analysis (where the contribution or significance of each attribute is assessed with respect to the classification task).
- The problem of finding the minimal subsets (reducts) of attributes that can describe all the concepts in the given data set is NP-hard.
- However, algorithms to reduce the computation intensity have been proposed.
- In one method, for example, a discernibility matrix is used that stores the differences between attribute values for each pair of data tuples.
- Rather than searching on the entire training set, the matrix is instead searched to detect redundant attributes.



A rough set approximation of class C’s set of tuples using lower and upper approximation sets of C. The rectangular regions represent equivalence classes.

FUZZY SET APPROACHES

Rule-based systems for classification have the disadvantage that they involve sharp cutoffs for continuous attributes.

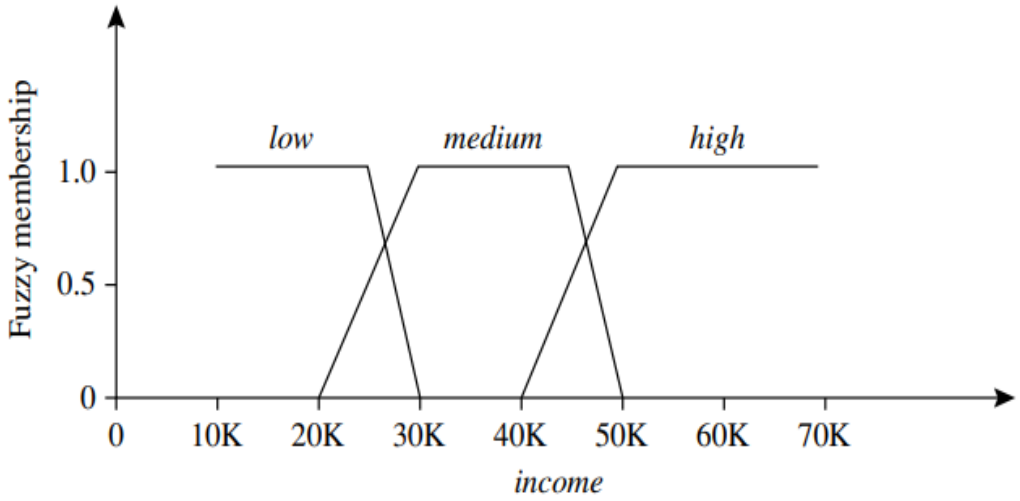
For example, consider the following rule for customer credit application approval. The rule essentially says that applications for customers who have had a job for two or more years and who have a high income (i.e., of at least $50,000) are approved:

IF (years employed ≥ 2) AND (income ≥ 50,000) THEN credit = approved.

By above rule, a customer who has had a job for at least two years will receive credit if her income is, say, $50,000, but not if it is $49,000. Such harsh thresholding may seem unfair.

- Instead, we can discretize income into categories (e.g., {low income, medium income, high income}) and then apply fuzzy logic to allow “fuzzy” thresholds or boundaries to be defined for each category (Figure 9.15). Rather than having a precise cutoff between categories.
- Fuzzy logic uses truth values between 0.0 and 1.0 to represent the degree ofmembership that a certain value has in a given category.
- Each category then represents afuzzy set.
- Fuzzy logic systemstypically provide graphical tools to assist users in converting attribute values to fuzzy truthvalues.
- Fuzzy set theory is also known as possibility theory.
- It was proposed by LotfiZadeh in1965 as an alternative to traditional two-value logic and probability theory.
- It lets uswork at a high level of abstraction and offers a means for dealing with imprecise measurementof data. Most important, fuzzy set theory allows us to dealwith vague or inexact facts.
- Unlike the notion of traditional —crispll sets where anelement either belongs to a set S or its complement, in fuzzy set theory, elements canbelong to more than one fuzzy set.
- Fuzzy set theory is useful for data mining systems performing rule-based classification.
- It provides operations for combining fuzzy measurements.
- Several procedures exist for translating the resulting fuzzy output into a defuzzifiedor crisp value that is returned by the system.
- Fuzzy logic systems have been used in numerous areas for classification, including market research, finance, health care, and environmental engineering.

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Fuzzy truth values for income, representing the degree of membership of income values with respect to the categories {low, medium, high}. Each category represents a fuzzy set. Note that a given income value, x, can have membership in more than one fuzzy set. The membership values of x in each fuzzy set do not have to total to 1.

PREDICTION

ACCURACY AND ERROR MEASURES

In data mining and machine learning, accuracy and error measures are crucial metrics used to evaluate the performance of predictive models. Here's an overview of some common accuracy and error measures:

Accuracy Measures:

1. **Accuracy**:
   - **Definition**: Accuracy measures how often the model's predictions are correct.
   - **Usage**: It's straightforward and commonly used but can be misleading in imbalanced datasets where one class dominates.

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2. **Precision**:
   - **Definition**: Precision measures the proportion of true positive predictions among all positive predictions made by the model.
   - **Usage**: It's important in applications where minimizing false positives is critical, like in medical diagnoses.

3. **Recall (Sensitivity)**:
   - **Definition**: Recall measures the proportion of true positives that are correctly identified by the model.
   - **Usage**: Useful when identifying false negatives is more critical, such as in detecting fraudulent transactions.

4. **F1 Score**:
   - **Definition**: The F1 score is the harmonic mean of precision and recall, providing a balance between the two.

- **Usage**: Particularly useful when there's an uneven class distribution (class imbalance).

Error Measures:

1. **Mean Absolute Error (MAE)**:
   - **Definition**: MAE measures the average magnitude of the errors in a set of predictions, without considering their direction (positive or negative).
   - **Usage**: Provides a straightforward interpretation of the average error magnitude.

2. **Mean Squared Error (MSE)**:
   - **Definition**: MSE measures the average of the squares of the errors—that is, the average squared difference between the predicted values and the actual values.
   - **Usage**: Emphasizes larger errors due to squaring the difference; commonly used in regression problems.

3. **Root Mean Squared Error (RMSE)**:
   - **Definition**: RMSE is the square root of MSE, providing a measure of the average magnitude of error with the same unit as the target variable.
   - **Usage**: Particularly useful when the errors are assumed to be normally distributed.

EVALUATING THE ACCURACY OF A CLASSIFIER OR PREDICTOR

Evaluating the accuracy of a classifier or predictor in data mining involves assessing how well the model performs in making predictions on unseen data. Here's a structured approach to evaluating classifier accuracy:

Here's a breakdown of the evaluation process:

1. Data Splitting:
The data is divided into two sets:
- **Training Set:** Used to build the classifier. The model learns patterns and relationships between features and class labels from this data.
- **Testing Set:** Used to evaluate the classifier's performance. This data is unseen by the model during training.

2. Training the Model:
The classifier is trained on the training data. This involves algorithms learning the underlying structure of the data.

3. Testing the Model:
The trained classifier makes predictions on the testing set. It assigns class labels to each data point based on what it learned during training.

4. Measuring Performance:
Various metrics assess how well the classifier performed on the testing set. A common metric is **accuracy**, which is the percentage of correctly classified instances.

Beyond Accuracy:

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While accuracy is popular, it can be misleading, especially for imbalanced datasets (where some classes have significantly more examples than others). To get a more complete picture, data miners use additional metrics like:

- **Precision:** Measures the proportion of true positives among all predicted positives (avoiding false positives).
- **Recall:** Measures the proportion of actual positives that were correctly identified (avoiding false negatives).
- **F1 Score:** Combines precision and recall into a single metric.

Evaluation Techniques:

- **Holdout Method:** A simple approach where the data is split into training and testing sets once. This can be efficient but might not be the most reliable due to dependence on the specific split.
- **Cross-Validation:** This method involves dividing the data into multiple folds. The classifier is trained on a combination of folds (k-1) and tested on the remaining fold. This process is repeated k times, providing a more robust estimate of performance.

By using these techniques and metrics, data miners can comprehensively assess their classifiers, identify areas for improvement, and ensure the models make accurate predictions on new data, leading to better decision-making.

ENSEMBLE METHODS

Ensemble modeling is the process of running two or more related but different analytical models and then synthesizing the results into a single score or spread. This improves the accuracy of predictive analytics and data mining applications.

Ensemble methods in data mining are powerful techniques that combine multiple base models to improve predictive performance. They work by leveraging the diversity among individual models to achieve better overall predictions. Here's a detailed overview of ensemble methods:

1. Introduction to Ensemble Methods:

- **Purpose**: Combine predictions from multiple models to improve generalizability and robustness.
- **Types**: Ensemble methods can be categorized into two main types:
  - **Bagging (Bootstrap Aggregating)**: Builds multiple models independently and combines them (e.g., Random Forest).
  - **Boosting**: Builds models sequentially to correct errors of previous models (e.g., AdaBoost, Gradient Boosting).

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2. Popular Ensemble Methods:

Bagging Methods:

- **Random Forest**:
  - **Concept**: Builds multiple decision trees using random subsets of features and averages their predictions.
  - **Advantages**: Reduces overfitting, handles large datasets with high dimensionality, and provides feature importance.

Boosting Methods:

- **AdaBoost (Adaptive Boosting)**:
  - **Concept**: Iteratively trains weak learners (e.g., decision trees) by focusing on examples that previous models misclassified.
  - **Advantages**: Improves accuracy by giving more weight to difficult cases, generally less prone to overfitting.

- **Gradient Boosting Machines (GBM)**:
  - **Concept**: Builds an ensemble of trees sequentially, where each subsequent tree corrects errors made by the previous one.
  - **Advantages**: Often yields better performance than AdaBoost, allows for optimization via learning rate, and handles missing data.

- **XGBoost (Extreme Gradient Boosting)**:
  - **Concept**: Optimizes gradient boosting with parallelized tree building, regularization, and handling of missing values.
  - **Advantages**: Efficient computation, high performance, and support for custom optimization objectives.

- **LightGBM and CatBoost**:
  - **LightGBM**: Focuses on gradient-based learning with tree-based algorithms, prioritizes splits with the largest gradient.
  - **CatBoost**: Handles categorical features naturally, incorporates ordered boosting, and improves training speed.

MODEL SELECTION

Model selection in prediction involves choosing the best predictive model from a set of candidate models based on performance and suitability for a given task. This process ensures that you select a model that accurately predicts future outcomes based on historical data. Here's a simplified guide to model selection specifically for prediction tasks:

1. Define the Prediction Task
- **Objective:** Clearly outline what you want to predict.
- **Examples:** Forecasting sales, predicting stock prices, or estimating customer lifetime value.

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2. Understand the Data
- **Purpose:** Analyze and preprocess your data to ensure it's suitable for modeling.
- **Tasks:**
  - **Data Types:** Identify the types of data (numerical, categorical, time series).
  - **Data Quality:** Check for missing values, outliers, and noise.
  - **Feature Engineering:** Create new features or modify existing ones to enhance model performance.

3. Choose Evaluation Metrics
- **Purpose:** Decide how to measure the model's performance.
- **Metrics for Regression:**
  - **Mean Absolute Error (MAE):** Average of absolute errors.
  - **Mean Squared Error (MSE):** Average of squared errors.
  - **Root Mean Squared Error (RMSE):** Square root of MSE.
  - **R-squared:** Proportion of variance explained by the model.
- **Metrics for Time Series:**
  - **Mean Absolute Percentage Error (MAPE):** Average percentage error.

- **Mean Squared Prediction Error (MSPE):** Average squared errors in time series predictions.

4. Select Candidate Models

- **Purpose:** Identify a range of models suitable for your prediction task.
- **Examples:**
  - **Linear Models:** Linear Regression, Ridge Regression, Lasso Regression.
  - **Tree-Based Models:** Decision Trees, Random Forests, Gradient Boosting Machines.
  - **Ensemble Methods:** Bagging, Boosting, Stacking.
  - **Advanced Models:** Neural Networks, Support Vector Machines (SVM), XGBoost.

5. Split the Data

- **Purpose:** Divide your data to train, validate, and test the models effectively.
- **Typical Splits:**
  - **Training Set:** Used to train the models.
  - **Validation Set:** Used to tune hyperparameters and select the best model.
  - **Test Set:** Used to assess the final performance of the selected model.

6. Train Models

- **Purpose:** Fit each candidate model to the training data.
- **Tasks:** Train models with different hyperparameters and configurations.

7. Validate Models

- **Purpose:** Evaluate the models using the validation set to fine-tune parameters and select the best performer.
- **Techniques:**
  - **Cross-Validation:** Use k-fold cross-validation to assess performance on different data subsets.
  - **Grid Search/Random Search:** Find the best hyperparameters by systematically exploring combinations.

8. Compare Model Performance

- **Purpose:** Assess and compare the performance of different models based on evaluation metrics.
- **Tasks:** Use metrics like MAE, MSE, or RMSE to compare how well each model predicts outcomes.

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9. Select the Best Model

- **Purpose:** Choose the model that best meets your performance criteria and business objectives.
- **Tasks:** Balance model accuracy with interpretability and complexity.

10. Test the Model

- **Purpose:** Evaluate the selected model on the test set to ensure it generalizes well to new data.
- **Tasks:** Verify performance metrics on unseen data.

11. Deploy the Model

- **Purpose:** Implement the chosen model in a production environment where it can make real-time predictions.
- **Tasks:** Integrate the model into applications or systems for live predictions.

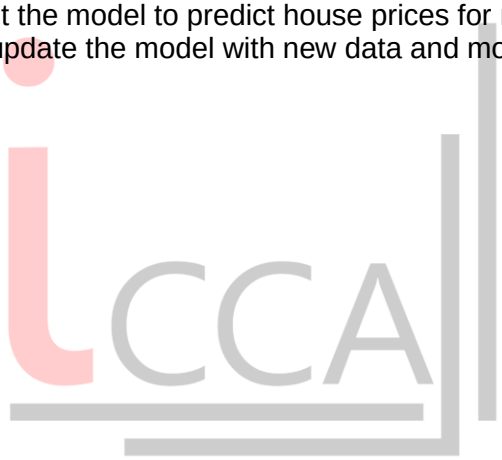
12. Monitor and Maintain

- **Purpose:** Continuously track the model's performance and update it as needed.
- **Tasks:** Regularly check metrics, retrain with new data, and adjust based on performance changes.

Example Scenario

Predicting House Prices:

1. **Objective:** Predict the price of houses based on features like size, location, and age.
2. **Data Understanding:** Analyze historical house prices and features. Handle missing values and outliers.
3. **Metrics:** Use RMSE and R-squared to evaluate model performance.
4. **Models:** Consider Linear Regression, Random Forest, and Gradient Boosting Machines.
5. **Data Split:** Divide data into training (70%), validation (15%), and test (15%) sets.
6. **Training:** Train each model using the training data.
7. **Validation:** Use cross-validation to find the best hyperparameters.
8. **Comparison:** Compare models based on RMSE and R-squared from the validation set.
9. **Selection:** Choose the model with the lowest RMSE and highest R-squared.
10. **Testing:** Evaluate the selected model on the test set to confirm its performance.
11. **Deployment:** Implement the model to predict house prices for new listings.
12. **Monitoring:** Regularly update the model with new data and monitor its accuracy



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UNIT – 4

CLUSTER ANALYSIS

- Cluster analysis or simply clustering is the process of partitioning a set of data objects (or observations) into subsets.
- Each subset is a cluster, such that objects in a cluster are similar to one another, yet dissimilar to objects in other clusters.
- The set of clusters resulting from a cluster analysis can be referred to as a clustering.
- In this context, different clustering methods may generate different clusterings on the same data set.
- The partitioning is not performed by humans, but by the clustering algorithm.
- Hence, clustering is useful in that it can lead to the discovery of previously unknown groups within the data

TYPES OF DATA IN CLUSTER ANALYSIS

In cluster analysis, the data being clustered can come in various forms, and the type of data often influences the choice of clustering algorithm or approach. Here are the common types of data that can be used in cluster analysis:

1. **Numerical Data:**
   - This type of data consists of numeric values that can be measured on a continuous or discrete scale.
   - Examples: Heights, weights, temperatures, ages, etc.
   - Clustering algorithms that work well with numerical data include K-means, hierarchical clustering, DBSCAN, and EM clustering.

2. **Categorical Data:**
   - Categorical data consists of categories or labels that do not have a natural numerical order.
   - Examples: Gender (male/female), marital status (married/single/divorced), occupation (teacher/engineer/doctor), etc.
   - Clustering categorical data often requires specific algorithms that can handle categorical variables, such as k-modes clustering.

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3. **Mixed Data:**
   - Data sets often contain a mix of numerical and categorical variables.
   - Examples: Customer data with both numerical (age, income) and categorical (gender, occupation) attributes.
   - Clustering algorithms for mixed data include methods that can handle both types of variables appropriately, such as k-prototypes clustering.

4. **Binary Data:**
   - Binary data consists of variables that have only two possible values (0 or 1).
   - Examples: Yes/no responses, presence/absence indicators, etc.
   - Clustering algorithms for binary data may include methods like binary partitioning or specialized algorithms designed for binary data clustering.

5. **Text Data:**
   - Textual data consists of words or documents where clustering is based on the similarity of text content.

- Examples: Document clustering, topic modeling.
- Clustering algorithms for text data include methods like hierarchical clustering, k-means with appropriate text representations (e.g., TF-IDF vectors), or more advanced methods like Latent Dirichlet Allocation (LDA).

6. **Time-Series Data:**
   - Time-series data consists of observations over time where clustering is based on patterns or trends over time.
   - Examples: Stock prices, sensor data, weather data, etc.
   - Clustering time-series data often involves techniques that consider temporal dependencies, such as dynamic time warping (DTW) clustering or methods adapted for time-series clustering.

7. **Spatial Data:**
   - Spatial data consists of geographic coordinates or data points located in space.
   - Examples: GPS coordinates, geographical regions, etc.
   - Clustering spatial data may involve spatial clustering algorithms like spatial DBSCAN, spatial k-means, or methods tailored for geographic clustering.

CATEGORIZATION OF MAJOR CLUSTERING METHODS

Major clustering methods can be categorized into several types based on their underlying principles and approaches to clustering data. Here's a categorization of major clustering methods:

1. **Partitioning Methods:**
   - **K-means Clustering:** Divides data into non-overlapping clusters based on centroids. Each data point belongs to the cluster with the nearest mean.
   - **K-medoids (PAM - Partitioning Around Medoids):** Similar to K-means but uses actual data points as cluster centers (medoids), which makes it more robust to outliers.
   - **CLARA (Clustering LARge Applications):** An extension of K-medoids suitable for large datasets by sampling.

2. **Hierarchical Methods:**
   - **Agglomerative Clustering:** Starts with each data point as its cluster and iteratively merges the closest clusters until only one cluster remains.
   - **Divisive Clustering:** Starts with all data points in one cluster and recursively splits it into smaller clusters until each data point is in its own cluster.
   - **BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies):** Hierarchical clustering method designed for large datasets using a tree structure.

3. **Density-Based Methods:**
   - **DBSCAN (Density-Based Spatial Clustering of Applications with Noise):** Clusters dense regions of points based on a minimum number of points (density) within a specified radius.
   - **OPTICS (Ordering Points To Identify the Clustering Structure):** Extends DBSCAN to determine the density-based clustering structure from the data.

4. **Model-Based Methods:**
   - **Expectation-Maximization (EM) Clustering:** Assumes that the data are generated by a mixture of several Gaussian distributions and iteratively finds the parameters of these distributions.

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- **Gaussian Mixture Models (GMM):** Represents the distribution of data points as a mixture of several Gaussian distributions.

5. **Grid-Based Methods:**
   - **STING (Statistical Information Grid):** Divides the data space into a finite number of cells to form a grid structure and then clusters the cells into high-density regions.

6. **Graph-Based Methods:**
   - **Spectral Clustering:** Uses the eigenvalues and eigenvectors of a similarity matrix derived from the data to reduce the dimensionality before clustering in fewer dimensions.
   - **Markov Clustering (MCL):** Uses flow simulations on graphs to partition graphs into clusters based on flow dynamics.

7. **Fuzzy Clustering Methods:**
   - **Fuzzy C-means (FCM):** A soft clustering method where each data point belongs to every cluster with a certain probability or degree of membership.
   - **Possibilistic C-means (PCM):** Extends FCM by allowing more flexibility in membership degrees.

8. **Constraint-Based Methods:**
   - **COBWEB (COnstraints-Based WEb Classifier):** Uses constraints or domain knowledge to guide the clustering process, forming attribute-value trees to represent clusters.

PARTITIONING METHODS

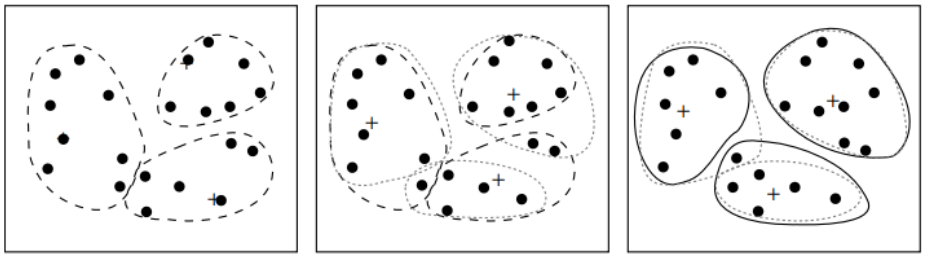
The simplest and most fundamental version of cluster analysis is partitioning, which organizes the objects of a set into several exclusive groups or clusters. To keep the problem specification concise, we can assume that the number of clusters is given as background knowledge.

1. K-means Clustering:

- **Description:** K-means is one of the most widely used clustering algorithms. It partitions the data into K clusters by iteratively assigning each data point to the nearest cluster centroid (mean) and then updating the centroids based on the current assignments.

- **Algorithm Steps:**
  - Initialize K centroids randomly.
  - Assign each data point to the nearest centroid.
  - Update the centroids by computing the mean of all data points assigned to each centroid.
  - Repeat the assignment and update steps until convergence (when centroids no longer change significantly or a maximum number of iterations is reached).

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(a) Initial clustering (b) Iterate (c) Final clustering

2. K-medoids (PAM - Partitioning Around Medoids):

- **Description:** K-medoids is similar to K-means but instead of centroids, it uses actual data points (medoids) as cluster centers. Medoids are representative points in the cluster that minimize the dissimilarity to other points within the same cluster.

- **Algorithm Steps:**
  - Select K initial medoids randomly.
  - Assign each data point to the nearest medoid.
  - For each cluster, evaluate the total dissimilarity (e.g., using distance metrics like Euclidean distance) between the medoid and all other points in the cluster.
  - Swap the medoid with another point in the cluster that minimizes the total dissimilarity if it decreases the total dissimilarity.
  - Repeat the assignment and swapping steps until convergence.

HIERARCHICAL METHODS

Hierarchical clustering is a method of cluster analysis in data mining that creates a hierarchical representation of the clusters in a dataset.

A hierarchical clustering method works by grouping data objects into a hierarchy or “tree” of clusters.

Representing data objects in the form of a hierarchy is useful for data summarization and visualization.

Agglomerative versus Divisive Hierarchical Clustering

A hierarchical clustering method can be either agglomerative or divisive, depending on whether the hierarchical decomposition is formed in a bottom-up (merging) or topdown (splitting) fashion.

Let’s have a closer look at these strategies.

- An agglomerative hierarchical clustering method uses a bottom-up strategy.
- It typically starts by letting each object form its own cluster and iteratively merges clusters into larger and larger clusters, until all the objects are in a single cluster or certain termination conditions are satisfied.
- The single cluster becomes the hierarchy’s root.
- For the merging step, it finds the two clusters that are closest to each other (according to some similarity measure), and combines the two to form one cluster.
- Because two clusters are merged per iteration, where each cluster contains at least one object, an agglomerative method requires at most n iterations.
- A divisive hierarchical clustering method employs a top-down strategy.
- It starts by placing all objects in one cluster, which is the hierarchy’s root.
- It then divides the root cluster into several smaller subclusters and recursively partitions those clusters into smaller ones.

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- The partitioning process continues until each cluster at the lowest level is coherent enough—either containing only one object, or the objects within a cluster are sufficiently similar to each other.

BIRCH: Multiphase Hierarchical Clustering Using Clustering Feature Trees

Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH) is designed for clustering a large amount of numeric data by integrating hierarchical clustering (at the initial microclustering stage) and other clustering methods such as iterative partitioning (at the later macroclustering stage).

- It overcomes the two difficulties in agglomerative clustering methods:
- (1) Scalability and (2) the inability to undo what was done in the previous step.
- BIRCH uses the notions of clustering feature to summarize a cluster, and clustering feature tree (CF-tree) to represent a cluster hierarchy.

Probabilistic Hierarchical Clustering

- Algorithmic hierarchical clustering methods using linkage measures tend to be easy to understand and are often efficient in clustering. They are commonly used in many clustering analysis applications.
- One way to look at the clustering problem is to regard the set of data objects to be clustered as a sample of the underlying data generation mechanism to be analyzed or, formally, the generative model.
- For example, when we conduct clustering analysis on a set of marketing surveys, we assume that the surveys collected are a sample of the opinions of all possible customers.
- Here, the data generation mechanism is a probability distribution of opinions with respect to different customers, which cannot be obtained directly and completely.
- The task of clustering is to estimate the generative model as accurately as possible using the observed data objects to be clustered.

DENSITY –BASED METHODS

- Partitioning and hierarchical methods are designed to find spherical-shaped clusters.
- They have difficulty finding clusters of arbitrary shape such as the “S” shape and oval clusters.
- Given such data, they would likely inaccurately identify convex regions, where noise or outliers are included in the clusters.
- To find clusters of arbitrary shape, alternatively, we can model clusters as dense regions in the data space, separated by sparse regions.
- This is the main strategy behind density-based clustering methods, which can discover clusters of nonspherical shape.

DBSCAN: Density-Based Clustering Based on Connected Regions with High Density

- The density of an object o can be measured by the number of objects close to o. DBSCAN (Density-Based Spatial Clustering of Applications with Noise) finds core objects, that is, objects that have dense neighborhoods. It connects core objects and their neighborhoods to form dense regions as clusters.
- A user-specified parameter > 0 is used to specify the radius of a neighborhood we consider for every object.
- The -neighborhood of an object o is the space within a radius centered at o. Due to the fixed neighborhood size parameterized by , the density of a neighborhood can be measured simply by the number of objects in the neighborhood.

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- To determine whether a neighborhood is dense or not, DBSCAN uses another user-specified parameter, MinPts, which specifies the density threshold of dense regions.
- An object is a core object if the -neighborhood of the object contains at least MinPts objects.
- Core objects are the pillars of dense regions.

OPTICS: Ordering Points to Identify the Clustering Structure

OPTICS (Ordering Points To Identify the Clustering Structure) is a density-based clustering algorithm that is useful for finding clusters of different densities and shapes in large, high-dimensional datasets. It is similar to the DBSCAN algorithm, but it does not require a fixed number of clusters as an input parameter. Instead, OPTICS creates an ordered list of points called the reachability plot, which can be used to identify clusters of different densities.

- The main idea behind OPTICS is to extract the clustering structure of a dataset by identifying the density-connected points.
- The algorithm builds a density-based representation of the data by creating an ordered list of points called the reachability plot.
- Each point in the list is associated with a reachability distance, which is a measure of how easy it is to reach that point from other points in the dataset.
- Points with similar reachability distances are likely to be in the same cluster.

- **Density-based clustering**: OPTICS groups points together based on their density, rather than their distance from each other.
- Reachability distance: The reachability distance between two points is the distance from one point to the other, plus the core distance of the other point. The core distance is the minimum number of points that must be within a certain radius of a point for it to be considered a core point.
- Reachability plot: A 2D plot that shows the reachability distance of each point on the y-axis and the order in which the points were processed on the x-axis. Clusters appear as valleys in the reachability plot.

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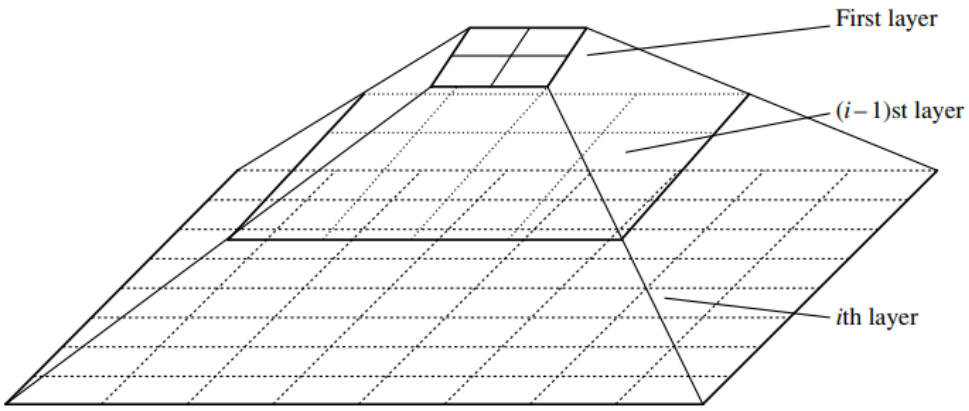
GRID-BASED METHODS

The grid-based clustering approach uses a multiresolution grid data structure. It quantizes the object space into a finite number of cells that form a grid structure on which all of the operations for clustering are performed.

STING: Statistical Information Grid

- The grid-based clustering methods use a multi-resolution grid data structure. It quantizes the object areas into a finite number of cells that form a grid structure on which all of the operations for clustering are implemented.
- The benefit of the method is its quick processing time, which is generally independent of the number of data objects, still dependent on only the multiple cells in each dimension in the quantized space.
- The grid-based clustering uses a multi-resolution grid data structure and uses dense grid cells to form clusters.
- There are several interesting methods are STING, wave cluster, and CLIQUE.

- STING − A statistical information Grid approach.
- The spatial area is split into rectangular cells.
- There are various levels of cells corresponding to different methods of resolution.
- Each cell at a high level is separated into multiple smaller cells in the next lower level.
- Statistical data of each cell is computed and stored beforehand and can answer queries.
- The specification of higher-level cells can be simply computed from the specification of lower-level cells.



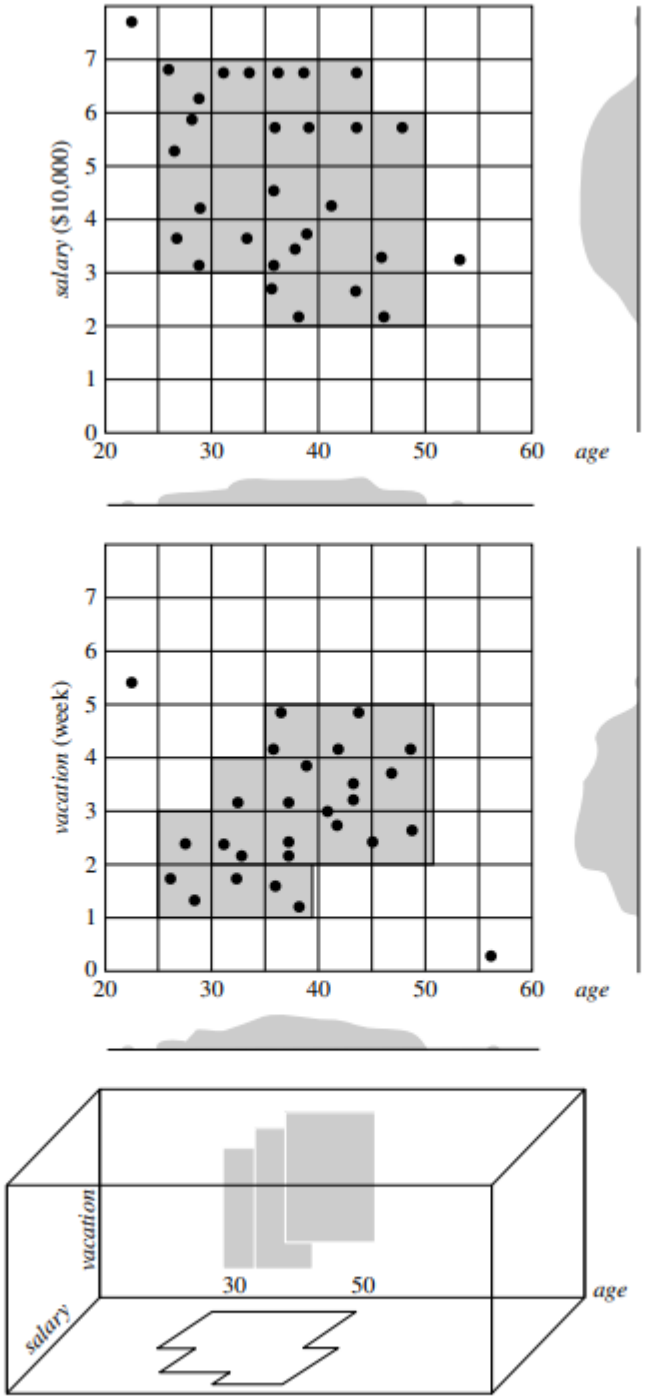
CLIQUE: An Apriori-like Subspace Clustering Method

- A data object often has tens of attributes, many of which may be irrelevant. The values of attributes may vary considerably. These factors can make it difficult to locate clusters that span the entire data space.
- It may be more meaningful to instead search for clusters within different subspaces of the data.

CLIQUE (CLustering In QUEst) is a simple grid-based method for finding densitybased clusters in subspaces.

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- CLIQUE partitions each dimension into nonoverlapping intervals, thereby partitioning the entire embedding space of the data objects into cells. It uses a density threshold to identify dense cells and sparse ones.
- A cell is dense if the number of objects mapped to it exceeds the density threshold.
- The main strategy behind CLIQUE for identifying a candidate search space uses the monotonicity of dense cells with respect to dimensionality. This is based on the Apriori property used in frequent pattern and association rule mining.
- CLIQUE performs clustering in two steps. In the first step, CLIQUE partitions the d-dimensional data space into nonoverlapping rectangular units, identifying the dense units among these. CLIQUE finds dense cells in all of the subspaces.
- In the second step, CLIQUE uses the dense cells in each subspace to assemble clusters, which can be of arbitrary shape. The idea is to apply the Minimum Description Length (MDL) principle (Chapter 8) to use the maximal regions to cover connected dense cells, where a maximal region is a hyperrectangle where every cell falling into this region is dense, and the region cannot be extended further in any dimension in the subspace.



Dense units found with respect to *age* for the dimensions *salary* and *vacation* are intersected to provide a candidate search space for dense units of higher dimensionality.

CLUSTERING HIGH-DIMENSIONAL DATA

Clustering is the task of dividing the population or data points into a number of groups such that data points in the same groups are more similar to other data points in the same group and dissimilar to the data points in other groups.

Challenges of Clustering High-Dimensional Data:

- Clustering of the High-Dimensional Data return the group of objects which are clusters.
- It is required to group similar types of objects together to perform the cluster analysis of high-dimensional data, But the High-Dimensional data space is huge and it has complex data types and attributes.
- A major challenge is that we need to find out the set of attributes that are present in each cluster.
- A cluster is defined and characterized based on the attributes present in the cluster. Clustering High-Dimensional Data we need to search for clusters and find out the space for the existing clusters.
- The High-Dimensional data is reduced to low-dimension data to make the clustering and search for clusters simple.
- some applications need the appropriate models of clusters, especially the high-dimensional data.
- clusters in the high-dimensional data are significantly small. the conventional distance measures can be ineffective.
- Instead, To find the hidden clusters in high-dimensional data we need to apply sophisticated techniques that can model correlations among the objects in subspaces.

Subspace Clustering Methods:

There are 3 Subspace Clustering Methods:

- **Subspace search methods**
- **Correlation-based clustering methods**
- **Biclustering methods**

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Subspace Search Methods:

Subspace clustering or subspace search methods aim to identify clusters within specific subspaces or combinations of attributes (dimensions) rather than the entire feature space. This is particularly useful when clusters exist only in certain dimensions or when different subsets of attributes define distinct clusters.

Correlation-Based Clustering Methods:

Correlation-based clustering methods focus on identifying clusters based on the correlation or similarity between attributes (features) rather than the distances between data points. These methods are useful when the relationships between features are of interest in defining clusters.

Biclustering Methods:

Biclustering methods, also known as co-clustering or two-mode clustering, simultaneously cluster rows and columns of a data matrix. These methods are particularly useful for identifying subsets of data points (rows) that exhibit similar behavior across subsets of attributes (columns).

CONSTRAINT-BASED CLUSTER ANALYSIS

Methods For Clustering With Constraints:

There are various methods for clustering with constraints and can handle specific constraints:

- **Handling Hard Constraints:** There is a method for handling the hard constraints by regarding the constraint in a cluster assignment procedure. It is a very important method for handling the difficult constraints we can regard the constraints in the assignment procedure of cluster.
- **Generating the super instances for must-link constraints**: There are must-link constraints that have transitive closure that can be calculated by it. so that we can say that must-link constraints are known as an equivalence relation. The subset can be defined by it. In subset, there are some objects which can be replaced by the mean.
- **Handling the soft constraints:** In the clustering process of soft constraints, there is always an optimization process. There is always a penalty requires in the clustering process. Hence the optimization in this process's aim is to optimize the constraint violation and decreasing the clustering aspect. For example, if we take two sets one is of data sets and the other is a set of constraints, CVQE stands for Constrained Vector Quantization Error. In the CVQE algorithm, K-means clustering is enforced to constraint violation penalty.
- The main objective of CVQE is the total distance used for K-means which are used as follows:

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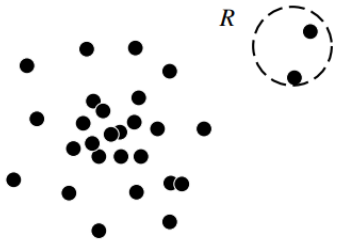
  - **Penalty in must-link violation:** This penalty occurs due to when there is a must-link constraint present on objects y, x. They are created to the given two centers C1, C2 by which the constraint can be violated hence the distance that lies between C1 and C2 is inserted but as a penalty.

  - **Penalty in cannot-link violation:** This type of penalty is different from a must-link violation as in this penalty there is one center created to a common center C when cannot link is present on objects x, y. Therefore the constraints are violated and hence the distance that lies between (C, C) can be inserted in the objective function and it is recognized as a penalty.

OUTLIER ANALYSIS

Outlier analysis in data mining is the process of identifying and examining data points that significantly differ from the rest of the dataset. An outlier can be defined as a data point that deviates significantly from the normal pattern or behavior of the data.

most objects follow a roughly Gaussian distribution. However, the objects in region R are significantly different. It is unlikely that they follow the same distribution as the other objects in the data set. Thus, the objects in R a re outliers in the data set.



The objects in region *R* are outliers.

- Outliers are different from noisy data.
- noise is a random error or variance in a measured variable.
- In general, noise is not interesting in data analysis, including outlier detection.
- For example, in credit card fraud detection, a customer's purchase behavior can be modeled as a random variable.
- A customer may generate some "noise transactions" that may seem like "random errors" or "variance," such as by buying a bigger lunch one day, or having one more cup of coffee than usual.
- Such transactions should not be treated as outliers; otherwise, the credit card company would incur heavy costs from verifying that many transactions.
- The company may also lose customers by bothering them with multiple false alarms.
- As in many other data analysis and data mining tasks, noise should be removed before outlier detection.

Outliers are of three types

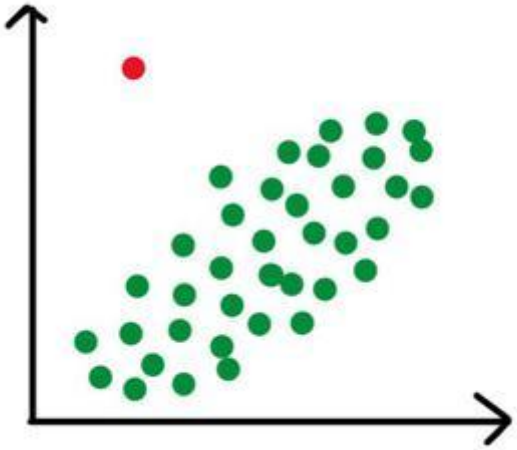
- **Global (or Point) Outliers**
- **Collective Outliers**
- **Contextual (or Conditional) Outliers**

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1. Global Outliers

1. Definition: Global outliers are data points that deviate significantly from the overall distribution of a dataset.
2. Causes: Errors in data collection, measurement errors, or truly unusual events can result in global outliers.
3. Impact: Global outliers can distort data analysis results and affect machine learning model performance.
4. Detection: Techniques include statistical methods (e.g., z-score, Mahalanobis distance), machine learning algorithms (e.g., isolation forest, one-class SVM), and data visualization techniques.
5. Handling: Options may include removing or correcting outliers, transforming data, or using robust methods.

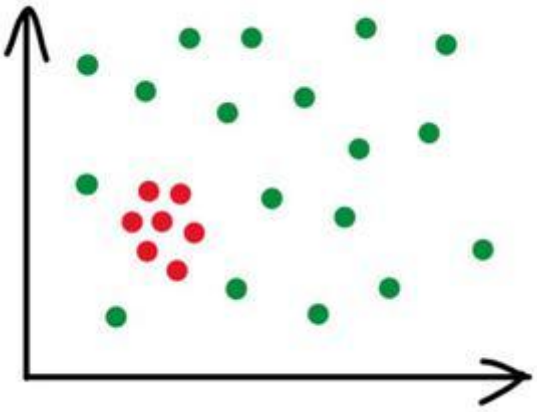
6. Considerations: Carefully considering the impact of global outliers is crucial for accurate data analysis and machine learning model outcomes.



2. Collective Outliers

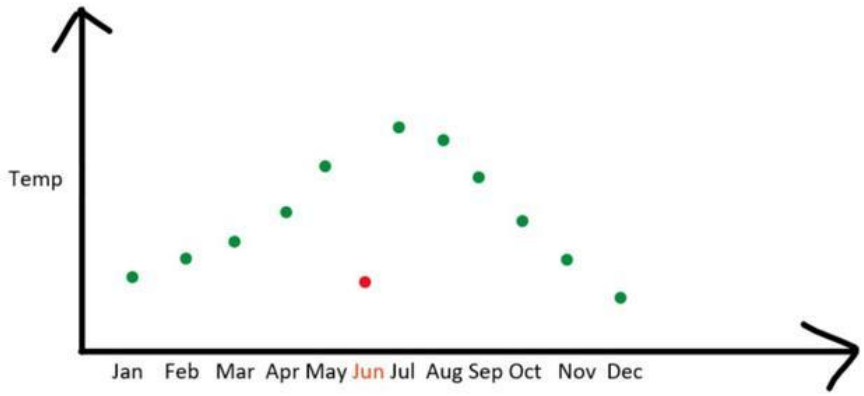
1. Definition: Collective outliers are groups of data points that collectively deviate significantly from the overall distribution of a dataset.
2. Characteristics: Collective outliers may not be outliers when considered individually, but as a group, they exhibit unusual behavior.
3. Detection: Techniques for detecting collective outliers include clustering algorithms, density-based methods, and subspace-based approaches.
4 Impact: Collective outliers can represent interesting patterns or anomalies in data that may require special attention or further investigation.
5. Handling: Handling collective outliers depends on the specific use case and may involve further analysis of the group behavior, identification of contributing factors, or considering contextual information.
6. Considerations: Detecting and interpreting collective outliers can be more complex than individual outliers, as the focus is on group behavior rather than individual data points. Proper understanding of the data context and domain knowledge is crucial for effective handling of collective outliers.

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3. Contextual Outliers

1. Definition: Contextual outliers are data points that deviate significantly from the expected behavior within a specific context or subgroup.
2. Characteristics: Contextual outliers may not be outliers when considered in the entire dataset, but they exhibit unusual behavior within a specific context or subgroup.
3. Detection: Techniques for detecting contextual outliers include contextual clustering, contextual anomaly detection, and context-aware machine learning approaches.
4. Contextual Information: Contextual information such as time, location, or other relevant factors are crucial in identifying contextual outliers.
5. Impact: Contextual outliers can represent unusual or anomalous behavior within a specific context, which may require further investigation or attention.
6. Handling: Handling contextual outliers may involve considering the contextual information, contextual normalization or transformation of data, or using context-specific models or algorithms.
7. Considerations: Proper understanding of the context and domain-specific knowledge is crucial for accurate detection and interpretation of contextual outliers, as they may vary based on the specific context or subgroup being considered.



Challenges of Outlier Detection

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- **Modeling normal objects and outliers effectively.** Outlier detection quality highly depends on the modeling of normal (nonoutlier) objects and outliers. Often, building a comprehensive model for data normality is very challenging, if not impossible. This is partly because it is hard to enumerate all possible normal behaviors in an application.

- **Application-specific outlier detection.** Technically, choosing the similarity/distance measure and the relationship model to describe data objects is critical in outlier detection. Unfortunately, such choices are often application-dependent. Different applications may have very different requirements.

- **Handling noise in outlier detection.** As mentioned earlier, outliers are different from noise. It is also well known that the quality of real data sets tends to be poor. Noise often unavoidably exists in data collected in many applications.

- **Understandability.** In some application scenarios, a user may want to not only detect outliers, but also understand why the detected objects are outliers. To meet the understandability requirement, an outlier detection method has to provide some justification of the detection.

UNIT – 5

MINING OBJECT, SPATIAL, MULTIMEDIA, TEXT AND WEB DATA

MULTIDIMENSIONAL ANALYSIS AND DESCRIPTIVE MINING OF COMPLEX DATA OBJECTS

Multi-Dimensional Analysis, also known as OLAP (Online Analytical Processing), is a data processing technique that enables businesses to analyze and explore data from multiple dimensions simultaneously. It allows users to slice and dice data along different dimensions to gain deeper insights and understand complex relationships within the data.

Descriptive mining is usually used to provide correlation, cross-tabulation, frequency, etc. These techniques are used to determine the data regularities and to reveal patterns. It targets the summarization and conversion of data into meaningful data for reporting and monitoring.

Data mining, complex data objects refer to data structures that go beyond traditional tabular or simple relational data. They often involve intricate relationships, multi-layered hierarchies, or multi-dimensional attributes, making them more challenging to analyse

Multidimensional Analysis and Descriptive Mining of Complex Data Objects

- Set-valued attribute
  - Generalization of each value in the set into its corresponding higher-level concepts
  - Derivation of the general behavior of the set, such as the number of elements in the set, the types or value ranges in the set, or the weighted average for numerical data
  - E.g., hobby = {tennis, hockey, chess, violin, nintendo_games} generalizes to {sports, music, video_games}
- List-valued or sequence-valued attribute
  - Same as set-valued attributes except that the order of the elements in the sequence should be observed in the generalization

Aggregation and Approximation in Spatial and Multimedia Data Generalization

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Aggregation and approximation should be considered another important means of generalization, which is especially useful for generalizing attributes with large sets of values, complex structure and spatial or multimedia data.

A multimedia data may contain complex texts,graphics,images,video fragments,maps, voice,music,and other forms of video/audio information.

Generalization of object identifies and Class/Subclass Hierarchies

Methods are an important component of object-oriented databases. Many behavioral data objects can be derived by the application of methods

- Spatial data:
  - Generalize detailed geographic points into clustered regions, such as business, residential, industrial, or agricultural areas, according to land usage
  - Require the merge of a set of geographic areas by spatial operations
- Image data:

- Extracted by aggregation and/or approximation
- Size, color, shape, texture, orientation, and relative positions and structures of the contained objects or regions in the image

➢ Music data:
- Summarize its melody: based on the approximate patterns that repeatedly occur in the segment
- Summarized its style: based on its tone, tempo, or the major musical instruments played

SPATIAL DATA MINING

Spatial data mining refers to the process of discovering hidden patterns and relationships in geospatial data. Geospatial data includes information that is associated with specific locations or geographic coordinates.

- Spatial data mining discovers patterns and knowledge from spatial data.
- In many cases, refer to geospace-related data stored in geospatial data repositories. The data can be in "vector" or "raster" formats, or in the form of imagery and geo-referenced multimedia.
- Recently, large geographic data warehouses have been constructed by integrating thematic and geographically referenced data from multiple sources.
- From these, we can construct spatial data cubes that contain spatial dimensions and measures, and support spatial OLAP for multidimensional spatial data analysis.
- Spatial data mining can be performed on spatial data warehouses, spatial databases, and other geospatial data repositories.
- Popular topics on geographic knowledge discovery and spatial data mining include mining spatial associations and co-location patterns, spatial clustering, spatial classification, spatial modeling, and spatial trend and outlier analysis.

Key Concepts in Spatial Data Mining

1. **Spatial Data:**
   - **Geographic Information Systems (GIS):** Systems designed to capture, store, manipulate, analyze, manage, and present spatial or geographic data.
   - **Spatial Objects:** Points (e.g., locations of stores), lines (e.g., roads, rivers), and polygons (e.g., land use areas, administrative boundaries).
2. **Spatial Data Types:**
   - **Vector Data:** Represented by points, lines, and polygons. Examples include city locations, road networks, and land parcels.
   - **Raster Data:** Gridded data with each cell representing a value. Examples include satellite imagery and elevation data.

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Techniques in Spatial Data Mining

1. **Spatial Clustering:**
   - **Purpose:** Group spatial objects into clusters based on their locations and attributes.
   - **Techniques:**
     - **K-Means Clustering:** Partitions spatial data into k clusters.
     - **DBSCAN (Density-Based Spatial Clustering of Applications with Noise):** Identifies clusters based on density and can handle noise.
2. **Spatial Association Rules:**
   - **Purpose:** Discover interesting relationships between spatial objects.
   - **Examples:** Identifying areas with high crime rates that are close to specific types of businesses.
3. **Spatial Classification:**

- **Purpose:** Classify spatial objects into predefined categories.
- **Techniques:**
  - **Decision Trees:** Classify spatial features based on their attributes.
  - **Support Vector Machines (SVMs):** Classify spatial objects based on a feature set.

4. **Spatial Regression:**
   - **Purpose:** Model and analyze spatial dependencies and relationships.
   - **Techniques:**
     - **Geographically Weighted Regression (GWR):** Assesses spatial variations in relationships.
     - **Spatial Autoregressive Models:** Account for spatial dependencies in regression analysis.
5. **Spatial Outlier Detection:**
   - **Purpose:** Identify spatial objects that significantly differ from their neighbors.
   - **Techniques:**
     - **Local Outlier Factor (LOF):** Detects spatial outliers based on local density.
6. **Spatial Trend Analysis:**
   - **Purpose:** Identify and analyze trends in spatial data over time.
   - **Techniques:** Time-series analysis of spatial data, such as tracking changes in land use over time.

Applications of Spatial Data Mining

1. **Urban Planning:**
   - **Land Use Planning:** Analyze land use patterns to make zoning decisions.
   - **Infrastructure Development:** Plan for new roads, utilities, and public services based on spatial data.
2. **Environmental Management:**
   - **Conservation:** Identify areas needing protection based on ecological patterns and threats.
   - **Disaster Management:** Analyze the impact of natural disasters on different regions and plan for emergency response.
3. **Transportation:**
   - **Traffic Management:** Optimize traffic flow and detect patterns in traffic congestion.
   - **Route Planning:** Develop efficient routes for delivery services or public transportation.
4. **Public Health:**
   - **Disease Tracking:** Analyze spatial patterns of disease outbreaks and their spread.
   - **Healthcare Accessibility:** Assess the accessibility of healthcare services in different regions.
5. **Retail and Marketing:**
   - **Market Analysis:** Identify optimal locations for new stores based on customer distribution and competitor locations.
   - **Customer Segmentation:** Understand spatial distribution of customer segments to tailor marketing strategies.

Tools and Technologies

- **GIS Software:** Tools such as ArcGIS, QGIS, and Google Earth provide functionalities for spatial analysis and data visualization.
- **Spatial Data Mining Libraries:** Libraries like GeoPandas, Shapely, and PySAL in Python support spatial data analysis and mining.

Challenges in Spatial Data Mining

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1. **Data Volume and Complexity:**
   - Handling large volumes of spatial data and managing complex spatial relationships can be challenging.
2. **Integration of Diverse Data Sources:**
   - Combining data from various sources (e.g., satellite imagery, survey data) requires alignment and integration.
3. **Scalability:**
   - Developing scalable algorithms and systems to handle extensive spatial datasets and perform real-time analysis.
4. **Spatial Dependency:**
   - Accounting for spatial autocorrelation (where nearby locations are more similar than distant ones) in analysis.

MULTIMEDIA DATA MINING

Multimedia data mining is the discovery of interesting patterns from multimedia databases that store and manage large collections of multimedia objects, including image data, video data, audio data, as well as sequence data and hypertext data containing text, text markups, and linkages.

- Multimedia data mining is an interdisciplinary field that integrates image processing and understanding, computer vision, data mining, and pattern recognition.
- Issues in multimedia data mining include content-based retrieval and similarity search, and generalization and multidimensional analysis.
- Multimedia data cubes contain additional dimensions and measures for multimedia information. Other topics in multimedia mining include classification and prediction analysis, mining associations, and video and audio data mining

Key Concepts in Multimedia Data Mining

1. **Multimedia Data Types:**
   - **Text:** Includes documents, web pages, social media posts, and more.
   - **Images:** Still photographs, graphics, and illustrations.
   - **Audio:** Spoken words, music, sound effects, and recorded conversations.
   - **Video:** Moving images with audio, including movies, video clips, and streaming content.
   - **3D Data:** Includes 3D models and environments used in virtual reality and simulations.
2. **Data Representation:**
   - **Feature Extraction:** Converting raw multimedia data into a structured form suitable for analysis (e.g., extracting key features from images or audio).
   - **Indexing and Retrieval:** Organizing and retrieving multimedia data based on its content and metadata.

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Techniques in Multimedia Data Mining

1. Text Mining

Objective: Extract meaningful information from textual data.
Techniques:

- **Natural Language Processing (NLP):** Analyzes and processes human language (e.g., sentiment analysis, topic modeling, named entity recognition).
- **Text Classification:** Categorizes text into predefined categories (e.g., spam detection in emails).

- **Information Retrieval:** Finds and retrieves relevant documents or text snippets based on queries.

Applications:

- **Sentiment Analysis:** Understanding public opinion from social media posts or reviews.
- **Topic Modeling:** Identifying themes or topics in a collection of documents.

2. Image Mining

Objective: Extract patterns and insights from images.

Techniques:

- **Image Classification:** Assigns categories to images based on their content (e.g., identifying objects or scenes in images).
- **Object Detection:** Locates and classifies objects within images (e.g., detecting faces or vehicles).
- **Image Segmentation:** Divides an image into meaningful regions (e.g., separating foreground from background).

Applications:

- **Medical Imaging:** Detecting diseases or anomalies in medical scans.
- **Surveillance:** Monitoring and identifying individuals or objects in security footage.

3. Audio Mining

Objective: Extract information from audio data.

Techniques:

- **Speech Recognition:** Converts spoken language into text (e.g., transcribing interviews or voice commands).
- **Audio Classification:** Categorizes audio signals based on content (e.g., music genre classification).
- **Speaker Identification:** Identifies or verifies speakers based on their voice.

Applications:

- **Voice Assistants:** Enabling interaction with virtual assistants through voice commands.
- **Music Recommendation:** Suggesting music based on audio features and user preferences.

4. Video Mining

Objective: Analyze and extract information from video data.

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Techniques:

- **Video Classification:** Categorizes videos into different types or genres (e.g., classifying video clips as sports or news).
- **Action Recognition:** Identifies specific actions or activities in video sequences (e.g., recognizing a person running).
- **Video Summarization:** Creates concise summaries or highlights from long video content.

Applications:

- **Content Moderation:** Filtering inappropriate or harmful content from video platforms.
- **Event Detection:** Monitoring and detecting significant events in surveillance videos.

5. 3D Data Mining

Objective: Analyze and interpret 3D data such as models and simulations.

Techniques:

- **3D Object Recognition:** Identifies and classifies objects within 3D models or scenes.
- **Shape Analysis:** Analyzes the shape and structure of 3D objects (e.g., comparing anatomical structures).
- **3D Reconstruction:** Reconstructs 3D models from 2D images or point clouds.

Applications:

- **Virtual Reality:** Creating realistic 3D environments for simulations or games.
- **Medical Visualization:** Analyzing 3D models of organs or tissues for diagnosis and treatment planning.

Multimedia Data Mining Process

1. **Data Collection:**
   - **Gather multimedia data** from various sources like databases, sensors, web content, or user-generated content.
2. **Data Preprocessing:**
   - **Clean and organize** multimedia data (e.g., noise reduction in audio, resizing images).
   - **Feature Extraction:** Convert multimedia content into features suitable for analysis (e.g., extracting color histograms from images).
3. **Data Analysis:**
   - **Apply mining techniques** such as classification, clustering, or regression to the extracted features.
   - **Pattern Discovery:** Identify patterns or anomalies in the data (e.g., detecting unusual behavior in video surveillance).
4. **Interpretation and Evaluation:**
   - **Interpret results** to derive meaningful insights (e.g., understanding customer sentiments from product reviews).
   - **Evaluate the performance** of mining algorithms using metrics like accuracy, precision, and recall.
5. **Visualization and Reporting:**
   - **Visualize findings** using charts, graphs, and multimedia representations (e.g., heatmaps for video analytics).
   - **Generate reports** to communicate results and insights to stakeholders.
6. **Decision Making:**
   - **Use insights** from multimedia data mining to inform decisions and actions (e.g., targeting advertising campaigns based on user behavior).

Challenges in Multimedia Data Mining

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1. **Data Complexity:**
   - Multimedia data is often high-dimensional and heterogeneous, making analysis complex.
2. **Scalability:**
   - Handling and processing large volumes of multimedia data requires efficient algorithms and computational resources.
3. **Integration:**
   - Combining and analyzing different types of multimedia data (e.g., text and video) can be challenging.
4. **Semantics:**
   - Understanding and interpreting the semantic meaning of multimedia content accurately can be difficult.
5. **Privacy and Security:**
   - Ensuring the privacy and security of sensitive multimedia data (e.g., personal videos or audio recordings) is crucial.

TEXT MINING

Text Mining is the process of extracting useful information and insights from unstructured text data. It involves converting text into structured data that can be analyzed and used to discover patterns, relationships, and knowledge. Text mining combines techniques from natural language processing (NLP), statistics, and machine learning to analyze and interpret textual information.

- Text mining is an interdisciplinary field that draws on information retrieval, data mining, machine learning, statistics, and computational linguistics.
- A substantial portion of information is stored as text such as news articles, technical papers, books, digital libraries, email messages, blogs, and web pages. Hence, research in text mining has been very active.
- An important goal is to derive high-quality information from text.
- This is typically done through the discovery of patterns and trends by means such as statistical pattern learning, topic modeling, and statistical language modeling.
- Text mining usually requires structuring the input text (e.g., parsing, along with the addition of some derived linguistic features and the removal of others, and subsequent insertion into a database).
- This is followed by deriving patterns within the structured data, and evaluation and interpretation of the output.
- "High quality" in text mining usually refers to a combination of relevance, novelty, and interestingness.
- Typical text mining tasks include text categorization, text clustering, concept/entity extraction, production of granular taxonomies, sentiment analysis, document summarization, and entity-relation modeling (i.e., learning relations between named entities).
- Examples include multilingual data mining, multidimensional text analysis, contextual text mining, and trust and evolution analysis in text data, as well as text mining applications in security, biomedical literature analysis, online media analysis, and analytical customer relationship management.
- Various kinds of text mining and analysis software and tools are available in academic institutions, open-source forums, and industry.
- Text mining often also uses WordNet, Sematic Web, Wikipedia, and other information sources to enhance the understanding and mining of text data.

Key Concepts in Text Mining

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1. **Unstructured Text Data:**
   - Text that lacks a predefined structure, such as articles, social media posts, emails, reviews, and documents.
2. **Structured Data:**
   - Text that has been converted into a structured format, often in the form of tables or databases, to facilitate analysis.

Steps in Text Mining Process

1. **Text Collection:**
   - **Objective:** Gather text data from various sources.
   - **Sources:** Websites, social media, databases, documents, email communications, etc.

2. **Text Preprocessing:**
   - **Objective:** Prepare text data for analysis by cleaning and normalizing it.
   - **Tasks:**
     - **Tokenization:** Splitting text into individual words or tokens.

- **Stopword Removal:** Removing common words that do not carry significant meaning (e.g., "the," "is").
- **Stemming and Lemmatization:** Reducing words to their base or root forms (e.g., "running" to "run").
- **Lowercasing:** Converting all text to lowercase to ensure uniformity.
- **Normalization:** Handling variations in text format, such as correcting spelling errors and expanding abbreviations.

3. **Feature Extraction:**
   - **Objective:** Convert text into a numerical or structured format that can be used for analysis.
   - **Techniques:**
     - **Bag of Words (BoW):** Representing text as a collection of word counts or frequencies.
     - **Term Frequency-Inverse Document Frequency (TF-IDF):** A statistical measure that evaluates the importance of a word in a document relative to a collection of documents.
     - **Word Embeddings:** Representing words in a continuous vector space (e.g., Word2Vec, GloVe) to capture semantic relationships between words.

4. **Text Analysis:**
   - **Objective:** Apply various mining techniques to extract patterns, trends, and insights.
   - **Techniques:**
     - **Text Classification:** Categorizing text into predefined categories (e.g., spam detection in emails).
     - **Sentiment Analysis:** Determining the sentiment expressed in text (e.g., positive, negative, neutral).
     - **Topic Modeling:** Identifying topics or themes in a collection of documents (e.g., Latent Dirichlet Allocation - LDA).
     - **Named Entity Recognition (NER):** Identifying and classifying entities like names, dates, and locations within text.
     - **Part-of-Speech Tagging:** Labeling words with their grammatical roles (e.g., noun, verb).

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5. **Pattern Discovery:**
   - **Objective:** Identify patterns and relationships in the text data.
   - **Techniques:**
     - **Association Rule Mining:** Discovering relationships between terms or phrases.
     - **Clustering:** Grouping similar documents or texts based on their content (e.g., K-Means clustering).
     - **Relationship Extraction:** Identifying relationships between entities mentioned in the text.

6. **Visualization and Interpretation:**
   - **Objective:** Present the findings in a comprehensible manner.
   - **Tasks:**
     - **Data Visualization:** Creating charts, word clouds, and graphs to visualize text data and analysis results.
     - **Reporting:** Summarizing insights and patterns in a report or presentation.

7. **Decision Making:**

- **Objective:** Use the insights gained from text mining to make informed decisions or take actions.
- **Applications:** Informing business strategies, improving customer service, developing new products, etc.

Applications of Text Mining

1. **Customer Feedback Analysis:**
   - Analyzing customer reviews and feedback to understand sentiment and improve products or services.
2. **Social Media Monitoring:**
   - Tracking social media conversations to gauge public opinion, detect trends, and identify potential issues.
3. **Information Retrieval:**
   - Enhancing search engines and recommendation systems by improving the relevance and accuracy of search results.
4. **Fraud Detection:**
   - Identifying fraudulent activities or patterns by analyzing transaction descriptions and communications.
5. **Healthcare:**
   - Analyzing medical records, research papers, and patient feedback to support diagnosis, treatment planning, and research.
6. **Legal Document Analysis:**
   - Reviewing legal documents to extract key information, identify relevant case laws, and support legal research.

Challenges in Text Mining

1. **Text Ambiguity:**
   - Handling ambiguities in language, such as homonyms or polysemous words (words with multiple meanings).
2. **Context Understanding:**
   - Understanding the context and nuance of language, including idiomatic expressions and domain-specific terminology.
3. **Data Volume:**
   - Managing and processing large volumes of text data efficiently.
4. **Data Quality:**
   - Dealing with noisy or incomplete text data, such as typos, slang, and informal language.
5. **Scalability:**
   - Ensuring that text mining techniques can scale to handle increasing amounts of data.

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Tools and Technologies

- **Natural Language Processing Libraries:** NLTK, SpaCy, and TextBlob for text processing and analysis.
- **Machine Learning Frameworks:** TensorFlow, PyTorch, and Scikit-learn for applying machine learning techniques.
- **Data Visualization Tools:** Tableau, Power BI, and D3.js for visualizing text data and analysis results.
- **Text Mining Platforms:** RapidMiner, KNIME, and IBM Watson for comprehensive text mining solutions.

MINING THE WORLD WIDE WEB

- The World Wide Web serves as a huge, widely distributed, global information center for news, advertisements, consumer information, financial management, education, government, and e-commerce.
- It contains a rich and dynamic collection of information about web page contents with hypertext structures and multimedia, hyperlink information, and access and usage information, providing fertile sources for data mining.
- Web mining is the application of data mining techniques to discover patterns, structures, and knowledge from the Web.
- According to analysis targets, web mining can be organized into three main areas: web content mining, web structure mining, and web usage mining.
- Web content mining analyzes web content such as text, multimedia data, and structured data (within web pages or linked across web pages).
- This is done to understand the content of web pages, provide scalable and informative keyword-based page indexing, entity/concept resolution, web page relevance and ranking, web page content summaries, and other valuable information related to web search and analysis.
- Web pages can reside either on the surface web or on the deep Web.
- The surface web is that portion of the Web that is indexed by typical search engines.
- The deep Web (or hidden Web) refers to web content that is not part of the surface web.
- Its contents are provided by underlying database engines.
- Web content mining has been studied extensively by researchers, search engines, and other web service companies.
- Web content mining can build links across multiple web pages for individuals; therefore, it has the potential to inappropriately disclose personal information.
- Studies on privacy-preserving data mining address this concern through the development of techniques to protect personal privacy on the Web.
- Web structure mining is the process of using graph and network mining theory and methods to analyze the nodes and connection structures on the Web.
- It extracts patterns from hyperlinks, where a hyperlink is a structural component that connects a web page to another location.
- It can also mine the document structure within a page (e.g., analyze the treelike structure of page structures to describe HTML or XML tag usage).
- Both kinds of web structure mining help us understand web content and may also help transform web content into relatively structured data sets.

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- Web usage mining is the process of extracting useful information (e.g., user click streams) from server logs.
- It finds patterns related to general or particular groups of users; understands users' search patterns, trends, and associations; and predicts what users are looking for on the Internet.
- It helps improve search efficiency and effectiveness, as well as promotes products or related information to different groups of users at the right time.
- Web search companies routinely conduct web usage mining to improve their quality of service.

Advantages of mining world wide web

1. Access to lots of data
2. Real-Time updates
3. Better decision-making
4. Cost-effective
5. Competitive edge

6. Deeper customer insights
7. Discover trends and patterns
8. Improved search results
9. Fraud detection
10. Innovation opportunities
11. Personalized experiences
12. Geographic and demographic insights.

Issues in mining world wide web

1. Data volume and complexity
2. Data quality
3. Dynamic content
4. Data privacy and security
5. Diverse data formats
6. Data integration
7. Scalability issues
8. Crawling and indexing challenges
9. Ethical consideration
10. Language and semantics
11. Legal and compliance issues
12. Server load and bandwidth
13. Content duplication.

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